SWINGERsim

Physics and Methods

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Contents

1 Hierarchies and orbits **3**
- 1.1 Hierarchical multiple stars . . . 3
- 1.2 Jacobi (hierarchical) coordinates . . . 5
- 1.3 Orbital elements and the reference frame . . . 5
- 1.4 Mutual inclination . . . 7
- 1.5 Nodal precession . . . 7
- 1.6 ZLK cycles . . . 8
- 1.7 Apsidal motion . . . 10
- 1.8 Resonances . . . 10
- 1.9 Coupling of the orbital planes of A and B . . . 11
- 1.10 Larger hierarchies and a fifth star . . . 11
- 1.11 Implementation . . . 11

2 Stars **12**
- 2.1 The Hertzsprung–Russell diagram . . . 12
- 2.2 Stellar models . . . 13
- 2.3 Structure constants: k_2 and r_g^2 . . . 16
- 2.4 Roche lobes . . . 16
- 2.5 Kinds of systems . . . 18
- 2.6 Implementation . . . 18

3 Dynamics **19**
- 3.1 Direct N-body integration . . . 19
- 3.2 Relativity . . . 19
- 3.3 Conservation checks . . . 20
- 3.4 Collisions . . . 20
- 3.5 Hierarchy detection . . . 20
- 3.6 Diagnostics of orbit pairs . . . 20
- 3.7 Backward integration . . . 22
- 3.8 Secular backends . . . 22
- 3.9 External perturbations . . . 25
- 3.10 Implementation . . . 26

4 Tides and rotation **27**
- 4.1 Equilibrium tides in the N-body backend . . . 28
- 4.2 Classical apsidal motion of a binary . . . 28
- 4.3 Tidal evolution: the constant-time-lag model . . . 29
- 4.4 Dissipation from the stellar structure . . . 29
- 4.5 Limits . . . 31
- 4.6 Implementation . . . 31

5 Mass transfer and evolution **32**
- 5.1 Coupling stellar evolution to the dynamics . . . 32
- 5.2 Winds . . . 32
- 5.3 Roche-lobe overflow and stable mass transfer . . . 33
- 5.4 Histories in the dynamics . . . 35
- 5.5 Semi-detached and contact binaries . . . 36

5.6 Tertiary mass transfer . . . 36
5.7 Common envelope . . . 37
5.8 Mergers . . . 38
5.9 Magnetic braking and activity . . . 38
5.10 Limits . . . 38
5.11 Implementation . . . 39

6 Stability and fates **40**
6.1 Algebraic criteria for nested triples . . . 40
6.2 Machine-learning classifiers . . . 41
6.3 The N-body ghost test . . . 42
6.4 Chaos maps . . . 42
6.5 Fates and their classification . . . 43
6.6 Survival in both directions of time . . . 44
6.7 Scattering experiments . . . 45
6.8 Statistical escape theory . . . 45
6.9 Implementation . . . 46

7 Observables **47**
7.1 Eclipse geometry . . . 47
7.2 Light curves . . . 49
7.3 Radial velocities . . . 52
7.4 Times of minima and O−C diagrams . . . 53
7.5 Pulsating stars as clocks . . . 56
7.6 Discs as occulters . . . 57
7.7 What is not modelled . . . 58
7.8 Implementation . . . 58

8 Fitting observed data **59**
8.1 Data . . . 60
8.2 Stage 1: analytic O−C model per pair . . . 60
8.3 Anchoring to the present . . . 61
8.4 Stage 2: the joint N-body model . . . 61
8.5 Errors per source, jitter and outliers . . . 63
8.6 MCMC uncertainties . . . 64
8.7 Predictions . . . 65
8.8 Limits . . . 65
8.9 Implementation . . . 65

9 Ensembles and inference **66**
9.1 What is varied . . . 67
9.2 Statistical error of a fraction . . . 68
9.3 Conditioning on observations . . . 68
9.4 Emulators . . . 70
9.5 Simulation-based inference . . . 70
9.6 Implementation . . . 71

10 Accuracy and limits **72**
10.1 Numerical accuracy . . . 72
10.2 Regime limits . . . 73
10.3 What is not modelled . . . 73
10.4 The validation principle . . . 74
10.5 Implementation . . . 74

11 References **75**

Chapter 1

Hierarchies and orbits

This volume describes the physics and the numerical methods of SWINGERsim, a simulator of hierarchical multiple stars. For each piece of physics it states what is modelled, the equations, the assumptions, the numerical method, the configuration keys that control it, how it was validated and where it stops being valid. The validation itself – the comparison of every module with published results – is documented in *Testing SWINGERsim, Part I: Validation against the literature*; here each method carries a one-sentence summary of its test. How to write a configuration, run the commands and read the outputs is the subject of the *User Guide* (chapters Configuration, Command line and GUI).

This first chapter defines the geometry: how a multiple star is described as a tree of orbits, which coordinates and orbital elements SWINGERsim uses, how the observer enters, and which slow (secular) phenomena change the orbits over time.

1.1 Hierarchical multiple stars

A *multiple star* is a gravitationally bound system of three or more stars. About half of the Sun-like stars and nearly all massive stars have at least one companion, and many systems contain three, four or more stars (Tokovinin 2018). Such a system survives for a long time only if it is *hierarchical*: it can be split into pairs whose orbits have very different sizes. Two stars close together form a *binary*; seen from a distance much larger than its separation, the binary acts on other bodies almost like a single star of the combined mass. It can therefore orbit another star or another binary on a much wider orbit. Each orbit is then nearly a fixed ellipse (a *Keplerian* orbit), slowly perturbed by the others. Without this separation of scales the stars exchange partners and eject each other within a few orbits (chapter Stability and fates).

A hierarchy is a **binary tree**: every node is an orbit that joins two components, and each component is either a star (a leaf) or another orbit. SWINGERsim accepts any such tree up to about ten stars. The common ones are presets of the key `topology`; any other tree is given explicitly with `topology: custom` and `hierarchy`, a map from the name of each orbit to its two components; Table 1 lists the presets.

Table 1. Hierarchies available as presets. *Preset* is the value of `topology`; *Stars* the number of stars; *Orbits* the names under which the orbits appear in the configuration and in all outputs; *Description* the arrangement. The notation “2+2” means two binaries orbiting each other, “3+1” a triple (a binary with a close third star) orbited by a distant fourth star; parentheses mark a sub-system inside a larger one.

Preset	Stars	Orbits	Description
`custom`	2–3	free names, e.g. `A`, `out`	a binary, or a triple (“2+1”)
`2+2`	4	`A`, `B`, `out`	two binaries A = (m1, m2) and B = (m3, m4) on a mutual orbit

Preset	Stars	Orbits	Description
`3+1`	4	`inner`, `mid`, `out`	binary (m1, m2), star m3 around it, star m4 around the triple
`2+2+1`, `3+1+1`	5	as above plus `out5`	a quadruple with a distant fifth star
`2+(2+1)`	5	`A`, `B`, `Bt`, `out`	binary A and a triple made of binary B and a close star m5
`2+2+2`	6	`A`, `B`, `C`, `AB`, `out`	a 2+2 quadruple AB and a third binary C
`3+3`	6	`A`, `B`, `T1`, `T2`, `out`	two triples on a mutual orbit
`custom`	up to ~10	free names	any tree of nested binaries

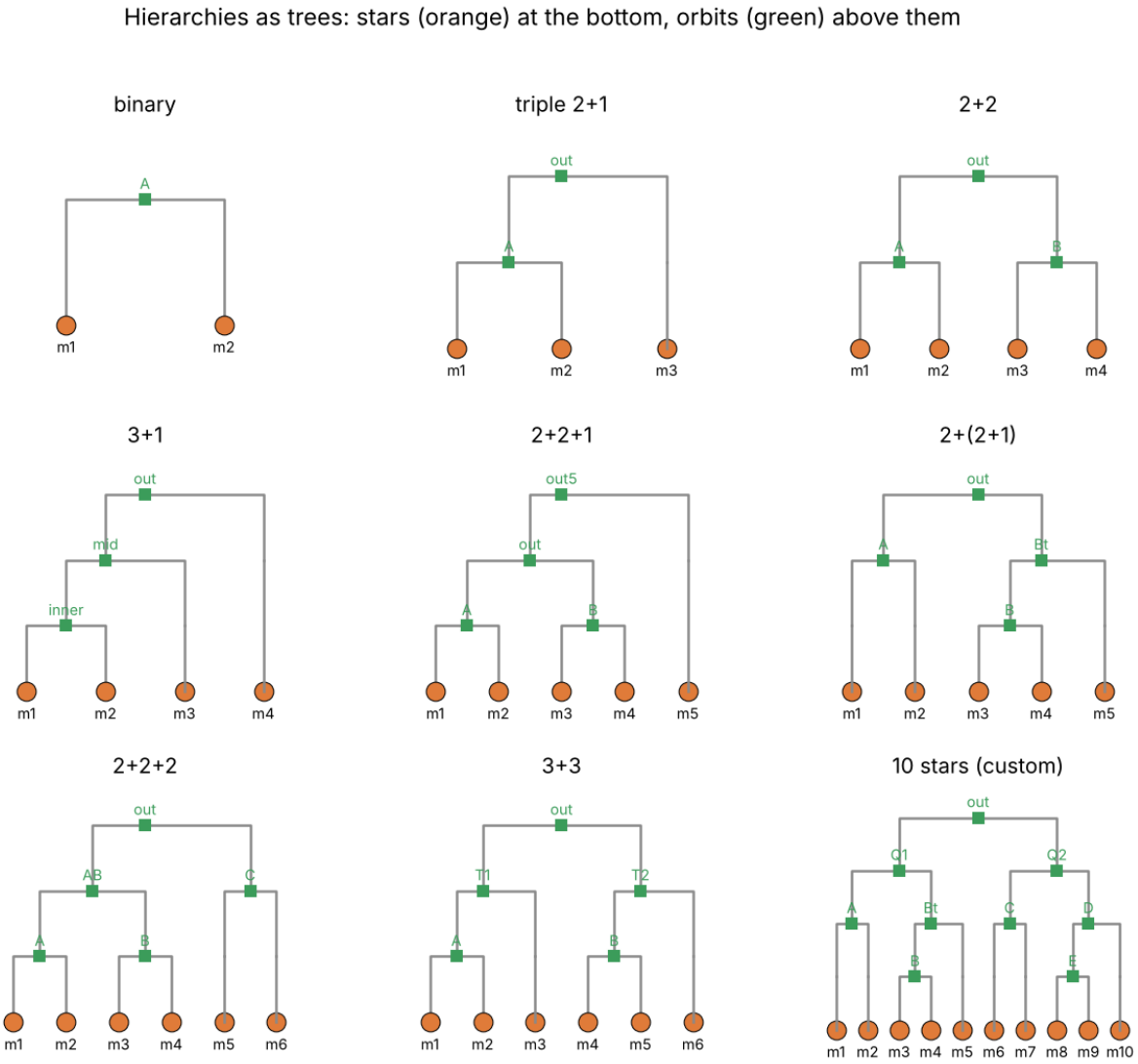


Figure 1. Hierarchies drawn as trees. Each panel is one hierarchy; its name (binary, triple 2+1, 2+2, 3+1, 2+2+1, 2+(2+1), 2+2+2, 3+3 and a custom tree of ten stars) is the title. Orange circles at the bottom are the stars m1, m2, …; green squares are the orbits, labelled with their names (A, B, out, inner, mid, …); the grey lines connect each orbit with its two components. The height of an orbit in the tree indicates its level: an orbit higher up is wider and slower. The figure shows that every hierarchy, however large, is built from the same element – an orbit of two components – which is why one set of equations serves all of them.

Figure 1 shows the trees of the presets. Read each panel from the bottom: two stars join into

an orbit (for example A), the orbit joins with another component into a wider orbit (out), and so on to the top. In the 2+2 panel both components of `out` are binaries; in the 3+1 panel the tree is a chain, each orbit containing the previous one. The last panel shows that a custom tree can mix both patterns. Because the model works with the tree and never with a fixed number of stars, the dynamics, the diagnostics, the observables and the fits are the same for every hierarchy; the presets only provide names.

Doubly eclipsing 2+2 systems, in which both binaries eclipse, are a particularly valuable class because the eclipses constrain the orientation of both inner orbits (Zasche et al. 2019, 2023b, 2024; Kolář et al. 2026). Many examples in this volume refer to them, but nothing in the methods is restricted to them.

1.2 Jacobi (hierarchical) coordinates

Every orbit of the tree describes the **relative motion of its second component around its first**, each component taken as its centre of mass. These are *Jacobi* (or *hierarchical*) coordinates. In a 2+2 system the orbit `out` is the motion of the centre of mass of B around the centre of mass of A; in a 3+1 system the orbit `mid` is the motion of m3 around the centre of mass of (m1, m2). The gravitational parameter of an orbit is $\mu = G(M_1 + M_2)$, where M_1 and M_2 are the total masses of its two components and G the gravitational constant.

Initial conditions. The configuration gives Keplerian elements for every orbit. SWINGERsim converts them to positions and velocities recursively from the outermost orbit inwards: the outermost orbit places the centres of mass of its two components, then each inner orbit places its two components around its own centre of mass, down to the stars. Finally the whole system is shifted so that its barycentre is at rest at the origin. Converting the resulting state back to elements returns the input to 10^{-12} (see *Testing, Part I*).

Osculating and mean elements. The elements reported by the N-body backend are *osculating*: the Keplerian ellipse the two components would follow if all other stars vanished at that instant. Because the other stars do not vanish, osculating elements oscillate on the time scale of the orbits even when nothing changes on average. The secular backends (Dynamics) work with *mean* (orbit-averaged) elements, which contain no such oscillations. The difference matters when a published photodynamical solution is used as a starting point: its osculating elements at an epoch give, in another code, a slightly different mean period, which accumulates into a drift of the eclipse times (Fitting).

1.3 Orbital elements and the reference frame

Each orbit is described by six elements, listed in Table 2.

Table 2. The orbital elements of one orbit. *Symbol* is the notation of this volume; *Key* the name under `orbits.<name>` in the configuration; *Meaning* the definition. The size is given either as a period or as a semi-major axis, never both.

Symbol	Key	Meaning
P or a	`P` [d] or `a` [au]	period, or semi-major axis; linked by Kepler's third law
e	`e`	eccentricity, $0 \le e < 1$
i	`i` [deg]	inclination of the orbital plane to the reference plane, 0°–180°
Ω	`Omega` [deg]	longitude of the ascending node in the reference plane
ω	`omega` [deg]	argument of periastron, measured in the orbital plane from the node
M_0	`M0` [deg]	mean anomaly at $t = 0$

Kepler's third law links the period and the semi-major axis,

$$a^3 = \frac{G\,(M_1 + M_2)}{4\pi^2}\,P^2,$$

where $M_1 + M_2$ is the total mass of the two components of the orbit. Two Suns with a period of one year are $2^{1/3} \approx 1.26$ au apart; with a period of 10 d about 0.11 au, or 24 $R_\odot$.

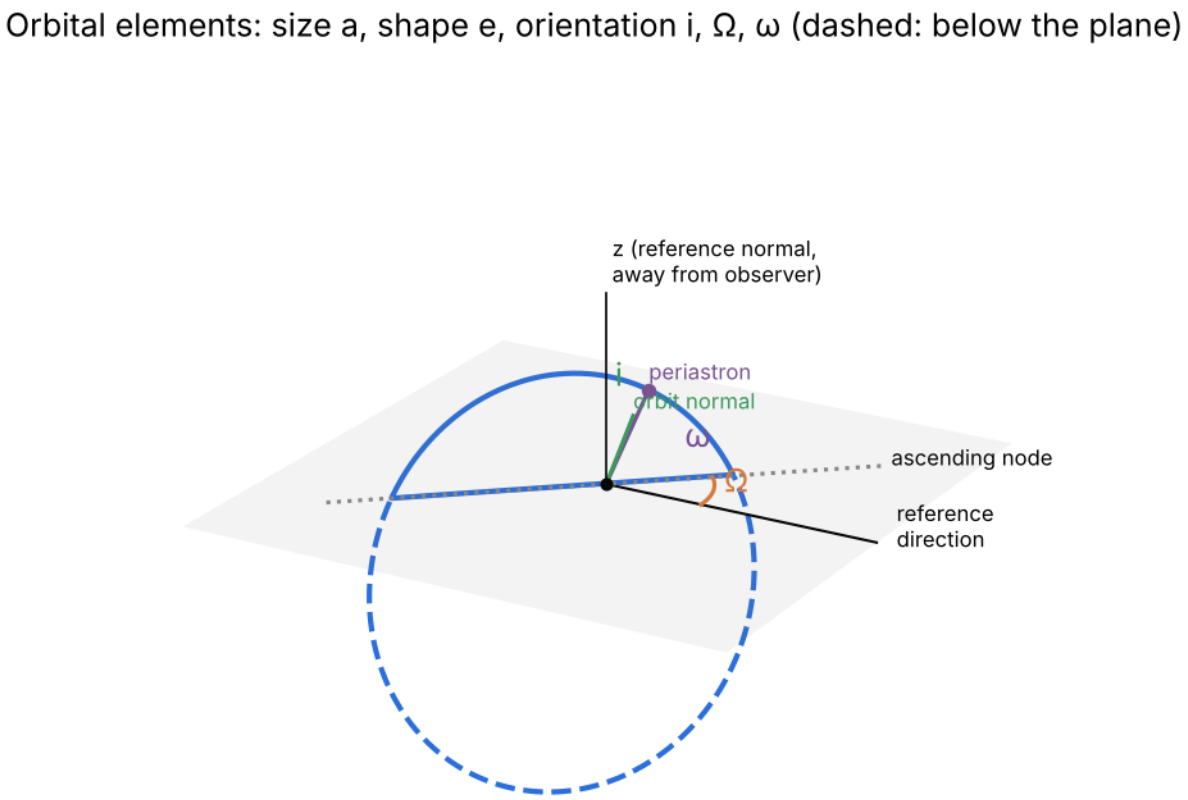


Figure 2. Orbital elements of one orbit. The grey plane is the reference plane (the plane of the sky when the elements are given relative to the sky). The black vertical axis is its normal z, which points **away from the observer**; the black line in the plane is the reference direction (x axis) from which the node is measured. The blue ellipse is the orbit, solid above the reference plane and dashed below it. The dotted line is the line of nodes; the orange arc Ω is the longitude of the ascending node, the green line the orbit normal tilted by the inclination i from z, and the violet arc ω the argument of periastron from the ascending node to the periastron (violet dot). Read together, i and Ω fix the orientation of the orbital plane, ω the orientation of the ellipse within that plane, and a, e its size and shape.

Figure 2 shows the geometry. SWINGERsim uses one **reference frame** for all orbits: the x–y plane is the reference plane, z its normal, and the rotation from the orbital frame to the reference frame is $R = R_z(\Omega)\,R_x(i)\,R_z(\omega)$. When the elements are given relative to the sky – the normal case for an observed system – the reference plane is the plane of the sky and **the line of sight is the z axis, pointing away from the observer**. Consequences of this convention, used throughout the volume:

- the radial velocity of a star is v_z, positive when the star recedes;
- light emitted at depth z arrives later by z/c, so the observed time of an event at the dynamical time t is $t_{\rm obs} = t + z/c$, with the barycentre of the system at $z = 0$ (Observables);

- $i = 90°$ means the orbit is seen edge-on and eclipses are possible; $i = 0°$ or $180°$ means face-on.

Inclination range 0°–180° and folding to 0°–90°. In the configuration i lies between 0° and 180°: an orbit with $i < 90°$ revolves counter-clockwise as seen from the observer's side of the sky plane, one with $i > 90°$ clockwise. The two senses matter dynamically, because the mutual inclination of two orbits depends on them (next section). An observer, however, cannot tell them apart from photometry: an eclipsing binary at i and at $180° - i$ has the same light curve, since eclipses depend only on $|\cos i|$. SWINGERsim therefore

- keeps the full range 0°–180° in the dynamics;
- folds inclinations to 0°–90° whenever they are compared with photometric measurements ($i \to 180° - i$ for $i > 90°$; Fitting);
- in ensembles, draws both senses with equal probability when `mirror: true` is set for an inclination (Ensembles);
- reports the *tilt* $|i - 90°|$ of an eclipsing binary, whose rate of change has the same sign for both senses of revolution (Ensembles).

Only the *relative* sense of revolution of two orbits is physical; one orbit (by convention the first binary) can always be kept at $i < 90°$.

The invariable plane. Besides the elements relative to the sky, SWINGERsim stores the inclination and node of every orbit relative to the *invariable plane*, the plane perpendicular to the conserved total angular momentum of the system (output columns `inc_inv`, `Omega_inv`). Secular theories are written in this plane, and the nodal precession periods in the run summary are measured in it.

1.4 Mutual inclination

The **mutual inclination** $i_{\rm mut}$ of two orbits is the angle between their orbital planes, computed from their orbital angular-momentum vectors $\mathbf{h}_1$ and $\mathbf{h}_2$:

$$\cos i_{\rm mut} = \frac{\mathbf{h}_1 \cdot \mathbf{h}_2}{|\mathbf{h}_1|\,|\mathbf{h}_2|}.$$

Unlike i, it does not depend on the reference plane, and it is the angle that governs the dynamics. Values below 90° mean that the orbits revolve in the same sense (prograde), above 90° in opposite senses (retrograde). Every pair of orbits has its own mutual inclination; SWINGERsim reports all of them (columns `imut_<pair>`).

For two orbits with sky inclinations i_1, i_2 and nodes Ω_1, Ω_2,

$$\cos i_{\rm mut} = \cos i_1 \cos i_2 + \sin i_1 \sin i_2 \cos(\Omega_1 - \Omega_2).$$

Eclipses measure i_1 and i_2 (up to the folding above) but not the nodes on the sky, so the mutual inclination of two eclipsing binaries is known only within the range allowed by the unknown $\Omega_1 - \Omega_2$. This is the main reason why observed systems are studied with ensembles (Ensembles and inference).

1.5 Nodal precession

When two orbits are not coplanar, the distant companion exerts a torque on an inner binary. The torque makes the inner orbital plane **precess**: its angular-momentum vector moves on a cone around the total angular momentum, keeping $i_{\rm mut}$ nearly constant. To lowest (quadrupole) order, for a circular inner orbit, the precession period is

$$P_{\rm prec} \approx \frac{4}{3}\,\frac{P_{\rm out}^2}{P_{\rm in}}\,\frac{m_{\rm tot}}{m_{\rm pert}}\,\left(1 - e_{\rm out}^2\right)^{3/2}\,\frac{1}{\cos i_{\rm mut}},$$

where $P_{\rm in}$ and $P_{\rm out}$ are the inner and outer periods, $e_{\rm out}$ the outer eccentricity, $m_{\rm tot}$ the total mass of the inner orbit and its perturber, and $m_{\rm pert}$ the mass of the perturber (for a 2+2 system the other binary). The derivation for triples goes back to Söderhjelm (1975); the 2+2 case is treated by Vokrouhlický (2016). SWINGERsim prints this estimate for every orbit pair (`t_prec`) next to the period measured in the run from the node in the invariable plane. The estimate is an orientation; the simulation is the result.

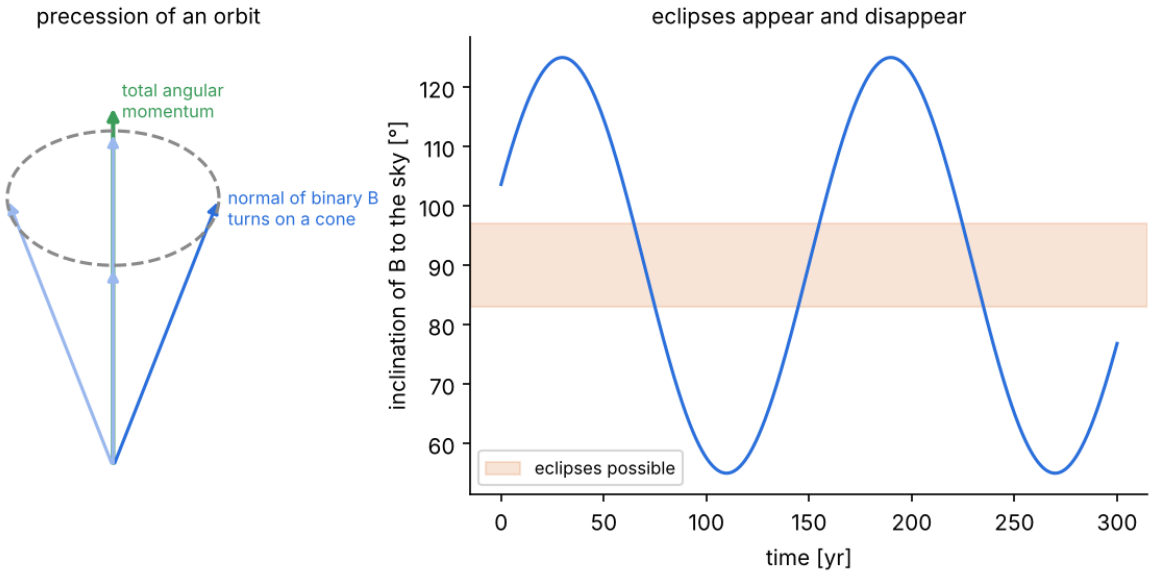


Figure 3. Nodal precession and its observable consequence (illustrative 2+2 system). Left: geometry. The green arrow is the total angular momentum; the blue arrows are three positions of the normal of binary B, which turns on a cone (dashed ellipse) around it. Right: the inclination of B to the sky (vertical axis, degrees) against time (horizontal axis, years). The blue curve oscillates with the precession period; the orange band, about 83°–97°, marks the inclinations at which this binary eclipses. Eclipses are seen only while the curve crosses the band – here for a few decades twice per precession period – and are absent in between.

Figure 3 shows why precession matters for observers. The cone on the left keeps its opening angle (the mutual inclination), but the orientation of B relative to the line of sight changes, so the sky inclination on the right oscillates between $90° - \theta$ and $90° + \theta$ (for a cone axis in the sky plane; θ the opening angle). The binary eclipses only inside the narrow orange band, whose width is set by the eclipse condition of chapter Observables. A slowly precessing binary therefore shows eclipses that deepen, become shallower and disappear over years to centuries; such changes are observed in several 2+2 systems (Zasche et al. 2025). In a 2+2 system both binaries precess, generally at different rates, so their mutual inclination oscillates as well. In larger hierarchies the same happens on every level, more slowly for wider orbits.

1.6 ZLK cycles

When the mutual inclination of an inner and an outer orbit lies between about 39.2° and 140.8°, the torque does more than turn the plane: it exchanges inclination for eccentricity. The inner eccentricity grows to a maximum while $i_{\rm mut}$ falls towards 39.2°, then both return. These are the **von Zeipel–Lidov–Kozai (ZLK) cycles** (Kozai 1962; Lidov 1962; review by Naoz 2016). The window follows from $\cos^2 i_{\rm mut} < 3/5$, i.e. $39.23° < i_{\rm mut} < 140.77°$.

For an initially circular inner orbit, a distant perturber and quadrupole order, the maximum eccentricity depends only on the initial mutual inclination i_0,

$$e_{\rm max} = \sqrt{1 - \tfrac{5}{3}\cos^2 i_0},$$

and the cycles repeat on a time scale of order

$$t_{\mathrm{ZLK}} \sim \frac{P_{\mathrm{out}}^2}{P_{\mathrm{in}}} \frac{m_{\mathrm{tot}}}{m_{\mathrm{pert}}} \left(1 - e_{\mathrm{out}}^2\right)^{3/2},$$

with the symbols of the previous section. The exact period depends on the initial conditions (Antognini 2015). SWINGERsim reports for every orbit pair whether it starts in the ZLK window, the quadrupole e_{max}, t_{ZLK}, and, from the double-averaged equations of Naoz et al. (2013) valid for any mass ratio, the predicted range of e and i_{mut} and the period of the cycle (Dynamics).

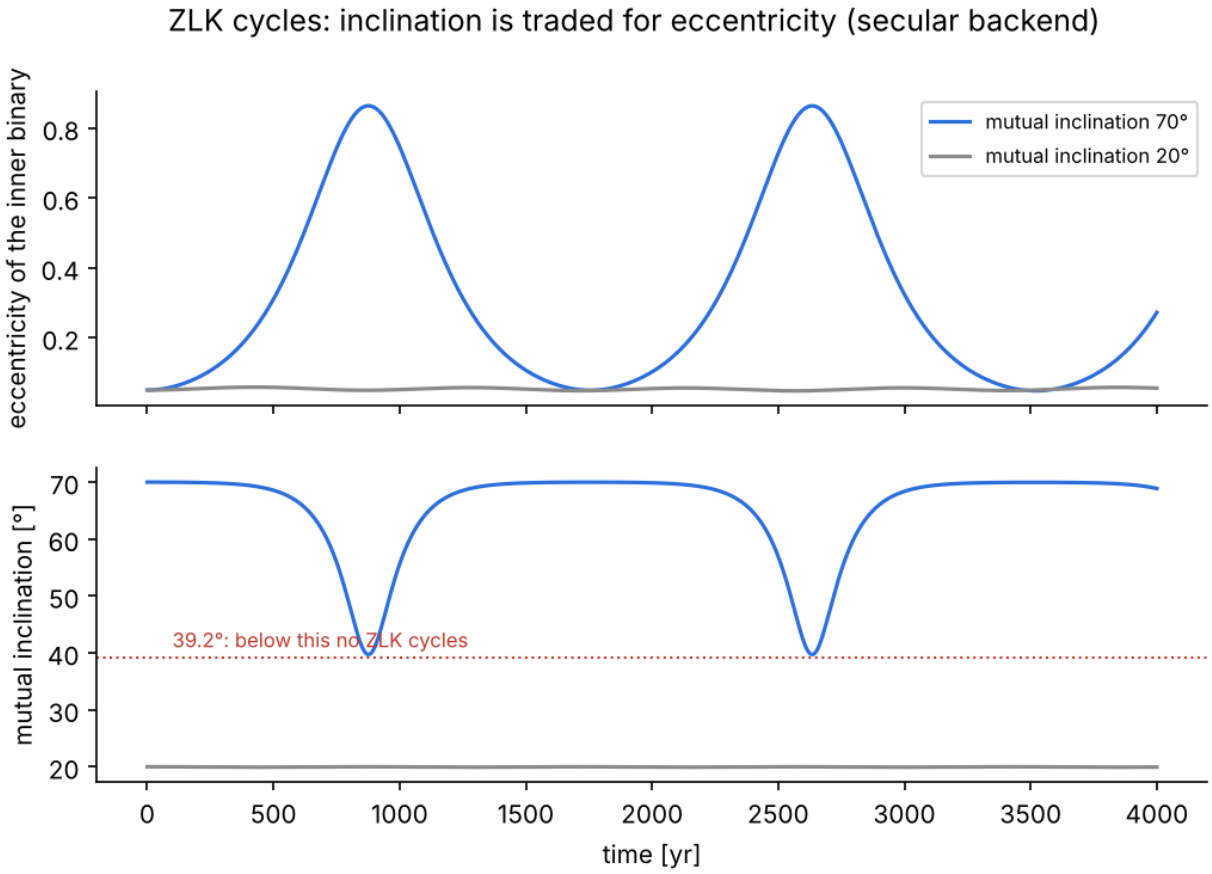


Figure 4. ZLK cycles of an inner binary computed with the secular backend. Top: eccentricity of the inner binary against time in years. Bottom: mutual inclination in degrees against the same time axis. Blue: initial mutual inclination 70°; grey: 20°. The red dotted line marks 39.2°, the lower edge of the ZLK window. At 70° the eccentricity climbs from 0.05 to about 0.86 while the inclination drops to the 39.2° line, and both return – inclination is traded for eccentricity. At 20°, outside the window, both stay almost constant.

Figure 4 shows the exchange. The two panels share the time axis, and the eccentricity maxima of the blue curve coincide with the inclination minima; the inclination minimum touches the 39.2° line, as the quadrupole theory predicts. The maximum eccentricity agrees with $\sqrt{1 - (5/3)\cos^2 70^\circ} = 0.81$ to within the effect of the small initial eccentricity and of the finite outer angular momentum. ZLK cycles matter because at maximum eccentricity the stars approach each other at periastron: tides can then shrink the orbit (Fabrycky & Tremaine 2007), the stars can fill their Roche lobes or collide.

Octupole order. When the masses of the inner binary differ and the outer orbit is eccentric, the next (octupole) term of the expansion breaks the regularity of the cycles and can drive

the eccentricity to values close to 1 and flip the orbit (Naoz 2016). Its strength is measured by

$$\varepsilon_{\rm oct} = \frac{|m_a - m_b|}{m_a + m_b} \frac{a_{\rm in}}{a_{\rm out}} \frac{e_{\rm out}}{1 - e_{\rm out}^2},$$

where m_a, m_b are the masses of the two components of the inner orbit and $a_{\rm in}$, $a_{\rm out}$ the inner and outer semi-major axes. SWINGERsim reports $\varepsilon_{\rm oct}$ for every pair; the N-body backend contains all orders automatically.

Suppression by short-range forces. Any additional apsidal precession of the inner orbit that is faster than the ZLK cycle – relativity, tidal and rotational distortion of the stars, a close third star – suppresses the cycles, because the periastron turns before the torque can build up eccentricity (Liu, Muñoz & Lai 2015; Naoz 2016). The run summary flags a pair when the relativistic apsidal period is shorter than $t_{\rm ZLK}$. Tidal and rotational distortion are often stronger: in the V994 Her setup of Vokrouhlický (2016), binary B reaches $e = 0.63$ with point masses but stays near its initial $e = 0.13$ with the stellar distortion included (see *Testing, Part I*). A pure point-mass model therefore overestimates eccentricity growth in close binaries; `physics.tides` should be switched on for them (Tides and rotation).

1.7 Apsidal motion

The periastron of an eccentric orbit turns slowly in its plane: **apsidal motion**, measured as the rate $\dot\omega$ or as the apsidal period $U = 2\pi/\dot\omega$. In an eclipsing binary it is directly observable, because primary and secondary minima move in opposite directions in the O−C diagram (Observables). Three causes contribute:

- **General relativity.** To first post-Newtonian (1PN) order,

$$\dot\omega_{\rm GR} = \frac{3\,G\,(m_1 + m_2)\,n}{c^2\,a\,(1 - e^2)}, \qquad n = \frac{2\pi}{P},$$

  with m_1, m_2 the masses, a the semi-major axis, e the eccentricity, n the mean motion and c the speed of light.
- **Tidal and rotational distortion of the stars** (Sterne 1939), which depends on the apsidal-motion constants k_2 and the radii of the stars (Tides and rotation).
- **The other stars** of the system, whose secular torque turns the periastron as well.

For every binary of two stars SWINGERsim gives the classical expectation (Sterne plus relativity) and the rate measured in the run from the longitude of periastron $\varpi = \Omega + \omega$; the difference is the contribution of the other stars. In a compact quadruple this contribution can be as large as the classical one. The command `swingersim apsidal` splits the rate into its tidal, rotational, relativistic and dynamical parts and gives the factor by which k_2 would have to change to match a measured rate.

1.8 Resonances

Two kinds of near-commensurability are typical of 2+2 systems:

- **Mean-motion (period) resonance.** When the periods of A and B are close to a ratio of small integers, the binaries can be locked in a resonance. The 1:1 case is treated by Breiter & Vokrouhlický (2018); the 3:2 resonance is observed in CzeV343 (Pejcha et al. 2022) and its statistical excess among 2+2 systems is discussed by Kolář et al. (2025). SWINGERsim reports the nearest ratio of small integers of the two periods. Mean-motion resonances are absent from the secular backends by construction and are followed only by the N-body backend.
- **Precession (secular) resonance.** When A and B precess at similar rates, their mutual torques add up coherently and can drive much larger eccentricity oscillations than in either binary alone (Pejcha et al. 2013; Hamers & Lai 2017).

The parameter β of Hamers & Lai (2017, their eqs. 32–33) compares the precession rate of the perturber's orbit with the ZLK rate of an inner orbit. For $\beta \ll 1$ the classical ZLK picture holds; for $\beta \gg 1$ the inner orbit feels a perturber averaged over its precession; for $\beta \sim 0.3$–3 the inner eccentricity can be excited strongly and chaotically, even at initial inclinations well below the ZLK window. SWINGERsim prints β for A and B (2+2) or for the inner orbit (3+1).

1.9 Coupling of the orbital planes of A and B

If the outer orbit held all the angular momentum, the planes of A and B would precess independently around it, and the angle J_{AB} between them would range from $|J_A - J_B|$ to $J_A + J_B$, where J_A and J_B are their tilts to the outer orbit. In reality the outer orbit precesses too, and the binaries are coupled through it. For circular A and B at quadrupole order Vokrouhlický (2016, Sect. 2.3) found two integrals of motion that limit J_{AB} to a maximum $J_{AB}^{\max}$, often much smaller than $J_A + J_B$: in his V994 Her setup $J_A = 11.6°$ and $J_B = 15.8°$ but $J_{AB}^{\max} = 5.2°$. The planes of two eclipsing binaries can therefore stay nearly parallel for a long time, which raises the chance that both eclipse. SWINGERsim prints $J_{AB}^{\max}$ for every 2+2 core where the bound is informative (`J_AB_max_analytic_deg`). It assumes circular binaries; eccentric binaries can exceed it slightly: for the observed eccentricities of V994 Her (Fig. 3 of the paper) Vokrouhlický finds J_{AB} between 1° and 6°, and SWINGERsim's N-body integration of the same set-up reaches 6.3° (see *Testing, Part I*).

1.10 Larger hierarchies and a fifth star

Real quadruples are often parts of quintuples, and an unseen fifth star can change the dynamics in both directions:

- **Destabilising.** A fifth star on a wide inclined orbit drives ZLK cycles of the orbit of the two binaries; its eccentricity grows and its periastron approaches them, which can push a quadruple that is stable on its own across the stability limit. The stability report lists the nested triple of `out` inside `out5` for this purpose.
- **Stabilising.** Any precession faster than the ZLK cycle suppresses it (Liu, Muñoz & Lai 2015). A close third star around one binary makes that binary precess quickly and shields it; likewise the quadrupole of the two binaries makes their mutual orbit precess and limits a ZLK cycle driven by a fifth star. In a test setup with real binaries (1 + 1 $M_\odot$, a = 1 au) instead of point masses, the eccentricity of `out` reaches 0.73 instead of the point-mass value 0.80.
- **Observable consequences.** A companion of only one binary changes only that binary's inclination and eclipse timings, as proposed for some doubly eclipsing systems whose binaries behave differently (Zasche et al. 2025).

Every method of this volume works with any tree. SWINGERsim remains a code for *hierarchical* systems: a non-hierarchical (democratic) configuration is followed by the N-body backend until it breaks up, as in the scattering experiments (Stability and fates), but it is not the target of the diagnostics.

1.11 Implementation

Everything in this chapter is SWINGERsim's own code; no external package is involved apart from NumPy and SciPy for linear algebra and ODE integration. Own implementations: the hierarchy tree and the recursive conversion of the elements of all orbits into positions and velocities and back (Kepler's equation solved by Newton iteration to machine precision); the mutual inclinations and the invariable plane; the analytic diagnostics of every orbit pair (quadrupole $e_{\max}$, $t_{\rm ZLK}$, $t_{\rm prec}$, $\varepsilon_{\rm oct}$, $\varepsilon_{\rm SA}$, the β parameter of Hamers & Lai 2017, period ratios) coded from the cited formulae; the integration of the double-averaged quadrupole equations of Naoz et al. (2013, eqs. A27, A29, A36), with the SciPy integrator DOP853; and the bound $J_{AB}^{\max}$ coded from Vokrouhlický (2016, eqs. 26–36).

Chapter 2

Stars

The dynamics of a multiple star needs more from each star than its mass. Its radius decides collisions, Roche-lobe overflow and eclipses; its internal structure, through the apsidal-motion constant k_2 and the moment of inertia, decides how strongly it is distorted by tides and rotation; its luminosity and effective temperature decide its light. All of these change as the star evolves. This chapter describes where SWINGERsim takes these quantities from, how they evolve, and how a star's size compares with its Roche lobe. Mass loss and mass transfer, which change the masses themselves, are the subject of chapter Mass transfer and evolution.

2.1 The Hertzsprung–Russell diagram

The *Hertzsprung–Russell (HR) diagram* plots the luminosity L of a star (the energy it radiates per second, in units of the solar luminosity $\mathrm{L}_\odot$) against its effective temperature T_{eff} (the temperature of a black body with the same surface flux; hot stars on the left). The two are linked to the radius R by $L = 4\pi R^2 \sigma T_{\mathrm{eff}}^4$, with σ the Stefan–Boltzmann constant. Most stars lie on the **main sequence**, where they burn hydrogen in their cores: the heavier, the hotter, brighter and shorter-lived. Before it, a young star contracts on the **pre-main sequence**; after it, the core contracts and the envelope swells into a **giant**, later an **asymptotic-giant-branch (AGB)** star. Massive stars lose their hydrogen envelopes in strong winds and become hot **Wolf–Rayet** stars.

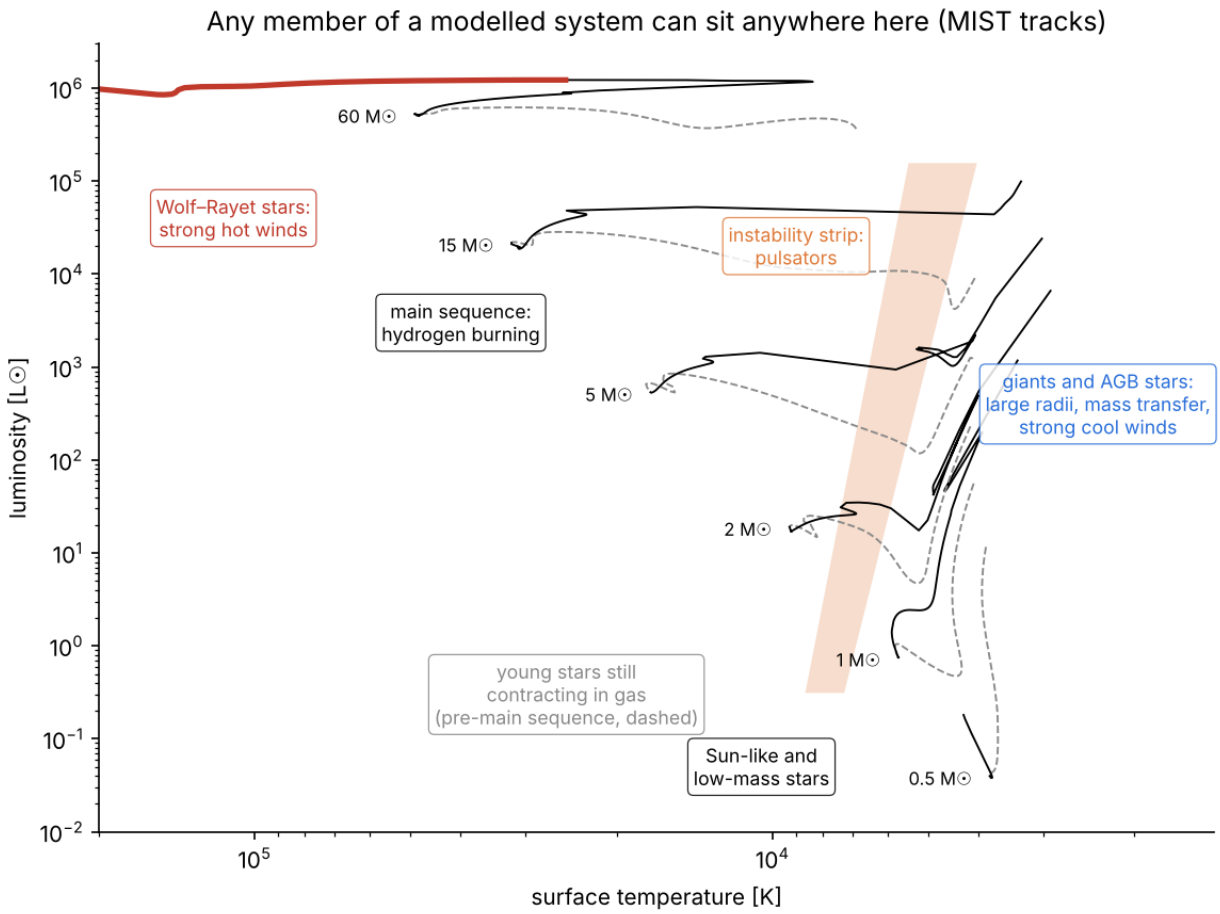


Figure 5. Where the members of a modelled system can lie in the HR diagram. Horizontal axis: effective temperature in kelvin, decreasing to the right (logarithmic); vertical axis: luminosity in $L_\odot$ (logarithmic). Black lines: MIST evolutionary tracks of stars of 0.5, 1, 2, 5, 15 and 60 $M_\odot$ (labelled at their zero-age main-sequence points), from the main sequence to the end of the track; grey dashed lines: the same stars on the pre-main sequence; red: the Wolf–Rayet phase of the 60 $M_\odot$ star. The orange band is the classical instability strip in which stars pulsate. Boxes name the regions. Any star of a modelled system can sit anywhere on these tracks, so one system can combine, for example, a young binary with a giant.

Figure 5 summarises the stellar side of the model. Each black track shows how one star moves in the diagram with age: nearly fixed on the main sequence, then to the right and upwards as it becomes a giant. The radius, which matters most for the orbits, grows by a factor of 100 and more along the giant branch and the AGB. The instability strip marks the region where pulsations make a star a clock (Observables).

2.2 Stellar models

A *stellar model* maps the mass of a star (and its age and metallicity, for evolving models) to its radius, luminosity, effective temperature, apsidal-motion constant k_2 and gyration constant $r_g^2 = I/(MR^2)$ (I the moment of inertia). It is chosen with `stellar_model.kind`; Table 3 lists the options.

Table 3. Stellar models. *Kind* is the value of `stellar_model.kind`; *Source* the model or data behind it; *Range* the masses, metallicities and phases covered; *Structure* where k_2 and r_g^2 come from. Values given for a single star under `stars.<name>` (`R`, `Teff`, `apsidal_k2`, `rg2`, `lag_s`) always override the model.

Kind	Source	Range	Structure (k_2, r_g^2)
`zams` (default)	zero-age main-sequence radius of Tout et al. (1996)	0.1–100 $M_\odot$, solar metallicity, no ageing	polytrope, $n = 1.5$ below 0.35 $M_\odot$, $n = 3$ above
`sse`	single-star evolution formulae of Hurley, Pols & Tout (2000)	0.1–8 $M_\odot$, Z = 0.004–0.03; main sequence, Hertzsprung gap, giant branch	Claret (2023) for 0.2–8 $M_\odot$, polytrope outside
`mist`	MIST v1.2 tracks (Choi et al. 2016; Dotter 2016)	0.1–300 $M_\odot$, [Fe/H] −1.5 to +0.5, $v/v_{\rm crit}$ = 0 or 0.4; pre-main sequence to the end of the AGB	Claret (2023) for 0.2–8 $M_\odot$, Claret (2004) above 8 $M_\odot$
`table`	a CSV table of the user (mass, radius, k_2, r_g^2, optional lag)	as tabulated, linear in mass	from the table
`python`	a function of the user returning the same quantities	any	from the function

With `structure: mesa:<grid>` the structure constants come from a grid of the user's own MESA models (Jermyn et al. 2023) instead (section Own MESA grids).

Radius, temperature and luminosity of a single star. When only the radius of a star is given, the luminosity of the model is kept and the effective temperature follows from $L = 4\pi R^2 \sigma T_{\rm eff}^4$; when R and $T_{\rm eff}$ are both given, they fix the luminosity. `radius_inflation` enlarges a star relative to the model at fixed luminosity. Magnetically active low-mass stars are larger than standard models predict – CM Dra by 3–4 % at solar metallicity and 5–7 % at a lower metallicity (Feiden & Chaboyer 2014) – and MIST shows the same deficit.

2.2.1 The default: zero-age main-sequence stars

The `zams` model gives every star the zero-age main-sequence (ZAMS) radius of the analytic fit of Tout et al. (1996) for solar metallicity (1 $M_\odot$ → 0.89 $R_\odot$, 3 $M_\odot$ → 1.98 $R_\odot$). The stars do not age. The structure constants are those of a polytrope – a star in which pressure and density are related by $P \propto \rho^{1+1/n}$ – with index $n = 1.5$ for fully convective stars below 0.35 $M_\odot$ and $n = 3$ above. SWINGERsim integrates the Lane–Emden equation for the polytrope together with Radau's equation for its tidal distortion and obtains k_2 and r_g^2; it reproduces the analytic values for $n = 1$ and the tabulated $k_2 = 0.1433$ ($n = 1.5$) and 0.0144 ($n = 3$) of Brooker & Olle (1955) (see *Testing, Part I*).

The polytropes are a rough approximation. Real main-sequence stars above about 1.5 $M_\odot$ are more centrally condensed, with k_2 of a few 10^{-3}, and k_2 falls further as the star evolves; the $n = 3$ polytrope overestimates k_2 of 2–3 $M_\odot$ stars by about a factor of three. For any quantitative apsidal motion the structure should come from stellar models (`sse`, `mist`, a table or per-star values).

2.2.2 Evolving stars: SSE

With `kind: sse` the stars have an age `age_myr` (counted from the ZAMS, all stars coeval unless `stars.<name>.age_offset_myr` shifts one) and a metallicity `z` ($Z = 0.02$ is solar in this calibration). Radius, luminosity, effective temperature and the convective envelope follow from the formulae of Hurley, Pols & Tout (2000), fitted to the detailed models of Pols et al. (1998), for the main sequence, the Hertzsprung gap and the first giant branch. The mass and depth of the convective envelope come from the binary-code extension of Hurley, Tout & Pols (2002), which is calibrated for $Z = 0.02$.

The formulae are not evaluated at run time. They were evaluated once with the published SSE routines on a grid of 320 masses and 5 metallicities; every phase is sampled at the same fractions of the phase (*equivalent evolutionary points*, as in MIST), and a star of any mass and metallicity is interpolated at fixed phase fraction, linearly in $\log M$ and $\log Z$. Against direct SSE evaluations the interpolation is accurate to 0.8 % in radius on the main sequence, 1.7 % in the Hertzsprung gap and 4 % on the giant branch (95th percentiles for 0.8–3 $M_\odot$, Z = 0.01–0.03) and to 0.2 % in the phase boundaries; for massive metal-poor giants the errors reach 30 %. Beyond helium ignition (core helium burning, AGB, remnants) SSE stars are not followed; a run stops there. For the Sun (1 $M_\odot$, Z = 0.02, 4.57 Gyr) the model gives R = 0.985 $R_\odot$, L = 0.958 $L_\odot$ and the base of the convective envelope at 0.734 R, against the helioseismic 0.713 R (Christensen-Dalsgaard et al. 1991).

2.2.3 Evolving stars: MIST

With `kind: mist` SWINGERsim uses the MESA tracks of MIST v1.2 (Choi et al. 2016; Dotter 2016) as an independent model family. They cover 0.1–300 $M_\odot$, [Fe/H] from −1.5 to +0.5 in steps of 0.25 and the two initial rotations of MIST, $v/v_{\rm crit}$ = 0 and 0.4 (the default, the MIST standard). Every track is stored at the *equivalent evolutionary points* (EEPs) 1–1409: pre-main sequence (1–201), main sequence (202–453), subgiant and red giant branch (454–604), helium flash and core helium burning (605–707), early AGB (708–807) and thermally pulsing AGB to the departure from the AGB (808–1409); massive stars end with core carbon burning. Post-AGB stars and white dwarfs are not followed. A star is interpolated at fixed EEP between the neighbouring masses (linearly in $\log M$) and metallicities, as in the MIST isochrones, and in age between the EEPs. With twice the grid spacing the interpolation reproduces the radius at the end of the main sequence to 1 % and the ages to 1–2 %. Ages count from the start of the MIST pre-main-sequence models, not from the ZAMS as in SSE.

MIST tracks contain no convective-envelope structure, which the tidal dissipation needs (Tides and rotation). SWINGERsim derives it: giants on the giant branch and the AGB, and core-helium-burning stars cooler than 5500 K, have a convective envelope containing all the mass above the helium core, as deep as the star (Hurley et al. 2002), shrinking linearly to zero between 5500 and 7000 K on the blue loop; main-sequence and Hertzsprung-gap stars take the envelope mass and depth fractions of the SSE star of the same mass at the same fraction of its phase; pre-main-sequence stars of 0.1–2.5 $M_\odot$ are fully convective on the Hayashi track, and from the minimum of their effective temperature (where the radiative core appears) the envelope shrinks linearly in EEP to its ZAMS size. For the Sun the [Fe/H] = 0 track gives R = 1.025 $R_\odot$ and L = 1.11 $L_\odot$ at 4.57 Gyr, the SSE track 0.985 and 0.958: the two families bracket the Sun.

Systematic differences between the families. Against MIST the SSE main-sequence lifetimes are about 10 % longer and the radii at the end of the main sequence up to 10 % smaller. This is the systematic uncertainty of any age derived with one family.

Massive stars. Above 8 $M_\odot$ three properties of MIST matter for the dynamics. (1) The winds are strong and depend on rotation: a 100 $M_\odot$ star ([Fe/H] = 0) leaves the main sequence with 68 $M_\odot$ without rotation and with 25 $M_\odot$ at $v/v_{\rm crit}$ = 0.4, so `vvcrit` is a choice of the mass history (Mass transfer and evolution). (2) The hydrogen-burning lifetimes agree with the Geneva models of Ekström et al. (2012) to 4–12 %, but the masses at the end of the main sequence differ by up to 30 %; Choi et al. (2016) call 1D models of the radiation-dominated envelopes of very massive stars uncertain. The masses of evolved massive stars are therefore a model choice, not a prediction. (3) From the late main sequence on, the envelopes of 40–70 $M_\odot$ stars inflate and contract within a few $M_\odot$ of initial mass; there the grid does not resolve the radius (errors of 20–40 %), and grid masses should be used where the radius matters.

Present masses. Masses from an orbit solution are present masses, whereas tracks are labelled by initial masses. With `physics.mass_loss: {source: track, masses: current}` SWINGERsim finds the initial mass of every star whose track has the given mass at `age_myr`. Massive stars can be ambiguous (at 2.5 Myr a 26 $M_\odot$ star is either a 26.4 $M_\odot$ main-sequence star or a 257 $M_\odot$ star stripped to a Wolf–Rayet star); by default the least evolved solution is taken with a warning, and a mass that no track reaches at that age is refused as a contradiction.

2.3 Structure constants: k_2 and r_g^2

The **apsidal-motion constant** k_2 measures how strongly a star responds to a tidal or rotational perturbation; it is half the potential *Love number* $k_{2,\mathrm{Love}} = 2k_2$ used in tidal theory. A centrally condensed star (mass concentrated in the core) has a small k_2: the $n = 3$ polytrope has 0.0144, a homogeneous body 0.75. The **gyration constant** $r_g^2 = I/(MR^2)$ sets how much angular momentum a spinning star stores. Both enter the tidal forces, the spin evolution and the apsidal motion (Tides and rotation).

With `sse` and `mist` both come from the MESA models of Claret (2023): 69 tracks, 0.2–8 $M_\odot$, [Fe/H] = −0.5, 0, +0.5, from the pre-main sequence to the white dwarf. As Claret recommends, the tables are interpolated in the surface gravity $\log g$ of the star – which carries the evolutionary state independently of the track code – along a monotonic segment of each track (pre-main sequence, main sequence and giant branch, core helium burning), then linearly in $\log M$ and [Fe/H]; the [Fe/H] of an SSE star is $\log(Z/0.02)$. The values reproduce the table rows to the published precision. On the main sequence k_2 decreases as the star becomes more centrally condensed (for 2 $M_\odot$ from 0.005 to 0.003); on the giant branch the deep convective envelope raises it by more than an order of magnitude. Above 8 $M_\odot$ (MIST) the constants come from Claret (2004) (0.8–125 $M_\odot$, Z = 0.02, ZAMS to carbon exhaustion), interpolated in the same way and agreeing with Claret (2023) to 1 % at 8 $M_\odot$; above 125 $M_\odot$ the 125 $M_\odot$ track is used. AGB stars take the values at the tip of the giant branch, because Claret's grids have no AGB models.

Rotation. A rotating star is more centrally condensed than a static one. With `physics.k2_rotation` SWINGERsim corrects the model constants, $\delta \log k_2 = -0.87\,\lambda_s$ with $\lambda_s = 2\Omega^2 R^3/(3GM) = 2V^2/(3gR)$, where Ω is the angular velocity of rotation, V the equatorial rotation velocity, g the surface gravity, R the radius and M the mass (Claret 1999; Claret & Giménez 2010 apply this correction to observed binaries), with the spin of each star at that time.

2.3.1 Own MESA grids

For structure quantities that published grids do not provide, SWINGERsim can drive a local installation of MESA (Paxton et al. 2011; Jermyn et al. 2023). A grid file lists masses, metallicities and the physics (mixing length, overshooting, end point); every track starts on the pre-main sequence and writes structure profiles. The reduction computes from each profile the base of the convective envelope, the convective turnover time, k_2 from Radau's equation (the method of Claret 2023), $I/(MR^2)$ and the inertial-wave dissipation coefficient (Tides and rotation). On polytropes written in the MESA format the reduction reproduces k_2 to 1 %, $I/(MR^2)$ to 0.3 % and the inertial-wave coefficient of a fully convective star to 1 %.

2.4 Roche lobes

In a binary that rotates with its orbit, the effective potential (gravity of both stars plus the centrifugal term) has a saddle point between the stars, the inner Lagrange point L1. The equipotential surface through L1 encloses two teardrop-shaped regions, the **Roche lobes**: material inside a lobe belongs to its star. A star that swells beyond its lobe pours gas through L1 onto its companion.

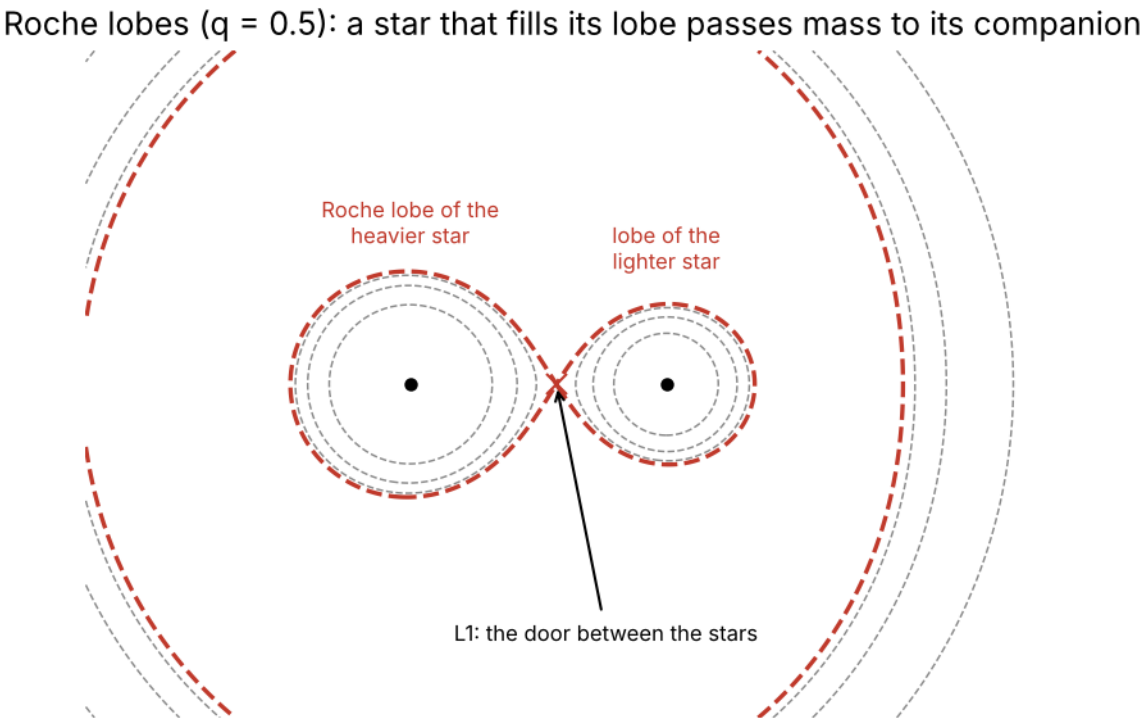


Figure 6. Roche lobes of a circular synchronised binary with mass ratio $q = 0.5$, cut in the orbital plane. The black dots are the centres of the heavier (left) and the lighter (right) star. Grey dashed lines are equipotentials of the effective potential; the thick red dashed figure-of-eight is the critical equipotential through the inner Lagrange point L1 (red cross), which bounds the two Roche lobes. The large curves at the edges are equipotentials far from the binary, where the centrifugal term dominates. A star whose surface reaches its lobe passes mass to the companion through L1.

Figure 6 shows the geometry. Close to each star the equipotentials are nearly spheres, so a star much smaller than its lobe is almost spherical. Near the lobe the surfaces become pointed towards L1: a star that nearly fills its lobe is visibly distorted, which matters for its light curve (Observables). The lobe of the heavier star is larger.

SWINGERsim uses the *volume-equivalent* lobe radius R_L (the radius of a sphere with the lobe's volume) of Eggleton (1983),

$$\frac{R_L}{d} = \frac{0.49\, q^{2/3}}{0.6\, q^{2/3} + \ln(1 + q^{1/3})}, \qquad q = \frac{M}{M_{\text{companion}}},$$

where d is the separation, M the mass of the star and $M_{\text{companion}}$ that of its companion; for equal masses $R_L/d \approx 0.38$. For every binary of two stars the *filling factor* R/R_L is evaluated at periastron, $d = a(1 - e)$ of the osculating orbit, as is usual for eccentric orbits (Sepinsky et al. 2007). A star that orbits a sub-system – a giant tertiary around an inner binary – has a lobe against the total mass of that sub-system (Eggleton 1983). The largest filling factor of every star is stored; with `run.stop_on_roche_overflow` a run stops at $R/R_L = 1$, and with `physics.mass_transfer` the transfer is followed instead (Mass transfer and evolution).

Limits. The formula assumes a circular synchronised binary; in an eccentric or asynchronous orbit the lobe changes along the orbit and its value at periastron is approximate (Sepinsky et al. 2007). For the shapes of distorted stars in light curves SWINGERsim solves the full Roche potential instead (Observables).

2.5 Kinds of systems

The same machinery describes systems at very different stages of life. Table 4 is a guide to the physics each kind needs; the chapters named in the last column describe it.

Table 4. Kinds of multiple stars and the physics they need. *Kind* describes the system; *Model* the recommended `stellar_model.kind`; *Physics* the switches under `physics` (and per-star keys) that matter; *Backend* the integration method of chapter Dynamics; *Chapter* where the physics is described.

Kind	Model	Physics	Backend	Chapter
young, still in gas	`mist`, age < 1 Myr	`gas`, `tides`, `tidal_lag: hurley2002`	secular	Dynamics, Tides
Sun-like, main sequence	`zams` or `mist`	`gr`, `tides`, `rotation`; `magnetic_braking`	either	Tides, Mass transfer
close eclipsing binaries in a multiple	`mist` or `sse`	`gr`, `tides`, `rotation`	N-body for timings	Tides, Observables
giants, AGB	`mist`	`evolution`, `mass_loss`, `tides`, `tidal_lag: hurley2002`	secular	Mass transfer
Algols, giant tertiaries	`sse` or `mist`	`evolution`, `mass_transfer`	secular	Mass transfer
contact binaries, rotating stars	`mist` or `sse`	`mass_transfer.contact`; `shape`, `contact_fillout`	secular for evolution, N-body for light	Mass transfer, Observables
pulsating components	`mist`	`stars.<m>.pulsation`, `evolution`	N-body for light	Observables
massive stars, Wolf–Rayet	`mist`, `vvcrit: 0.4`	`evolution`, `mass_loss`, `tides`	secular	Mass transfer
very wide (> 1000 au)	any	`galactic_tide`, `encounters`	either	Dynamics
compact remnants	point masses	`gr`, `gw`	either	Dynamics

A useful habit, borrowed from the PHOEBE documentation: switch the effects on one at a time and compare the runs. Each switch shows what that piece of physics does to the system, and a result that changes much with an effect thought negligible deserves a closer look.

2.6 Implementation

External data. The SSE formulae were evaluated with the published SSE Fortran routines of Hurley, Pols & Tout (2000) once, offline, into a stored grid; the MIST v1.2 tracks (Choi et al. 2016; Dotter 2016) are converted from the published EEP archives; the structure constants are the tables of Claret (2023) and Claret (2004); MESA (Jermyn et al. 2023) is an external program run by the user for own grids. **Own code.** The ZAMS radius fit of Tout et al. (1996) and the polytropes (Lane–Emden and Radau equations integrated with SciPy); the EEP and phase-fraction interpolation of both track families; the $\log g$ interpolation of the Claret tables along monotonic segments; the convective envelopes of MIST stars (giants, blue loop, pre-main sequence) derived as described above; the search for initial masses from present masses; the rotation correction of k_2; the reduction of MESA profiles (k_2 by Radau's equation, $I/(MR^2)$, turnover time, inertial-wave coefficient); and the Eggleton (1983) lobes evaluated at periastron for binaries and for stars orbiting sub-systems.

Chapter 3

Dynamics

SWINGERsim computes the motion of the stars in two complementary ways, called *backends*. The **N-body backend** integrates the equations of motion of all stars directly; it is exact up to the included physics, sees every orbit and every close encounter, and is slow when the inner orbits are short and the run is long. The **secular backends** average every orbit over its revolution and follow only the slow changes of the averaged ellipses; they are thousands of times faster and reach billions of years, but they assume that the hierarchy stays intact. The backend is chosen with `run.backend` (`nbody` or `secular`). This chapter describes both, the diagnostics computed during a run, and the weak external perturbations (gas, Galactic tide, passing stars, gravitational waves).

3.1 Direct N-body integration

All stars (2 to about 10) are integrated as point masses – extended bodies when tides are on (Tides and rotation) – with the IAS15 integrator of REBOUND (Rein & Liu 2012; Rein & Spiegel 2015). IAS15 is a 15th-order Gauss–Radau integrator with an adaptive step. Its error stays at the level of floating-point round-off, about 10^{-15} per step, and it handles close encounters and high eccentricities without special settings. The price is speed: the step follows the shortest orbit, so the computing time grows with $t_{\mathrm{max}}/P_{\mathrm{in}}$, the run length divided by the shortest period.

Because the full N-body problem is solved, all orders of the interaction (quadrupole, octupole and higher), the short-period terms that orbit averaging removes, mean-motion resonances, the coupling between the two binaries of a 2+2 system and any break-down of the hierarchy are included automatically.

3.2 Relativity

With `physics.gr: true` (the default) SWINGERsim adds the first post-Newtonian (1PN) equations of motion for all bodies with the `gr_full` effect of REBOUNDx (Tamayo et al. 2020). The simpler REBOUNDx effects `gr` and `gr_potential` assume one dominant central mass, which is not true in a multiple star, and are not used. The main consequence is the periastron advance $\dot{\omega}_{\mathrm{GR}}$ of each orbit (Hierarchies and orbits); the simulated advance of an isolated binary (1.5 $\mathrm{M}_{\odot}$, a = 0.05 au, e = 0.3) agrees with the analytic formula to 1 % (see *Testing, Part I*).

Relativity is a small correction for most stellar binaries, with apsidal periods of thousands to tens of thousands of years at periods of days. It becomes decisive for binaries with periods of about a day or shorter and a slowly acting distant companion, where it can suppress ZLK cycles completely (Hierarchies and orbits).

3.3 Conservation checks

The relative energy error $|E/E_0 - 1|$ is recorded at every output. Without relativity it stays near 10^{-14}. With relativity the checked quantity is the conserved 1PN Hamiltonian of all bodies, which already contains the Newtonian kinetic and potential energy; it stays at 10^{-13}–10^{-11}, also at high eccentricities and with tides. With tides the spin and tidal energies are included. The total angular momentum is checked in its Newtonian form, which the 1PN equations do not conserve exactly: with relativity it fluctuates at about 10^{-7} without growing. A run stops with the event `energy_error` when the error exceeds `run.max_energy_error` (default 10^{-5}), far above both levels. When energy is lost physically – tidal dissipation, mass loss, gravitational waves – the energy is reported but no longer used as a stop criterion; with tidal dissipation the conservation of the total angular momentum of orbits and spins is checked instead (`angmom_error`).

3.4 Collisions

Every star has a radius, from the stellar model or the configuration. With `run.collisions` (default on) REBOUND tests at every step whether the surfaces of two stars touch; the run then stops with the event `collision`, the time and the stars involved – or, with `physics.mergers`, continues with the merged star (Mass transfer and evolution). The default radii of the `zams` model are those of zero-age main-sequence stars; for any system in which collisions matter, evolved radii should be used.

3.5 Hierarchy detection

At every output SWINGERsim determines which stars are bound to which. Starting with every star as a separate object, it repeatedly joins the pair whose two-body relative orbit is bound and has the smallest semi-major axis, replaces the pair by its centre of mass, and stops when no bound pair remains. A stable 2+2 system gives one tree, `((m1, m2), (m3, m4))`. A new structure must persist for `run.hierarchy_persistence` consecutive outputs (default 3) before it is confirmed, so a single unusual snapshot near a close passage does not count. Confirmed changes are classified as exchange, reconfiguration, ejection, break-up or capture (Stability and fates). The detection is purely energetic (bound or not); it does not judge long-term stability, which the subsequent dynamics shows.

3.6 Diagnostics of orbit pairs

For every orbit that has a parent in the tree (A and B in `out` for a 2+2; `inner` in `mid` and `mid` in `out` for a 3+1) SWINGERsim compares analytic expectations with the run. Table 5 lists them.

Table 5. Diagnostics of an orbit pair (inner orbit and the orbit around it). *Quantity* is what is reported; *Basis* the formula or theory; *Meaning* how to read it. All analytic values are computed from the initial elements and are estimates for orientation; the simulated values are the result.

Quantity	Basis	Meaning
ZLK window, $e_{\max}$, t_{ZLK}	quadrupole test-particle theory (Hierarchies and orbits)	whether and how strongly ZLK cycles are expected
range of e, i_{mut}, cycle period	double-averaged quadrupole equations, any mass ratio (Naoz et al. 2013)	prediction of the secular cycle
t_{prec}	quadrupole nodal precession (Hierarchies and orbits)	expected precession period

Quantity	Basis	Meaning
GR apsidal period, ZLK suppression flag	1PN periastron advance	relativity suppresses ZLK if faster
$\varepsilon_{\mathrm{oct}}$	Naoz (2016)	strength of the octupole term
$\varepsilon_{\mathrm{SA}}$	Luo, Katz & Dong (2016)	validity of orbit averaging; warning above 0.01
Mardling–Aarseth and Vynatheya et al. ratios	stability criteria (Stability and fates)	> 1 stable
β	Hamers & Lai (2017)	coupling of precession and ZLK
$J_{AB}^{\max}$, period ratio	Vokrouhlický (2016); nearest small-integer ratio	2+2 plane coupling, resonances
measured e range, i_{mut} range, number and period of ZLK cycles, measured precession period	the run	the simulated result

3.6.1 Secular prediction at quadrupole order

For every orbit pair SWINGERsim integrates the orbit-averaged quadrupole equations of Naoz et al. (2013, their eqs. A27, A29 and A36), with the perturber treated as a point mass. Unlike the test-particle formula, these equations hold for any mass ratio, conserve the total angular momentum exactly and let the outer orbit change its inclination. They predict the range of e and i_{mut} and the period of the eccentricity cycle. The implementation reproduces the published curve of Naoz et al. (2013, Fig. 4) to 0.04° and the Kozai limit to 5×10^{-5} (see *Testing, Part I*).

3.6.2 Extremes between outputs

Eccentricity peaks of ZLK cycles can be much shorter than the spacing of the outputs; in an eccentric ZLK test $1 - e$ drops to about 10^{-5} for a few centuries. The N-body backend therefore samples the state internally every `run.extremes_every_orbits` periods of the shortest inner orbit (default 10) and keeps for every orbit the largest eccentricity and the smallest periastron distance $q = a(1-e)$ with their times, and for every binary of two stars the largest Roche-lobe filling factor at periastron. A spacing of a few orbits suffices because these peaks are secular and last many inner orbits; collisions are caught separately at every integrator step. The extra cost is a few per cent.

3.6.3 Chaos indicator MEGNO

With `run.megno: true` SWINGERsim computes the Mean Exponential Growth factor of Nearby Orbits (MEGNO; Cincotta & Simó 2000) from REBOUND's variational equations. MEGNO tends to 2 for regular (quasi-periodic) motion and grows linearly with time for chaotic motion; its slope gives the Lyapunov time, the e-folding time of the separation of neighbouring trajectories. Chaos means that a single run cannot be predicted beyond a few Lyapunov times and only statistics over many runs are meaningful (Ensembles and inference). REBOUND's variational equations cover only Newtonian gravity, so MEGNO is computed in a parallel Newtonian point-mass copy of the run, without relativity, tides or mass loss (MEGNO is refused together with mass loss); the cost roughly doubles.

3.7 Backward integration

A negative `run.t_max_yr` integrates into the past, with both backends. Gravity and the relativistic terms are time-reversible: integrating a run forward and its end state back returns the initial elements to 10^{-12}. Three limits apply to the past:

- **Chaos.** Beyond a few Lyapunov times the uncertainties of the observed elements grow so much that a single past has no meaning; then only an ensemble of runs within the uncertainties gives the distribution of possible pasts.
- **Tidal dissipation** run backward makes orbits wider and more eccentric; the result depends directly on the poorly known dissipation rate.
- **Stellar evolution** runs backward too: the stars become smaller towards the ZAMS (SSE) or the start of the pre-main sequence (MIST), where the run stops (`stellar_birth`).

Survival in both directions is a constraint used in ensembles (Stability and fates).

3.8 Secular backends

Orbit averaging replaces each Keplerian orbit by a ring of mass spread along it in proportion to the time spent there. The interaction energy averaged over the inner and the outer orbit (*double averaging*) depends only on the shapes and orientations of the orbits, not on the positions of the stars, so the semi-major axes are constant (in the point-mass problem) and the equations contain only the slow, secular time scales. The integrator step then follows these time scales instead of the orbital periods.

3.8.1 Vectorial elements and equations of motion

Each orbit X is described by two vectors, the normalised angular momentum $\mathbf{K}_X = \sqrt{1-e_X^2}\,\mathbf{k}_X$ and the Laplace vector $\pi_X = e_X\,\mathbf{p}_X$, where $\mathbf{k}_X$ is the unit orbit normal and $\mathbf{p}_X$ the unit vector to the periastron (Vokrouhlický 2016). These elements have no singularities at zero eccentricity or inclination. Their equations of motion (his eqs. 12–17) are

$$\frac{d\mathbf{K}_X}{dt} = -\frac{1}{\Lambda_X}\left(\pi_X \times \frac{\partial H}{\partial \pi_X} + \mathbf{K}_X \times \frac{\partial H}{\partial \mathbf{K}_X}\right), \qquad \frac{d\pi_X}{dt} = -\frac{1}{\Lambda_X}\left(\mathbf{K}_X \times \frac{\partial H}{\partial \pi_X} + \pi_X \times \frac{\partial H}{\partial \mathbf{K}_X}\right),$$

with $\Lambda_X = \mu_X n_X a_X^2$ the circular angular momentum of the orbit (μ_X its reduced mass, n_X its mean motion, a_X its semi-major axis) and H the averaged interaction energy of the whole system. The equations are integrated with the eighth-order Runge–Kutta method DOP853 of SciPy with relative tolerance `run.secular_rtol` (default 10^{-10}).

3.8.2 2+2 quadruples

For a 2+2 system H is the sum of the quadrupole and octupole terms of two triples: binary A perturbed by the centre of mass of B, and B perturbed by the centre of mass of A (Vokrouhlický 2016, eqs. 18–25). The octupole terms can be switched off with `run.secular_octupole: false`. Added to H are terms that depend only on the eccentricity of each orbit and reproduce the relativistic apsidal advance (with `physics.gr`) and the apsidal advance from the tidal and rotational distortion of the stars after Sterne (1939) (with `physics.tides`). Because all these terms are conservative, H stays exactly conserved and serves as a check of the integration. With tidal dissipation the semi-major axes, eccentricities and spin rates of A and B evolve in addition by the constant-time-lag theory (Tides and rotation). The Fig. 2 run of Vokrouhlický (2016) over 50 kyr takes about 5 s instead of about an hour with the N-body backend.

3.8.3 3+1 and other hierarchies

For every other hierarchy (3+1, and trees with five or more stars) SWINGERsim uses the general nested-binary formulation of Hamers et al. (2015) and Hamers & Portegies Zwart

(2016), written in the same vectors and equations of motion. H is a sum of *binary-pair* terms, one for each orbit k and each orbit l that contains it:

$$H_{kl} = -\frac{G\,\mu_k M_o a_k^2}{8\,a_l^3}\, f_2(\mathbf{K}_k, \pi_k, \mathbf{K}_l) - \frac{15}{64}\,\frac{G\,\mu_k M_o a_k^3}{a_l^4}\,\frac{m_{k1}-m_{k2}}{M_k}\, s\, f_3(\mathbf{K}_k, \pi_k, \mathbf{K}_l, \pi_l),$$

where μ_k is the reduced mass of orbit k, a_k and a_l the semi-major axes, M_o the mass of the component of l that does not contain k, m_{k1}, m_{k2} the masses of the two components of k and M_k their sum, $s = \pm 1$ says in which component of l the orbit k lies, and f_2, f_3 are the quadrupole and octupole functions of the vectorial elements (Hamers et al. 2015). For a 2+2 quadruple these are exactly the terms of the 2+2 backend, and both backends give the same result. Not included are the *binary-triplet* terms (octupole order, $(a_k/a_m)^2(a_m/a_l)^3$ for k inside m inside l) and terms of hexadecupole and higher order; the comparison with the code MSE (Hamers et al. 2021), which has them, shows differences below 0.001 in $e_{\max}$ for the tested systems. Relativity, the stellar distortion and tidal dissipation in every orbit of two single stars are added as for the 2+2 backend; for a 2+2 system both backends agree to the integration tolerance and the circularisation rate equals that of Hut (1981).

3.8.4 Output of the secular backends

The output has the same structure as that of the N-body backend. The elements are mean elements; the mean anomaly is not followed and is stored as 0. The hierarchy cannot change and collisions cannot occur. Stellar spins stay aligned with their orbits; with rotating stars this causes a small error in the angular-momentum check, of the order of 10^{-5} in a typical example.

3.8.5 When orbit averaging fails

Orbit averaging assumes that the outer orbit is much slower than the inner one. The size of the neglected terms is measured by the single-averaging parameter (Luo, Katz & Dong 2016)

$$\varepsilon_{\mathrm{SA}} = \left(\frac{a_{\mathrm{in}}}{a_{\mathrm{out}}(1-e_{\mathrm{out}}^2)}\right)^{3/2} \frac{m_{\mathrm{pert}}}{\sqrt{m_{\mathrm{in}}\,m_{\mathrm{tot}}}},$$

where a_{in}, a_{out} are the semi-major axes, e_{out} the outer eccentricity, m_{in} the mass of the inner orbit, m_{pert} that of the perturber and m_{tot} their sum. Table 6 shows the effect for one binary and perturber as the outer orbit widens.

Table 6. ZLK cycle period of the same binary and perturber computed with the N-body backend and with the double-averaged theory, for three outer semi-major axes. a_{out} is given relative to the closest case, which has $\varepsilon_{\mathrm{SA}} = 0.03$; *difference* is (N-body − averaged)/averaged. The difference falls quickly as the hierarchy becomes stronger.

a_{out}	N-body period	averaged period	difference
1 ×	12.8 yr	18.9 yr	−32 %
2 ×	153.8 yr	151.4 yr	+1.5 %
4 ×	1211 yr	1213 yr	−0.2 %

At $\varepsilon_{\mathrm{SA}} = 0.03$ the eccentricity range still agrees but the cycle period is a third shorter; at twice the separation ($\varepsilon_{\mathrm{SA}}$ eight times smaller) the difference is at the per-cent level. The run summary therefore warns when $\varepsilon_{\mathrm{SA}} > 0.01$. Near very high eccentricities the short-period terms neglected by the averaging can push the osculating eccentricity higher than the mean one (Luo, Katz & Dong 2016); in a reproduction of the population study of Pejcha et al. (2013)

the N-body backend found three times more triples reaching $1 - e < 10^{-3}$ than the secular one, while for the quadruples both agreed.

Validity in summary. The secular backends are first-order secular theories. They are reliable for strong hierarchies ($\varepsilon_{\mathrm{SA}}$ well below 0.01) far from mean-motion resonances. They neglect the short-period terms and, in a 2+2, the direct coupling of A and B of order ε^4; they cannot describe close encounters, exchanges, ejections or collisions. In chaotic regions they follow the N-body solution only for a while: for the Fig. 4 setup of Vokrouhlický (2016) both agree for about 4 kyr and then diverge, as in the paper.

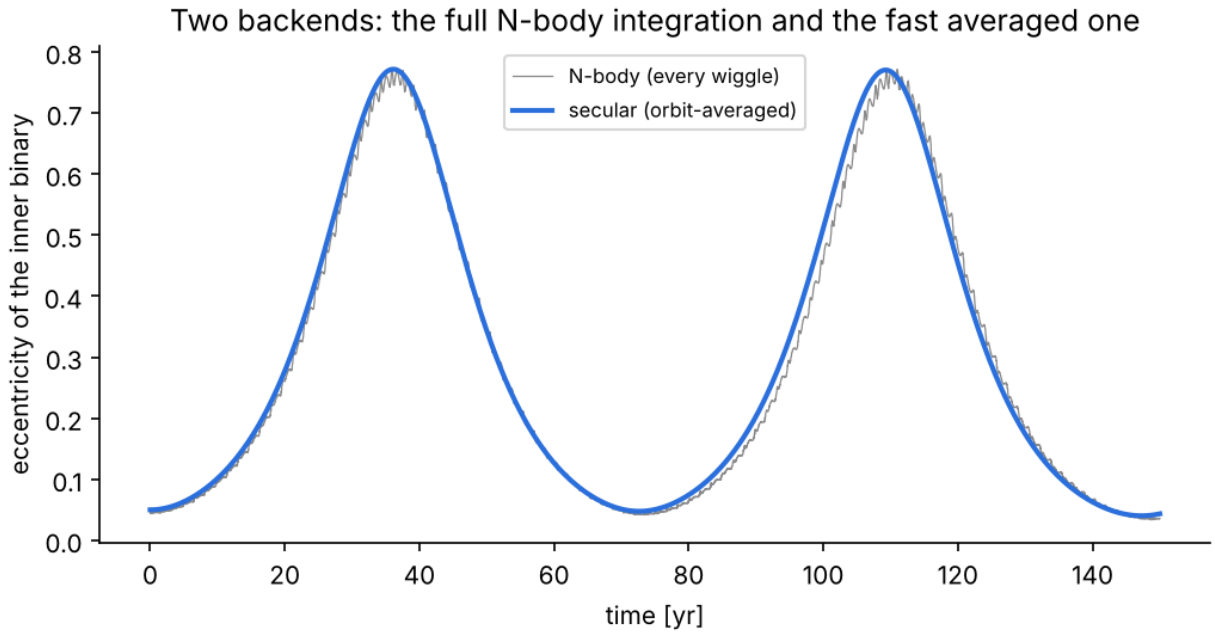


Figure 7. The same triple computed by both backends. Horizontal axis: time in years; vertical axis: eccentricity of the inner binary. The thin grey curve is the N-body solution, which shows every short-period oscillation of the osculating eccentricity; the thick blue curve is the secular (orbit-averaged) solution. Both show the same ZLK cycles of about 70 yr with a maximum near 0.77; the secular curve follows the average of the N-body curve.

Figure 7 shows the relation of the two backends for a well-separated triple. The grey curve oscillates on the time scale of the outer orbit around the blue one but neither drifts away: in this regime the secular backend reproduces the slow evolution at a small fraction of the cost. Table 7 summarises when to use which.

Table 7. Choice of backend. *Situation* is the question or the system; *Backend* the recommended one; *Reason* the physics behind the choice.

Situation	Backend	Reason
stability, break-up, exchanges, collisions	N-body	only it follows non-hierarchical motion
orbits not well separated ($\varepsilon_{\mathrm{SA}} > 0.01$, stability ratio near 1)	N-body	averaging fails for mild hierarchies
eclipse timings, light curves, radial velocities	N-body	needs the positions on each orbit
inner periods of days, runs of Myr or longer	secular	an N-body run would take years of computing
evolution over the stellar lifetime (tides, winds, mass transfer)	secular	billions of years
large ensembles of stable systems	secular	seconds per member

Situation	Backend	Reason
unsure	both	run one system with both and compare

3.9 External perturbations

3.9.1 Gas phase of formation

Multiple stars form in gas, and their orbits keep changing as long as the gas is present. SWINGERsim describes this phase with the parametric damping of Papaloizou & Larwood (2000): for every orbit listed in `physics.gas.orbits`

$$\frac{da}{dt} = -\frac{f(t)\,a}{\tau_a}, \qquad \frac{de}{dt} = -\frac{f(t)\,e}{\tau_e}, \qquad \frac{d\mathbf{k}}{dt} = \frac{f(t)}{\tau_i}\left[\mathbf{n} - (\mathbf{n}\cdot\mathbf{k})\,\mathbf{k}\right],$$

where τ_a, τ_e, τ_i are the migration, circularisation and alignment time scales (`tau_a_yr`, `tau_e_yr`, `tau_i_yr`), $\mathbf{k}$ the orbit normal, $\mathbf{n}$ the normal of the gas disc (the initial invariable plane or a given plane), and $f = 1$ until `t_end_yr`, $f = \exp[-(t - t_{\rm end})/t_{\rm decay}]$ afterwards. The last equation turns the orbit plane towards the gas plane without changing e, so that $\tan(i/2)$ to the gas plane decays as e^{-t/τ_i}. The semi-major axes become state variables. The phase runs in the secular backend. The time scales are inputs: they depend on the mass and structure of the gas, which SWINGERsim does not model.

3.9.2 Galactic tide

For orbits wider than about 1000 au the tide of the Galactic disc matters. In the solar neighbourhood its vertical part dominates, with the potential $2\pi G\rho_0 z_{\rm G}^2$ ($z_{\rm G}$ the height along the Galactic pole, ρ_0 the local disc density, default 0.185 $\mathrm{M}_\odot$ pc^{-3}; Heisler & Tremaine 1986; Grishin & Perets 2022, eqs. 3–4). Averaged over an orbit of reduced mass μ,

$$H = \pi G\rho_0\,\mu\,a^2\left[(1-e^2) - (\mathbf{j}\cdot\hat{\mathbf{z}})^2 + 5(\mathbf{e}\cdot\hat{\mathbf{z}})^2\right],$$

with $\mathbf{j} = \sqrt{1-e^2}\,\mathbf{k}$, the eccentricity vector $\mathbf{e}$ and the Galactic pole $\hat{\mathbf{z}}$. It drives ZLK-like cycles above the critical inclination $\cos^2 i < 4/5$ to $e_{\rm max} = \sqrt{1 - (5/4)\cos^2 i}$, on the time scale $\tau_{\rm GT} = 0.89\ \mathrm{Gyr}\,(M/\mathrm{M}_\odot)^{1/2}(a/10^4\,\mathrm{au})^{-3/2}$ (Grishin & Perets 2022, eqs. 6–7). In the secular backends these terms are added to H of every orbit; in the N-body backend the tide is applied as velocity kicks $\Delta\mathbf{v}_i = -4\pi G\rho_0\left[(\mathbf{r}_i - \mathbf{r}_{\rm cm})\cdot\hat{\mathbf{z}}\right]\hat{\mathbf{z}}\,\Delta t$ every 1/400 of the outer period (operator splitting), and its potential energy enters the energy check. A 60°-inclined wide binary reaches $e_{\rm max}$ within 10^{-4} of the analytic value.

3.9.3 Passing stars

A star passing at a periapsis distance Q much larger than an orbit changes the orbit's eccentricity and orientation but hardly its energy. To first order in the perturbation the changes of $\mathbf{e}$ and $\mathbf{j}$ follow from Hamers & Samsing (2019, eq. 9), with the small parameter $\epsilon = [M^2/(m(m+M))\,(a/Q)^3]^{1/2}(1+E)^{-3/2}$ (M the perturber mass, m the mass of the orbit, E the eccentricity of the perturber's hyperbola). With `physics.encounters` SWINGERsim draws a Poisson sequence of encounters (rate $n\pi b_{\rm max}^2\langle v\rangle$, relative speeds from a Maxwellian weighted by v, impact parameters uniform in area, isotropic orientations). The secular backends apply the first-order kicks to every orbit for which the passage is distant ($Q > 3a(1+e)$) and count closer passages without applying them. The N-body backend integrates every passing star as a particle from 20 periapsis distances before to 20 after its periapsis, so close passages are followed too. At $Q = 20a$ the first-order kick agrees with a direct N-body passage to 0.6 %.

3.9.4 Gravitational waves

With `physics.gw` every orbit of two single stars loses energy and angular momentum by gravitational radiation, orbit-averaged after Peters (1964):

$$\dot{a} = -\frac{64}{5}\frac{G^3 m_1 m_2 M}{c^5 a^3 (1-e^2)^{7/2}}\left(1+\tfrac{73}{24}e^2+\tfrac{37}{96}e^4\right), \qquad \dot{e} = -\frac{304}{15}\frac{e\,G^3 m_1 m_2 M}{c^5 a^4 (1-e^2)^{5/2}}\left(1+\tfrac{121}{304}e^2\right),$$

with m_1, m_2 the masses and $M = m_1 + m_2$. The secular backends integrate these rates with the orbits; the N-body backend applies them once per output step at fixed orientation and mean anomaly. The orbital decay of the Hulse–Taylor pulsar is reproduced to 0.02 % (Weisberg & Huang 2016; see *Testing, Part I*). The effect matters only for compact remnants and very close binaries.

3.10 Implementation

External libraries. REBOUND (Rein & Liu 2012) with its IAS15 integrator (Rein & Spiegel 2015), collision search and variational equations for MEGNO; REBOUNDx (Tamayo et al. 2020) with the effects `gr_full` (1PN for all bodies), `tides_spin` (Lu et al. 2023; next chapter) and the operator `modify_mass` (mass loss); SciPy's DOP853 integrator for the secular equations; Numba, when installed, compiles SWINGERsim's 2+2 secular kernel. **Own code.** The secular equations of motion and the averaged energies for 2+2 (Vokrouhlický 2016, eqs. 12–25) and for any nested tree (binary-pair terms after Hamers et al. 2015 and Hamers & Portegies Zwart 2016) were coded for SWINGERsim, together with the eccentricity-only Hamiltonian terms of relativity and stellar distortion that keep H conserved, the constant-time-lag dissipation in the secular equations, the gas damping, the Galactic tide (secular terms and N-body kicks), the encounter sequence and first-order kicks, the orbit-averaged gravitational-wave rates, the hierarchy detection, the tracking of extremes between outputs, the pair diagnostics, the energy book-keeping (1PN Hamiltonian, spin and tidal energy, angular momentum) and the backward runs.

Chapter 4

Tides and rotation

A star in a close binary is not a point. The companion pulls harder on the near side than on the far side, so the star is stretched along the line of centres (a *tidal bulge*); if it rotates, it is also flattened at the poles (a *rotational bulge*). Both distortions add a quadrupole to the star's gravity, which turns the periastron (apsidal motion) and couples the spin of the star to the orbit. Internal friction makes the tidal bulge lag behind the companion; the lag transfers angular momentum between spin and orbit and dissipates energy, so that spins synchronise, orbits circularise and shrink. In a multiple star tides have a second role: the fast apsidal motion they cause in close binaries suppresses ZLK cycles (Hierarchies and orbits).

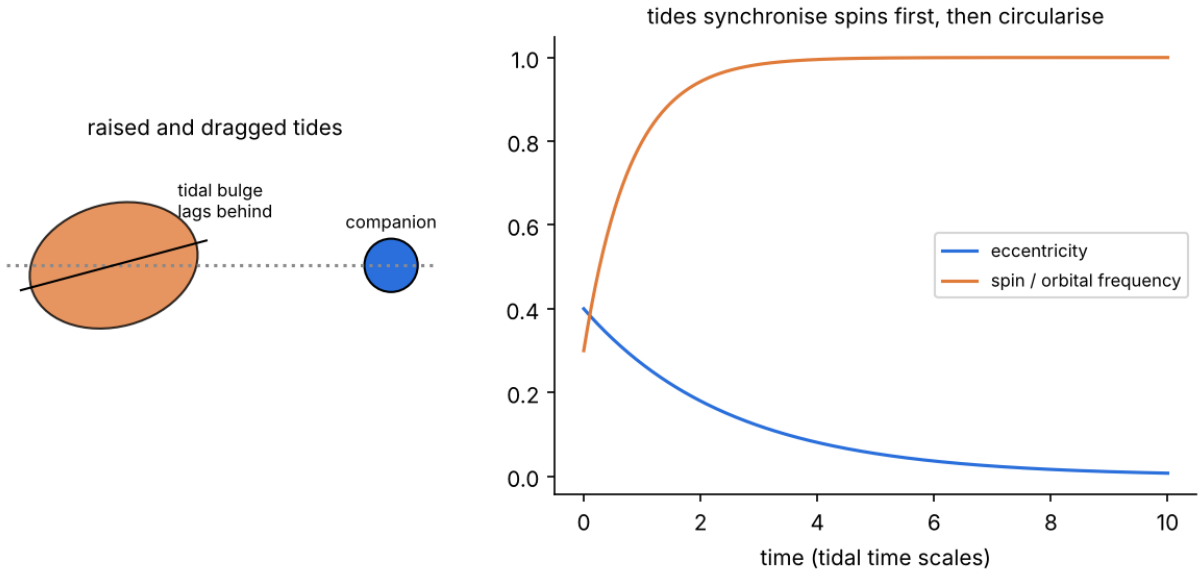


Figure 8. Tides, schematically. Left: a star (orange) distorted by its companion (blue); the black line is the axis of the tidal bulge, which lags behind the line of centres (grey dotted) because of internal friction. Right: evolution under tidal friction in units of the tidal time scale (horizontal axis). Orange: spin frequency of the star divided by the orbital frequency, rising from 0.3 to 1 (synchronisation); blue: orbital eccentricity, falling from 0.4 towards 0 (circularisation). Synchronisation is reached several times faster than circularisation because a star stores much less angular momentum in its spin than the orbit does.

Figure 8 summarises the order of events. The right panel is illustrative (arbitrary time unit), but its message holds for real binaries: the spin adjusts first, the orbit later. The equations behind it are given below.

4.1 Equilibrium tides in the N-body backend

With `physics.tides: true` the stars become extended bodies. SWINGERsim uses the REBOUNDx effect `tides_spin` (Lu et al. 2023), which implements the equilibrium-tide model of Eggleton, Kiseleva & Hut (1998) for any number of bodies:

- every star is distorted by its own rotation and by the tidal field of **all** other stars, and the distorted stars attract each other with the corresponding quadrupole forces; this conservative part makes the periastra advance and the orbital planes precess about the stellar spins;
- the spin vector of each star evolves under the torques of all other stars (for example the precession of a spin tilted to its orbit);
- with a time lag $\Delta t > 0$ the tidal bulge lags behind, energy is dissipated, orbits circularise and shrink and the spins approach pseudo-synchronism.

The strength of the distortion is set by the potential Love number $2k_2$ and by R^5, the spin evolution by the moment of inertia $I = r_g^2 M R^2$ (Stars). `physics.rotation` (which requires `physics.tides`) gives the stars a spin: synchronous with their binary by default (`stars.<name>.P_rot: sync`), pseudo-synchronous (`pseudosync`), or a given period in days, with an `obliquity` (spin–orbit angle) and a `spin_azimuth`. Without `physics.rotation` the stars start without spin, which leaves only the tidal distortion (the model used by Vokrouhlický 2016, his Fig. 7).

Time-lag convention. SWINGERsim expects the time lag `lag_s` (seconds) in the convention of Mignard (1979), Hut (1981) and Leconte et al. (2010). Reading the source of `tides_spin` and comparing its rates with these formulae showed that the same number given to REBOUNDx as its parameter τ produces rates exactly twice as large; SWINGERsim therefore passes $\tau = \Delta t/2$, and benchmarks against Leconte et al. (2010) check the result (see *Testing, Part I*). For orientation, a constant time lag corresponds to a tidal quality factor $Q \approx 1/(2\,|\Omega - n|\,\Delta t)$ at the tidal frequency, where Ω is the spin and n the orbital angular frequency.

Conservation. Without dissipation the energy including spin and tidal energy is conserved to $|dE/E| \sim 10^{-13}$ (without relativity). With dissipation the energy decreases physically, and the run checks the total angular momentum of orbits and spins instead.

4.2 Classical apsidal motion of a binary

For every binary of two stars SWINGERsim gives the apsidal advance expected from the binary itself, the classical tidal and rotational term of Sterne (1939):

$$\frac{\dot\omega}{n} = \sum_i k_{2,i} \left(\frac{R_i}{a}\right)^5 \left[15\,\frac{m_j}{m_i}\, f(e) + \left(1 + \frac{m_j}{m_i}\right) \left(\frac{\Omega_i}{n}\right)^2 g(e) \right],$$

$$f(e) = \frac{1 + \frac{3}{2}e^2 + \frac{1}{8}e^4}{(1-e^2)^5}, \qquad g(e) = \frac{1}{(1-e^2)^2},$$

where the sum runs over the two stars i (with companion j), $k_{2,i}$ is the apsidal-motion constant, R_i the radius, m_i the mass, Ω_i the spin angular frequency (aligned with the orbit), a the semi-major axis, e the eccentricity and n the mean motion. The first term in the brackets is the tidal, the second the rotational contribution; the relativistic term (Hierarchies and orbits) is added separately. The steep dependence on $(R/a)^5$ makes apsidal motion a probe of the radii and of the internal structure through k_2 (Claret & Giménez 2010).

In a multiple system the measured advance contains in addition the perturbation by the other stars; the run summary gives both, and `swingersim apsidal` splits the total into tidal, rotational, relativistic and dynamical parts and gives the factor by which k_2 of both stars would have to change to reach a target rate. Published photodynamical solutions often fit the apsidal rate as a free parameter instead of computing it from stellar structure; a difference against

SWINGERsim then measures the tension between the measured rate and stellar models, not a difference between two codes.

Pseudo-synchronisation. In an eccentric orbit tides cannot synchronise a spin with the orbit at every point; they drive it to the *pseudo-synchronous* rate at which the orbit-averaged tidal torque vanishes (Hut 1981, eq. 42), faster than the mean motion (1.24 times at $e = 0.2$). SWINGERsim uses it for `P_rot: pseudosync` and reports the pseudo-synchronous period in the tidal time scales below.

Rotation and k_2. A rotating star is more centrally condensed than a static one. With `physics.k2_rotation` the model constants are corrected by $\delta \log k_2 = -0.87\,\lambda_s$, $\lambda_s = 2\Omega^2 R^3/(3GM) = 2V^2/(3gR)$ (Claret 1999; Claret & Giménez 2010 apply this rotational correction to observed binaries), with the angular velocity Ω, equatorial velocity V, surface gravity g, radius R and mass M of each star at that time, in all backends.

4.3 Tidal evolution: the constant-time-lag model

The secular backends and the analytic time scales use the constant-time-lag theory (Mignard 1979; Hut 1981) in the form of Leconte et al. (2010, eqs. 1–3) for aligned spins. With the potential Love number $2k_{2,i}$ and the time lag Δt_i of star i,

$$K_i = \tfrac{3}{2}\,(2k_{2,i})\,\Delta t_i\,\frac{G m_j^2}{R_i}\left(\frac{R_i}{a}\right)^6 n^2,$$

$$\frac{da}{dt} = \frac{4a^2}{G m_1 m_2}\sum_i K_i\left[N(e)\frac{\Omega_i}{n} - N_{a1}(e)\right], \qquad \frac{de}{dt} = \frac{11\,a\,e}{G m_1 m_2}\sum_i K_i\left[\Omega_e(e)\frac{\Omega_i}{n} - \frac{18}{11}N_{e1}(e)\right],$$

$$\frac{d\Omega_i}{dt} = -\frac{2K_i}{C_i\,n}\left[\Omega(e)\frac{\Omega_i}{n} - N(e)\right], \qquad C_i = r_{g,i}^2 m_i R_i^2,$$

where N, N_{a1}, N_{e1}, Ω_e and $\Omega(e)$ are the polynomial functions of the eccentricity of Leconte et al. (2010) (the same functions appear in Hut 1981, eqs. 9–11), C_i the moment of inertia and the other symbols as above. The spin equation follows from the balance of the tidal torque with the change of the orbital angular momentum; for a circular orbit it reduces to Murray & Dermott (1999, eq. 4.159). The secular backends integrate these equations for every orbit of two single stars together with the gravitational dynamics.

For every binary the run summary lists the time scales at the start: circularisation $e/|\dot{e}|$, orbital decay $a/|\dot{a}|$, the e-folding time of each spin towards pseudo-synchronism and the pseudo-synchronous period. Realistic stellar lags give time scales of millions of years and longer, beyond direct N-body integrations; in N-body runs dissipation is mainly useful for strongly eccentric phases (ZLK peaks, where it shrinks orbits; Fabrycky & Tremaine 2007) and for tests.

4.4 Dissipation from the stellar structure

The time lag is the least known input of tidal evolution. With `physics.tidal_lag: hurley2002` SWINGERsim derives it for every star of a two-star binary from its structure after Hurley, Tout & Pols (2002, Sect. 2.3), instead of a fixed `lag_s`:

- a star with a **convective envelope** dissipates through turbulent viscosity (equilibrium tide; Zahn 1977; Rasio et al. 1996),

$$\left(\frac{k}{T}\right)_{\rm c} = \frac{2}{21}\,\frac{f_{\rm conv}}{\tau_{\rm conv}}\,\frac{M_{\rm env}}{M}, \qquad f_{\rm conv} = \min\left[1, \left(\frac{P_{\rm tid}}{2\tau_{\rm conv}}\right)^2\right],$$

with the envelope mass $M_{\rm env}$, the eddy turnover time $\tau_{\rm conv} = 0.4311\,[M_{\rm env}R_{\rm env}(R - R_{\rm env}/2)/(3L)]^{1/3}$ yr ($R_{\rm env}$ the depth of the envelope, L the luminosity, solar units) and the tidal pumping period $P_{\rm tid}$, $1/P_{\rm tid} = |1/P_{\rm orb} - 1/P_{\rm spin}|$ (Hurley et al. 2002, eq. 33; the factor 2 in $f_{\rm conv}$, their eq. 32, is there because the tide comes around twice in this period). The factor $f_{\rm conv}$ reduces the viscosity when the tidal period is shorter than the turnover time – quadratically (Goldreich & Nicholson 1977; Hurley et al. 2002) by default, or linearly (Zahn 1966, 1989; Zahn & Bouchet 1989) with `physics.tidal_convective_reduction: linear`;
- a star with a **radiative envelope** dissipates through the dynamical tide (Zahn 1975, 1977), $(k/T)_{\rm r} = 1.9782\times10^{4}\,(MR^2/a^5)(1+q)^{5/6}E_2$ yr⁻¹ with $E_2 = 1.592\times10^{-9}M^{2.84}$ (q the mass ratio, solar units).

Hut's (1981) (k/T) corresponds to the constant time lag through $T = R^3/(GM\tau_{\rm Hut})$ and $\tau_{\rm Hut} = 2\,\Delta t$ (SWINGERsim uses the Love number $2k_2$ of Leconte et al. 2010; the factor was checked numerically). The lag is computed from the initial orbit and spin and, for evolving stars, again in every segment; a `lag_s` given for a star is kept, and `physics.tidal_efficiency` multiplies the dissipation.

Calibration. This recipe is widely used (BSE, MSE, COSMIC), but its calibration is the weakest link of long-term evolution. On the main sequence alone, solar-type binaries circularise in 4 Gyr only up to a period of about 3.8 d, whereas the cluster M67 shows 12.1 +1.0/−1.5 d (Meibom & Mathieu 2005, Table 3); a `tidal_efficiency` of about 150 would be needed on the main sequence alone (100 gives 11.2 d, 200 gives 12.9 d).

4.4.1 Inertial waves

In a rotating convective envelope a tidal forcing with frequency below 2Ω excites *inertial waves*, restored by the Coriolis force. Their dissipation – the dynamical tide of convective zones – is much stronger than the damping of the equilibrium tide by turbulent viscosity, which is weak because the tidal period is short compared with the convective turnover time (Ogilvie 2013; Mathis 2015; Barker 2020, 2022). Averaged over the tidal frequency it depends only on the structure and the spin,

$$\frac{1}{\langle Q'_{\rm IW}\rangle} = \epsilon^2\,C, \qquad \epsilon^2 = \frac{\Omega^2R^3}{GM},$$

where Q' is the modified tidal quality factor, Ω the spin rate and C a structure coefficient. With `physics.inertial_waves: true` (requires `tidal_lag: hurley2002`) SWINGERsim computes C for a star with a radiative core of radius αR and mass βM (from the envelope model) with the two-layer formula of Ogilvie (2013), and for a fully convective star (below 0.35 $M_\odot$) for an $n = 1.5$ polytrope, $C = 1/130.9$, computed once with the method of Barker (2020, eqs. 12–38; 2022, App. A) for continuous profiles. The eccentricity damping $d\ln e/dt = -(225/16)(M_2/M)(R/a)^5\,n/Q'$ (Barker 2022, eq. 1) is converted into an equivalent constant time lag by matching the damping rate of the lag model for a synchronous star, $-(21/2)(k/T)\,q(1+q)(R/a)^8$ (Hut 1981), and added to the equilibrium tide. Inertial waves are switched off when the star rotates slower than half the orbital frequency (no forcing frequency below 2Ω).

The numerical method reproduces the two-layer formula in the form of Barker (2020, eq. 39) for a smoothed two-layer profile to 0.2 %, a check of the solver and its normalisation. For main-sequence stars of 0.3 and 1 $M_\odot$ rotating in 10 d it gives $Q' = 0.9$ and 1.0×10^7, as Barker (2022) finds with MESA profiles. For realistic profiles of solar-type stars the two-layer model under-predicts the dissipation by about a factor of two, and for F stars above 1.2 $M_\odot$ by an order of magnitude or more (Barker 2020).

4.4.2 Tides before the main sequence

Young stars are large and fully convective, and most of the circularisation of solar-type binaries happens before they reach the main sequence (Zahn & Bouchet 1989). With MIST

stars on the pre-main sequence (Stars) the equilibrium tide and the inertial waves act there too. Twin 1 $M_\odot$ binaries started at 1 Myr are circular up to an initial period of 7.9 d at 50 Myr (Zahn & Bouchet: 7.2–8.5 d). In the period observers use – the final period of the binaries just circularised – they are circular up to 6.6 d at 3 Myr (pre-main-sequence binaries: 7.1 ± 1.2 d; Meibom & Mathieu 2005, Table 3) and up to 7.5 d at 4 Gyr (M67: 12.1 +1.0/−1.5 d). With inertial waves the main sequence alone circularises solar-type twins up to 7.5 d in 4 Gyr (3.8 d with the equilibrium tide alone). The pre-main sequence is thus reproduced; the further growth on the main sequence is not. Barker (2022), with frequency-averaged inertial-wave dissipation computed from MESA profiles and synchronised spins, finds circularisation periods that grow with main-sequence age as observed. SWINGERsim's two-layer estimate with SSE and MIST structures and evolving spins does not reproduce this; the difference lies in the stellar structure (the two-layer model under-predicts the dissipation of realistic solar-type profiles, see above) and in the spin history, and it remains an open point. Resonance locking of gravity modes, which SWINGERsim does not model, acts mainly on the pre-main sequence: Zanazzi & Wu (2021) find circular periods of about 4–6 d from it and a negligible main-sequence contribution. The result also depends on the start of the calculation: a younger, larger star circularises more (started at 0.15 Myr, 11.6 d already at 3 Myr). On the pre-main sequence the inertial waves dominate, so the linear and quadratic reductions of the turbulent viscosity agree there within 1 %.

4.5 Limits

- The equilibrium-tide model assumes small, quasi-static distortions. Dynamical tides of radiative stars enter only through the Zahn recipe in the lag; resonance locking and non-linear tides are not modelled.
- In the secular backends spins stay aligned with their orbits; spin–orbit misalignment and spin precession are followed only by the N-body backend.
- The dissipation rates carry the calibration uncertainty discussed above; the circularisation periods of old clusters are not reproduced by any equilibrium-tide model, including this one.
- Without `physics.tides` the eccentricity growth in ZLK cycles of close binaries is overestimated (Vokrouhlický 2016).

4.6 Implementation

External library. The tidal and rotational forces and the spin evolution of the N-body backend are the REBOUNDx effect `tides_spin` (Lu et al. 2023; Tamayo et al. 2020). **Own code.** The conversion of the time-lag convention ($\tau = \Delta t/2$, found by reading the REBOUNDx source and verified against Leconte et al. 2010); the initial spins (synchronous, pseudo-synchronous after Hut 1981, given periods and obliquities); the Sterne (1939) apsidal rates and the split into contributions; the constant-time-lag equations of Leconte et al. (2010) in the secular backends and the tidal time scales; the structure-based lag of Hurley, Tout & Pols (2002) with both viscosity reductions; the rotation correction of k_2; and the inertial-wave dissipation – the two-layer formula of Ogilvie (2013) and an own solver of the continuous-profile method of Barker (2020, 2022) for polytropes and MESA profiles.

Chapter 5

Mass transfer and evolution

Stars change their masses: they lose mass in winds, give mass to a companion when they fill their Roche lobes, share an envelope in contact binaries, engulf a companion in a common envelope, or merge with it. Every such change acts on all orbits of the system that contain the star. This chapter describes how SWINGERsim models these processes and how it couples the evolution of the stars (chapter Stars) to the dynamics (chapter Dynamics).

5.1 Coupling stellar evolution to the dynamics

Stars change on nuclear time scales (Myr–Gyr), in most systems much more slowly than the secular dynamics. With `physics.evolution: true` the coupling is an **operator splitting** as in the code MSE (Hamers et al. 2021):

- **Secular backends.** The run is cut into segments within which no stellar radius, apsidal constant, moment of inertia (and, with mass loss, mass) changes by more than `run.evolution_max_dr` (default 1 %). Within a segment the stellar structure (radius, k_2, r_g^2, tidal lag) is held at its value in the middle of the segment; the orbital state is carried from segment to segment, and spins conserve their angular momentum when the moment of inertia changes. On the thermally pulsing AGB, where the radius peaks for only a few centuries at every pulse, the segment grid includes the nodes of the MIST tracks and is refined by bisection until no radius changes by more than `evolution_max_dr` between two grid points. Runs of millions to billions of years are streamed in chunks and reduced to their extremes, so they fit in memory.
- **N-body backend.** The stellar structure is updated at every output step (radius for collisions and Roche lobes, Love number, moment of inertia and lag for the tidal forces, spins at constant spin angular momentum), held at its value in the middle of the step. The output steps (`run.n_output`) must therefore be short compared with the evolution.

A run ends at the end of the track of any star (`stellar_evolution_end`: helium ignition with SSE, the departure from the AGB or the end of carbon burning with MIST), backwards in time at the birth of the stars (`stellar_birth`), and, unless mass transfer is followed, when a star fills its Roche lobe (`roche_overflow`).

5.2 Winds

With `physics.mass_loss` the masses change during the run, from one of two sources:

- `source: track` – the mass of every star follows its track. MIST masses contain the wind prescriptions of the MESA models behind MIST (Vink et al. 2001 for hot stars with full strength, Nugis & Lamers 2000 for Wolf–Rayet stars, Reimers 1975 on the giant branch and Blöcker 1995 on the AGB, boosted for rotating stars by $(1-\Omega/\Omega_{\rm crit})^{-0.43}$; Choi et al. 2016). The SSE tracks are computed without winds; for them SWINGERsim

adds the wind of the SSE code (Hurley, Pols & Tout 2000, Sect. 7.1): the larger of $9.6 \times 10^{-15}(Z/0.02)^{1/2} R^{0.81} L^{1.24} M^{0.16}$ $M_\odot$ yr^{-1} (Nieuwenhuijzen & de Jager 1990) above 4000 $L_\odot$ and $4\times10^{-13}\,\eta\,LR/M$ $M_\odot$ yr^{-1} (Reimers 1975) from the Hertzsprung gap on, with η = `reimers_eta` (default 0.5); R, L, M in solar units.

- `source: rates` – constant rates $\dot{M}$ per star, for parametric studies and benchmarks.

The wind is isotropic in the frame of the star and carries the star's momentum (the *Jeans mode*; Jeans 1924; Hadjidemetriou 1963). Its effect on an orbit X of total mass M_X is measured by the mass-loss index (Veras et al. 2011, eq. 15)

$$\Psi = \frac{|\dot{M}_X|}{n_X M_X},$$

the ratio of the orbital period to the mass-loss time scale divided by 2π (n_X the mean motion).

- **N-body backend.** The masses change during the integration with the REBOUNDx operator `modify_mass` ($\dot{m} = m/\tau$ with velocities unchanged), with τ of every star set at every output step so that the mass reaches its value at the end of the step. This is exact for any Ψ, from the adiabatic regime ($\Psi \ll 1$: aM constant, e constant on average) to the impulsive one ($\Psi \gg 1$: position and velocity are kept while the mass drops; a circular binary that loses half its mass at once is unbound; Blaauw 1961; Hills 1983).
- **Secular backends.** Orbit-averaged equations need the adiabatic response: every orbit keeps its eccentricity vector and orientation and $a_X M_X$ is constant (Hadjidemetriou 1963). A run is refused when Ψ of any orbit exceeds `physics.mass_loss.max_psi` (default 0.05; the transition to the non-adiabatic regime lies at $\Psi \sim 0.01$–0.1, Veras et al. 2011); the largest Ψ of every orbit is stored. The test-particle ZLK cycles with a perturber of changing mass then agree with the N-body backend to 0.2 % in timing.
- **Spins.** The wind carries away spin angular momentum at the rate $\dot{J} = \frac{2}{3}\dot{M}R^2\Omega$ (a thin spherical shell; Hurley et al. 2002, eq. 11), so $J \propto M^{2/(3r_g^2)}$.

Triple evolution dynamical instability. Mass loss from an inner binary widens the inner orbit by $M_{\mathrm{in},0}/M_{\mathrm{in}}$ and the outer one only by the smaller factor $M_{\mathrm{tot},0}/M_{\mathrm{tot}}$, so the hierarchy weakens and a stable triple can cross the stability limit (Perets & Kratter 2012; Toonen et al. 2022). In a benchmark the inner eccentricity starts to grow within 7 % of the time at which the adiabatically expanding orbits cross the Mardling & Aarseth (2001) limit; the break-up itself takes many outer orbits after that, as Toonen et al. found.

5.3 Roche-lobe overflow and stable mass transfer

A star of a two-star binary that fills its Roche lobe (Stars) gives mass to its companion. With `physics.mass_transfer` (which needs `physics.evolution` with SSE or MIST stars) SWINGERsim follows stable transfer with the binary-star evolution algorithm (BSE) of Hurley, Tout & Pols (2002, Sect. 2.6). Every two-star binary is treated as circular – tides circularise close binaries before the overflow – and its **history** (masses, separation, stellar states) is computed before the dynamics from the stellar tracks and the initial separation. The dynamics then follows that history (section Histories in the dynamics).

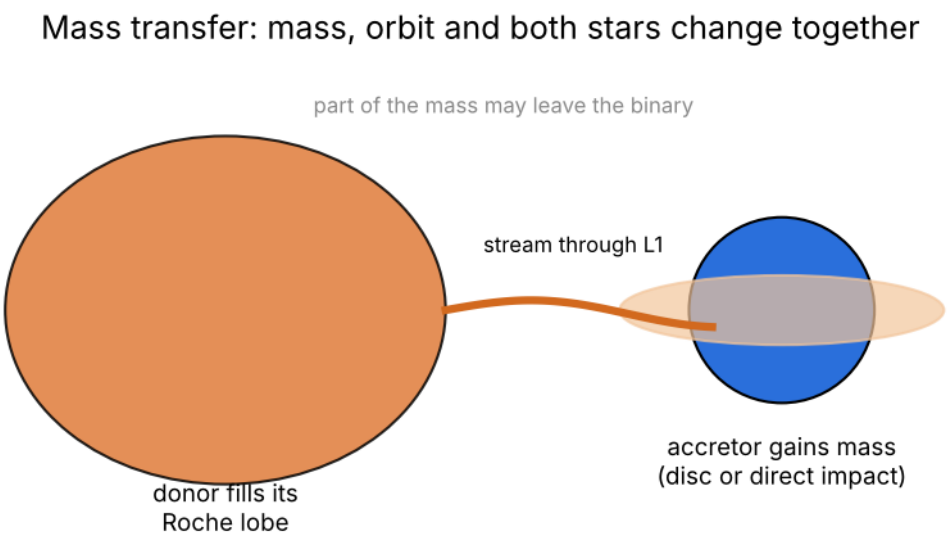


Figure 9. Mass transfer, schematically. The donor (orange, left) fills its Roche lobe; a stream of gas (dark orange line) leaves it through the inner Lagrange point L1 and reaches the accretor (blue, right), either directly or through an accretion disc (light orange ellipse). The grey note at the top recalls that part of the transferred mass may leave the binary. The masses of both stars, the orbit and both stars' structure change together.

Figure 9 names the ingredients that the equations below quantify: the rate at which the donor loses mass, the fraction the accretor keeps, the angular momentum carried by the mass that leaves, and the response of both stars.

Rate (Hurley et al. 2002, eqs. 58–63). A donor of radius R overfilling its lobe R_L transfers

$$\dot{M} = 3 \times 10^{-6} \, [\min(M, 5)]^2 \, [\ln(R/R_L)]^3 \ \mathrm{M}_\odot \, \mathrm{yr}^{-1}$$

(M in $\mathrm{M}_\odot$), so it stays just above its lobe while its expansion or the shrinking lobe drives the transfer. The rate is limited by the thermal rate M/τ_{KH} of giant-like donors (Hertzsprung gap and later; $\tau_{\mathrm{KH}} = 10^7 \, MM'/(RL)$ yr with the envelope mass $M' = M - M_c$, M_c the core mass, solar units) or by the dynamical rate.

Stability. Transfer is dynamically unstable and leads to a common envelope when the mass ratio $q = M_{\mathrm{donor}}/M_{\mathrm{accretor}}$ exceeds a critical value q_{crit} (Table 8).

Table 8. Critical mass ratios $q_{\mathrm{crit}} = M_{\mathrm{donor}}/M_{\mathrm{accretor}}$ above which transfer is dynamical (Hurley et al. 2002, Sect. 2.6). *Donor* is the evolutionary phase of the donor; $x = 0.3$ is the exponent of the giant mass–radius relation and M_c/M the core mass fraction.

Donor	q_{crit}	Note
main sequence below 0.7 $\mathrm{M}_\odot$ (deeply convective)	0.695	
main sequence above 0.7 $\mathrm{M}_\odot$, core helium burning	3.0	`q_crit_ms`; e.g. 1.6 of binary_c (de Mink et al. 2007)
Hertzsprung gap	4	
giant branch, AGB	$(1.67 - x + 2(M_c/M)^5)/2.13$	eq. 57

With `stability: zeta` giant donors are judged instead by the adiabatic response of a condensed polytrope (Hjellming & Webbink 1987; Soberman et al. 1997, eqs. 60–61),

$$\zeta_{\rm ad} = \frac{2}{3}\frac{m_c}{1-m_c} - \frac{1}{3}\frac{1-m_c}{1+2m_c} - 0.03\,m_c + \frac{0.2\,m_c}{1+(1-m_c)^{-6}}, \qquad m_c = \frac{M_c}{M},$$

where $\zeta_{\rm ad} = d\ln R/d\ln M$ of the donor on the dynamical time scale; transfer is unstable when $\zeta_{\rm ad}$ falls below the exponent $\zeta_L = d\ln R_L/d\ln M$ of the Roche lobe for the actual accretion fractions.

Accretion (eqs. 64–65). A main-sequence, Hertzsprung-gap or core-helium-burning accretor takes at most $10\,\tau_{\dot M}/\tau_{\rm KH,2}$ of the transferred mass ($\tau_{\dot M} = M_2/\dot M$, $\tau_{\rm KH,2}$ its thermal time scale); the rest leaves with the donor's specific orbital angular momentum. Giant accretors take everything. With `accretion: fractions` the fractions `alpha` and `beta` below are constant.

Orbit. Of the mass $|dM_d|$ lost by the donor a fraction α leaves from the donor's vicinity, a fraction β from the accretor's (isotropic re-emission), and $1-\alpha-\beta$ is accreted. With the orbital angular momentum $J = M_d M_a\sqrt{Ga/M}$ of a circular binary (M_d, M_a the donor and accretor masses, $M = M_d + M_a$),

$$\frac{d\ln a}{dM_d} = 2\,\frac{\alpha M_a^2 + \beta M_d^2}{M M_d M_a} + \frac{\alpha+\beta}{M} - \frac{2}{M_d} + \frac{2(1-\alpha-\beta)}{M_a}.$$

This reproduces the closed forms of Soberman et al. (1997) and Postnov & Yungelson (2014) to 10^{-12}: conservative transfer keeps $aM_d^2M_a^2$ constant, the Jeans mode aM.

Response of donors and accretors. Every star carries its mass, its age on the track it follows and the mass of that track (Hurley et al. 2000, Sect. 7.1; 2002, Sect. 2.6.6). A main-sequence star, or a Hertzsprung-gap star whose new mass would not shrink its helium core, follows the track of its present mass at the same evolutionary state (EEP for MIST, phase fraction for SSE): a donor shrinks with its mass and crosses the gap on the longer time scale of its new mass, an accretor brightens. With `rejuvenation: hurley` (default) an accretor with a convective core (above 1.25 $M_\odot$) or a fully convective one (below 0.35 $M_\odot$) mixes the fresh hydrogen into its burning region and is rejuvenated further, $t' = (M/M')\,(t'_{\rm MS}/t_{\rm MS})\,t$, where t and t' are the ages before and after, M, M' the masses and $t_{\rm MS}$, $t'_{\rm MS}$ the main-sequence lifetimes (Hurley et al. 2000). A gainer therefore looks younger and less evolved than a single star of its mass. Giants keep the track of the mass they had when they left the gap (their structure follows the core), and their radius follows the envelope mass with $\zeta_{\rm eq} = -x + 2(M_c/M)^5$, $x = 0.3$ (eq. 56). The helium core never shrinks.

Events that end a history: `common_envelope` (unless the common envelope is followed), `contact` (the accretor fills its own lobe, unless contact binaries are followed), `donor_stripped` (the donor is down to its helium core; helium stars are not followed).

Check. For the BSE Algol example (2.9 + 0.9 $M_\odot$, 2.7 d) SWINGERsim ends with the donor mass of BSE (0.404 $M_\odot$; BSE in COSMIC, Breivik et al. 2020: 0.401; the paper 0.42); the accreted mass is the least certain number of that example, and the published numbers are not self-consistent (see *Testing, Part I*).

5.4 Histories in the dynamics

In the dynamics the masses of the two stars of a transferring binary follow its history. The separation is scaled at every operator-splitting step (secular backends) or output step (N-body: the relative orbit $\mathbf{r} \to g\mathbf{r}$, $\mathbf{v} \to \mathbf{v}/\sqrt{g}$ at fixed eccentricity vector and phase) by the change beyond the adiabatic response aM = const, which the backends apply themselves. Outer orbits respond adiabatically to the mass that leaves the binary. During the transfer the donor's radius is its Roche-lobe radius (for eclipses, tides and diagnostics), and the overflow of the transferring star does not end the run. The re-modelled stars carry their radius, luminosity and temperature into the dynamics and the light curves. Winds and transfer

combine: the stars lose mass by their winds before, during and after the transfer, and the binary widens by aM = const for the wind part.

5.4.1 Transfer started by the dynamics

The histories are computed for circular binaries of the configured separations. The dynamics can drive a star over its lobe that no history expects: ZLK cycles raise the eccentricity of an inner binary until a star touches its lobe at periastron (Eggleton & Kisseleva-Eggleton 2006; Fabrycky & Tremaine 2007). Tides then circularise the binary quickly; triple codes assume that the orbit is circular at the onset (Toonen et al. 2016). With `physics.mass_transfer.dynamic_onset` SWINGERsim ends the run at the contact – the eccentricity e_c at which $a(1-e_c)\,r_L = R$, with r_L the lobe in units of the separation – and continues with the circularised binary: at constant orbital angular momentum, $a_c = a(1-e_c^2)$ (option `angular_momentum`; Hut 1981), or at the periastron, $a_c = a(1-e_c)$ (option `periastron`, the donor stays at its lobe). The pair keeps its centre of mass, plane and phase, its stars rotate synchronously, and a new mass-transfer history starts there. In a triple driven to $e_c = 0.89$ the N-body and the secular backend find the contact within 1.7 % in time and the same circular orbit.

5.5 Semi-detached and contact binaries

Imposed period change. Algols and W UMa binaries change their periods by mass transfer and by losing angular momentum; their minima then lie on a parabola, $O-C = \frac{1}{2}\dot{P}PE^2$ (E the cycle number; Kwee 1958; Hilditch 2001). When the cause need not be modelled, `orbits.<pair>.Pdot` imposes a constant $\dot{P}$ (d yr^{-1}) on a two-star binary inside the N-body run, $P(t) = P + \dot{P}\,t$. The time axis is cut into intervals of length Δt, and where the integration crosses a boundary (in either direction) the relative orbit is scaled to the period of the new interval, $a \to a(P_{\rm new}/P_{\rm old})^{2/3}$, at fixed eccentricity vector and phase and with the centre of mass of the pair unchanged. The eclipse times then deviate from the parabola by at most $(\dot{P}/P)\Delta t^2/8$, kept below 0.05 s. `Pdot_mode: conservative` changes the masses with the period as in conservative transfer, $\dot{P}/P = 3\dot{M}_1(M_1-M_2)/(M_1M_2)$. All observables and the third body see the same drifting orbit, so the parabola and the light-time effect can be fitted together (Fitting). The quadratic coefficient of the computed minima equals $\frac{1}{2}P\dot{P}$ to 5×10^{-5}. This option exists only in the N-body backend.

Contact binaries. When both stars fill their lobes, `physics.mass_transfer.contact` lets them share a common envelope (Lucy 1968) instead of ending the history: the less massive star stays at its lobe and feeds the primary through the envelope (conservatively), while the magnetised wind of the envelope removes angular momentum,

$$\dot{J} = -4.9\times10^{41}\,\frac{R_1^2M_1 + R_2^2M_2}{P}\ \mathrm{g\,cm^2\,s^{-1}\,yr^{-1}}$$

(R, M in solar units, P in days; Gazeas & Stępień 2008, eq. 19). The orbit shrinks and the mass ratio falls; the binary merges when it reaches the Darwin instability ($q < q_{\rm min}$, default 0.09; Rasio 1995) or overfills its outer critical surface through L2 (events `contact_merger_darwin`, `contact_merger_l2`). Thermal relaxation oscillations and energy transfer between the components beyond one common temperature are not modelled.

5.6 Tertiary mass transfer

A star that orbits a two-star binary (for example the giant tertiary of HD 181068) and fills its Roche lobe transfers mass onto that binary (`physics.mass_transfer.tertiary`, on by default, secular backend). SWINGERsim computes the history of the three stars before the dynamics, like that of a two-star binary: the donor sees the inner binary as a point mass, so its lobe is

that of a binary with $q = M_3/(M_1+M_2)$ at the periastron of the outer orbit (Eggleton 1983), and the rate and its limits follow BSE (Hurley et al. 2002; Hamers et al. 2022). The stream forms a circumbinary disc when the inner separation is below the circularisation distance $a_c = a_{\rm peri}(1-R_L)^2$, where R_L is the Roche-lobe radius of the donor in units of the outer separation (Eggleton 1983). Leigh et al. (2020, eq. 5) write $a_c = a_{\rm out}(1-R_L)^2$ with the outer semi-major axis; SWINGERsim evaluates it at the periastron $a_{\rm peri}$ of the outer orbit, where the donor overflows. With a disc, 90 % of the mass is accreted (two thirds by the more massive inner star), otherwise 20 % (Hamers et al. 2022). The rest leaves the inner binary isotropically; for the outer orbit it is re-emitted by the accretor (Soberman et al. 1997), and its ejection costs inner orbital energy (de Vries et al. 2014):

$$\mathrm{d}\ln a_{\rm in} = \frac{\mathrm{d}M_1}{M_1} + \frac{\mathrm{d}M_2}{M_2} - \frac{2M_{\rm in}\,\mathrm{d}M_{\rm ej}}{\alpha\lambda\, M_1 M_2},$$

with the inner masses M_1, M_2, their sum $M_{\rm in}$, the ejected mass $\mathrm{d}M_{\rm ej}$ and the efficiency $\alpha\lambda$ = 5 (de Vries et al. found 3–7 from hydrodynamics); `inner_orbit: angular_momentum` instead lets the accreted gas bring the specific angular momentum $h\sqrt{GM_{\rm in}a_{\rm in}}$. The history stops at a triple common envelope, when an inner star fills its own lobe, when the triple becomes dynamically unstable (Mardling & Aarseth 2001), when the donor is stripped to its core or at the end of its track. Not modelled: the drag of the stream on the inner orbit beyond the energy formalism, the change of the outer eccentricity during the transfer, and episodes at every periastron of an eccentric outer orbit (the rate is orbit-averaged). Outside the secular backend, or without this option, the overflow of a tertiary ends the run; ensembles record which star overflowed first and whether its companion is a star or a binary.

5.7 Common envelope

When the transfer becomes dynamically unstable, the donor's envelope engulfs both stars. With `physics.mass_transfer.common_envelope` SWINGERsim computes the outcome with the energy formalism of Hurley, Tout & Pols (2002, Sect. 2.7.1; after Webbink 1984 and Tout et al. 1997) instead of stopping:

- binding energy of the envelopes (eq. 69), $E_{\rm bind} = -\frac{G}{\lambda}\left(\frac{M_1 M_{\rm env,1}}{R_1} + \frac{M_2 M'_{\rm env,2}}{R_2}\right)$, with the structure parameter λ (`lam`, default 0.5) and a main-sequence companion counted as all core ($M'_{c,2} = M_2$, $M'_{\rm env,2} = 0$);
- the cores spiral in and give orbital energy to the envelope with the efficiency $\alpha_{\rm CE}$ (`alpha`, default 1; eqs. 70–72), $E_{\rm bind} = \alpha_{\rm CE}\left(-\frac{GM_{c,1}M'_{c,2}}{2a_f} + \frac{GM_{c,1}M'_{c,2}}{2a_i}\right)$, which gives the final separation a_f from the initial a_i;
- if a core would fill its Roche lobe at a_f, the cores merge where this happens, and the merger keeps the part of the envelope the released energy could not unbind (eqs. 73–77). Core radii: the white-dwarf radius for degenerate helium cores and AGB cores, the zero-age helium-star radius otherwise (Hurley et al. 2000).

The envelope leaves isotropically within `duration_yr`. In the N-body backend the binary's centre of mass keeps its position and velocity while the stars take their new masses and a circular orbit of separation a_f; the outer orbits feel an impulsive mass loss and widen, become eccentric ($e' = f/(1-f)$ for a circular orbit losing the fraction f of its mass) or unbound ($f \geq 1/2$; Hadjidemetriou 1963; Veras et al. 2011). The secular backends stop at the end of the envelope, because the impulsive response depends on the orbital phase. After the envelope the core is a point mass of constant mass (remnants are not evolved); the companion continues on the track of its present mass. The outcome is validated on published BSE examples and on the analytic impulsive response (see *Testing, Part I*). A tertiary that fills its lobe in an eccentric outer orbit can also lead to a triple common envelope (Glanz & Perets 2021; Hamers et al. 2022), which ends the run.

5.8 Mergers

With `physics.mergers` a run continues after two stars merge – a collision in the N-body backend, or a common envelope or contact binary that ends in a merger – with one merged star and the remaining orbits (a quadruple becomes a triple). Collisions in bound few-body systems are slow (relative velocity about the escape velocity), so the stars merge rather than destroy each other. The merger ejects the fraction $\phi = 0.3\,q/(1+q)^2$ of the total mass, $q = M_2/M_1 \leq 1$ (head-on collisions of main-sequence stars, Glebbeek et al. 2013: 2–7.5 %; Lombardi et al. 2002: at most about 8 %); a common-envelope merger keeps the mass of the energy formalism. The product sits at the centre of mass of the two stars with their total momentum; the mass leaves at once, so the outer orbits feel an impulsive mass loss. It mixes fresh hydrogen into its core: its fractional main-sequence age is $f = r\,(M_1 f_1 + M_2 f_2)/(M_1 + M_2)$, with r from 0.81 ($M \leq 6$ $M_\odot$) to 0.64 ($M \geq 60$ $M_\odot$) (Glebbeek & Pols 2008; Schneider et al. 2016) or 0.1 of BSE (Hurley et al. 2002). Born near break-up, the product starts at a fraction `spin_fraction` (default 0.3) of the break-up rate and spins down quickly (Schneider et al. 2019).

5.9 Magnetic braking and activity

Magnetic braking. With `physics.magnetic_braking` (requires `evolution` and `rotation`) the spins of stars with convective envelopes slow down by the torque of Matt et al. (2015),

$$T = -T_0 \left(\frac{\tau_{\rm cz}}{\tau_{\rm cz,\odot}}\right)^p \left(\frac{\Omega}{\Omega_\odot}\right)^{p+1} \text{ (unsaturated)}, \qquad T = -T_0\,\chi^p\,\frac{\Omega}{\Omega_\odot} \text{ (saturated)},$$

with $T_0 = 9.5 \times 10^{30}\,\mathrm{erg}\,(R/\mathrm{R}_\odot)^{3.1}(M/\mathrm{M}_\odot)^{0.5}$, $\chi = 10$, $p = 2$, $\Omega_\odot = 2.6 \times 10^{-6}$ s^{-1}, saturation below the Rossby number $\mathrm{Ro}_\odot/\chi$ ($\mathrm{Ro} = 1/(\Omega\tau_{\rm cz})$), and the convective turnover time of Cranmer & Saar (2011), $\tau_{\rm cz} = 314.24\,\exp[-T_{\rm eff}/1952.5\,\mathrm{K} - (T_{\rm eff}/6250\,\mathrm{K})^{18}] + 0.002$ d (12.9 d for the Sun, nearly zero above 6500 K). Within each segment the moment of inertia is constant and $d\Omega/dt = T/I$ is integrated exactly. With tides, a braked star in a close binary is spun up again by its companion and the binary loses orbital angular momentum: magnetic braking shrinks close binaries, a path to contact binaries in triples. The solar spin at the solar age and the Skumanich law $\Omega \propto t^{-1/2}$ (Skumanich 1972) are reproduced; for stars below 0.6 $\mathrm{M}_\odot$ the torque is known to be too weak (Douglas et al. 2017).

Applegate mechanism. A magnetically active star whose quadrupole moment changes through its activity cycle modulates the period of its binary (Applegate 1992). `swingersim applegate` compares the energy the modulation needs with the energy the star radiates in one modulation period (thin-shell model of Tian et al. 2009 as used by Völschow et al. 2016, $\Delta P/P = 2\pi\,(O-C)/P_{\rm mod}$ with the full O−C amplitude) and gives the largest O−C semi-amplitude that activity can produce. A timing variation larger than that cannot be activity; the two-zone models of Völschow et al. need one to two orders of magnitude more energy, so the thin-shell limit is generous. With `physics.activity` a cycle becomes part of the forward model: the eclipses of the binary are shifted by $K\sin(2\pi t/P_{\rm mod} + \varphi)$ in the synthetic timings and minima.

5.10 Limits

- Mass-transfer histories assume circular binaries; eccentric transfer is entered only through the dynamically started onset with instantaneous circularisation.
- Helium stars, white dwarfs, neutron stars and black holes are not evolved; a common-envelope core is a point mass of constant mass.
- Winds are isotropic and leave the system; wind accretion by a companion is not modelled.
- Rates and stability follow BSE recipes and carry their uncertainties (critical mass ratios, $\alpha_{\rm CE}\lambda$, accretion efficiency).

5.11 Implementation

External library and data. The N-body mass change uses the REBOUNDx operator `modify_mass` (Tamayo et al. 2020); MIST winds come with the MIST tracks (Choi et al. 2016).
Own code. Everything else was written for SWINGERsim from the cited papers: the operator splitting of stellar evolution and dynamics with adaptive segments and streaming; the SSE wind prescription integrated along the SSE tracks; the adiabatic response and the Ψ check of the secular backends; the BSE mass-transfer algorithm (rates, limits, critical mass ratios, accretion efficiency, the $d\ln a/dM_d$ equation, re-modelling of donors and accretors, rejuvenation); the ζ stability criterion; the dynamically started onset with circularisation; the imposed period change; contact binaries with angular-momentum loss and their merger criteria; tertiary mass transfer; the common-envelope energy formalism; mergers; magnetic braking; and the Applegate energy check and activity modulation.

Chapter 6

Stability and fates

A hierarchical multiple star is *dynamically stable* when its orbits stay well separated for a very long time; an unstable one breaks up, exchanges partners or ejects a star, usually within a number of outer orbits that is small compared with its age. Whether an observed configuration is stable is therefore a constraint on its unknown parameters: a solution that disrupts in a few thousand years cannot describe a system that exists today. SWINGERsim offers tools from cheap to definitive – algebraic criteria, machine-learning classifiers, a direct N-body test, chaos maps – and classifies the fates of the runs.

6.1 Algebraic criteria for nested triples

Each criterion applies to a *nested triple*: an inner orbit and the orbit around it, the rest of the system reduced to point masses at the centres of mass. A 2+2 quadruple has two (A with B as a point mass, B with A); a 3+1 also two (`inner` with m3, `mid` with m4); larger trees one per orbit pair. Both criteria are written as a ratio that exceeds 1 for stable systems.

Mardling & Aarseth (2001) (MA01), in the form of Vynatheya et al. (2022, eq. 2):

$$\frac{a_{\rm out}(1-e_{\rm out})}{a_{\rm in}} > 2.8\left[\frac{(1+q_{\rm out})(1+e_{\rm out})}{(1-e_{\rm out})^{1/2}}\right]^{2/5}\left(1-0.3\,\frac{i_{\rm mut}}{180^\circ}\right), \qquad q_{\rm out}=\frac{m_{\rm out}}{m_{\rm in}},$$

where $a_{\rm in}$, $a_{\rm out}$ are the semi-major axes of the inner and outer orbit, $e_{\rm out}$ the outer eccentricity, $m_{\rm in}$ the mass of the inner orbit, $m_{\rm out}$ that of the outer component and $i_{\rm mut}$ the mutual inclination. The left side is the outer periastron distance in units of the inner semi-major axis.

Vynatheya et al. (2022) (V22, their eqs. 3–4) add the eccentricity the inner orbit reaches in ZLK cycles and a better inclination dependence:

$$\widetilde{Y}_{\rm crit} = 2.4\left[\frac{1+q_{\rm out}}{(1+\tilde{e}_{\rm in})(1-e_{\rm out})^{1/2}}\right]^{2/5}\left[\frac{1-0.2\tilde{e}_{\rm in}+e_{\rm out}}{8}(\cos i_{\rm mut}-1)+1\right],$$

with $\widetilde{Y} = a_{\rm out}(1-e_{\rm out})/[a_{\rm in}(1+\tilde{e}_{\rm in})]$ (outer periastron over inner apastron), $\tilde{e}_{\rm in} = \max(e_{\rm in}, \frac{1}{2}e_{\rm max}^2)$ and $e_{\rm max}$ the quadrupole ZLK maximum (Hierarchies and orbits). The ratio $\widetilde{Y}/\widetilde{Y}_{\rm crit}$ is reported.

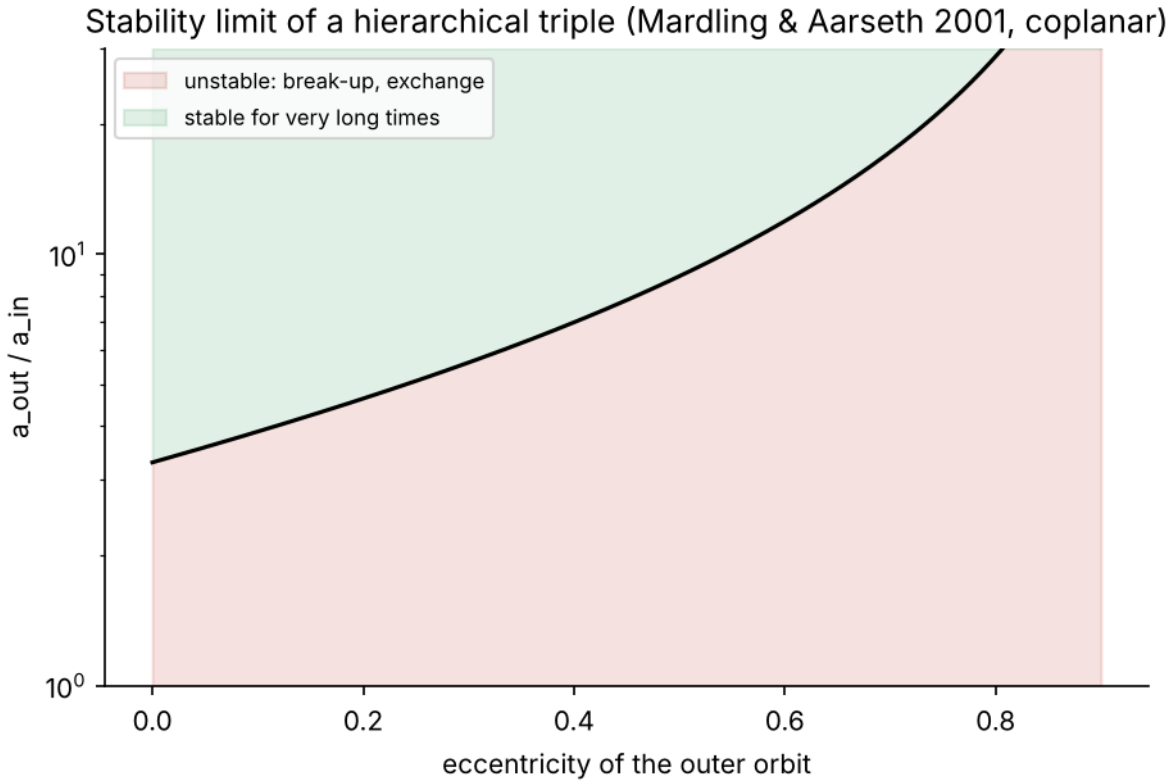


Figure 10. The Mardling & Aarseth (2001) stability limit of a coplanar triple. Horizontal axis: eccentricity of the outer orbit; vertical axis: ratio of the outer to the inner semi-major axis $a_{\rm out}/a_{\rm in}$ (logarithmic). The black curve is the limit for three equal masses ($q_{\rm out} = 0.5$); above it (green) the triple is stable for very long times, below it (red) it breaks up or exchanges stars. At $e_{\rm out} = 0$ the limit is $a_{\rm out}/a_{\rm in} \approx 3.3$; it rises steeply with the outer eccentricity because the outer periastron, not the semi-major axis, decides.

Figure 10 shows the content of the criterion: what matters is how close the outer component comes at periastron, so an eccentric outer orbit must be several times wider than a circular one. For inclined orbits the factor $(1 - 0.3\, i_{\rm mut}/180^\circ)$ lowers the limit slightly (retrograde orbits are more stable), while the V22 form raises it near the ZLK window, where the inner eccentricity grows.

Limits. Both criteria are fits to triple simulations. For quadruples, reducing a binary to a point mass is not always reliable, especially in 3+1 systems, where the fourth star excites the eccentricity of the middle orbit (Vynatheya et al. 2023). The run summary prints both ratios for every orbit pair as a first check.

6.2 Machine-learning classifiers

Vynatheya et al. (2023) trained multi-layer-perceptron classifiers on 5×10^5 N-body integrations each of 2+2 and 3+1 quadruples; they classify 94 % and 93 % of the systems correctly, against 88 % and 66 % for their triple MLP (Vynatheya et al. 2022) applied to the nested triples and 83 % and 56 % for the criterion of Mardling & Aarseth (2001) (Vynatheya et al. 2023, Tables 3–4). SWINGERsim uses the networks published by the authors when their repository is installed. Hierarchies of more than four stars are classified sub-quadruple by sub-quadruple: every orbit with two orbits as components gives a 2+2, every orbit inside an orbit inside an orbit a 3+1, deeper sub-systems being point masses at their centres of mass (a 2+2+2 gives four sub-quadruples). The system counts as stable if all are; outside the training set this is an approximation. SWINGERsim can also train its own classifiers on ensembles (Ensembles and inference).

6.3 The N-body ghost test

The definitive answer for one configuration is a direct integration with the definition of stability of Vynatheya et al. (2023, Sect. 2). The system and a *ghost* copy, in which the semi-major axis of the lighter inner binary (2+2) or of the innermost binary (3+1) is larger by 10^{-6} au, are integrated for 100 outer orbits as Newtonian point masses. The system is unstable if an orbit becomes unbound (for at least three consecutive checks) or if the relative difference of the two semi-major axes exceeds 10^{-2} – the divergence of neighbouring trajectories that signals chaos. The test takes seconds to a minute.

Limits. All three tools refer to dynamical stability over about 100 outer orbits. A system that passes can still evolve secularly (ZLK cycles, chaotic diffusion) on longer time scales; MEGNO and long runs check that. The ghost criterion is sensitive to the short-period oscillations of the osculating semi-major axis: for moderately hierarchical systems the two copies, which drift in phase, differ by about 1 % even when both stay bound for 1000 outer orbits. “Diverged” verdicts with a difference only slightly above 10^{-2} are therefore marginal.

6.4 Chaos maps

`swingersim chaosmap` integrates a grid of configurations in which one or two parameters vary (for example `orbits.out.a` and `orbits.out.e`) and records MEGNO (Dynamics) at the end of each run. A point counts as chaotic when MEGNO ≥ 4; a point where a star escapes (beyond 20 times the largest initial semi-major axis) or two stars collide is marked separately. The runs are Newtonian point masses because REBOUND’s variational equations cover only Newtonian gravity.

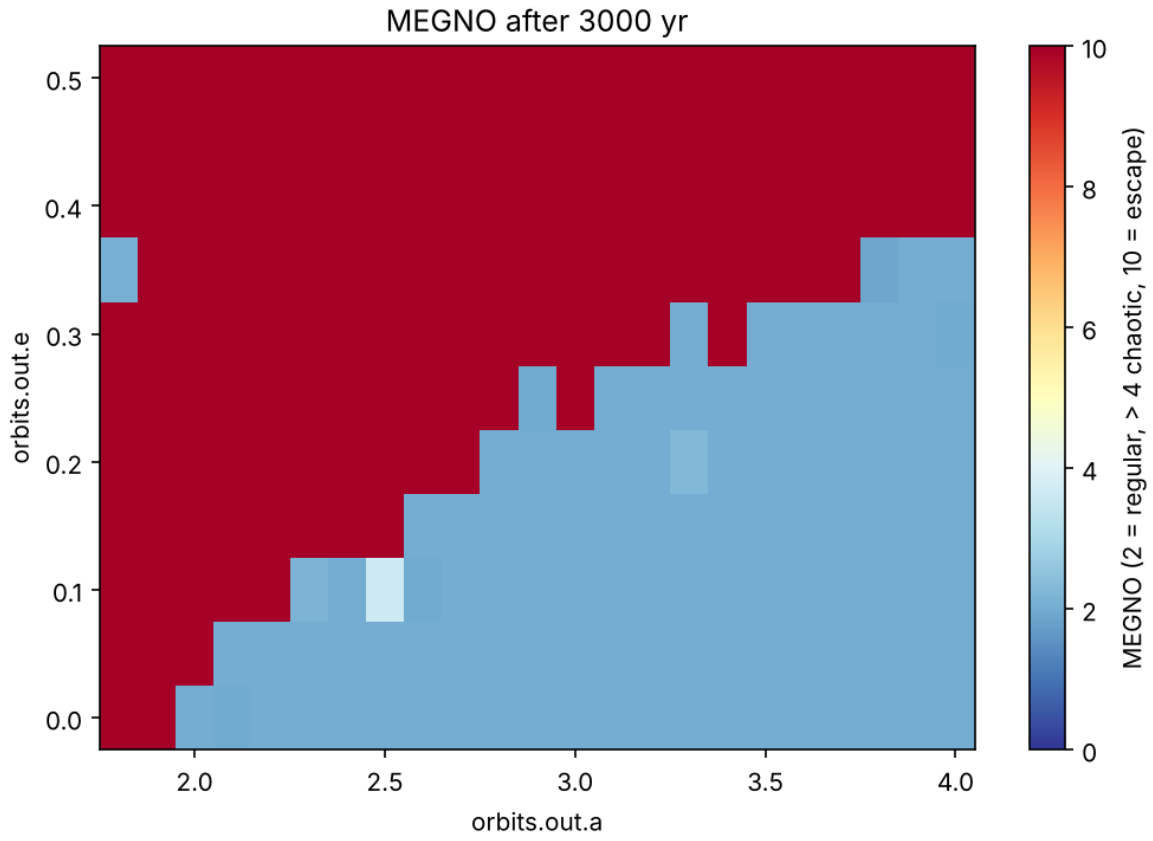


Figure 11. MEGNO chaos map of a test particle around an equal-mass circular binary of separation 1 au, after 3000 yr. Horizontal axis: semi-major axis of the particle's orbit (`orbits.out.a`, au, i.e. in units of the binary separation); vertical axis: its eccentricity (`orbits.out.e`). Colour: MEGNO, from blue (2, regular) through white (4) to dark red (10, used for escapes). The blue region at large distances and small eccentricities is regular; the red region close to the binary and at high eccentricity is chaotic or unstable; the boundary rises with eccentricity, as the stability criteria predict.

Figure 11 reads like Figure 10 with the axes exchanged: stable orbits lie far from the binary and on nearly circular orbits, and the boundary moves outwards as the eccentricity grows. The isolated red cells inside the blue region and blue cells inside the red one are chaotic islands and resonances, which a single criterion cannot represent. With the initial phase (`orbits.<orbit>.M0`) as the second parameter, the smallest distance beyond which every phase is regular is a stability limit: for a test particle around an equal-mass circular binary SWINGERsim finds 2.30 binary separations, against 2.27 in the published grid of Quarles et al. (2018) (see *Testing, Part I*).

6.5 Fates and their classification

Every run, and every member of an ensemble, ends with a *fate*. Table 9 lists them.

Table 9. Fates of a run. *Fate* is the value stored in the outputs (column `fate` of ensembles); *Meaning* what happened; *Detected by* the mechanism that recognises it.

Fate	Meaning	Detected by
`intact`	the hierarchy survived the whole run	no event
`exchange`	a star swapped partner	hierarchy detection

Fate	Meaning	Detected by
`reconfiguration`	the tree changed shape without losing a star	hierarchy detection
`ejection`	a star left the system	hierarchy detection
`breakup`	the system split into unbound parts	hierarchy detection
`capture`	a new bound structure formed	hierarchy detection
`hierarchy_change`	another confirmed change of the tree	hierarchy detection
`collision`	two stellar surfaces touched	REBOUND collision search
`merger`	two stars merged (common envelope, contact binary)	mass-transfer history
`roche_overflow`	a star filled its Roche lobe and transfer was not followed	Roche filling factor
`e_reached`	a requested eccentricity was reached	`run.stop_when_e_reached`
`energy_error`, `wall_time`, `gw_unresolved`, `error`	the run stopped for a numerical or computing reason	run control

A change of the hierarchy is confirmed only after it has persisted for `run.hierarchy_persistence` outputs (Dynamics). The last row is not physics: such members should be counted separately and, if frequent, the settings changed.

6.6 Survival in both directions of time

A system observed today has existed since its formation, so a configuration that disrupts in the recent past cannot describe it. Because the dynamics without dissipation is time-reversible, the same holds for the near future: an unstable configuration is unstable in both directions. In a 10 000-member N-body ensemble of CzeV4315 integrated ±60 yr, 6.0 % of the members disrupt forward and as many backward, and two thirds of those that fail in one direction fail within the same 60 years in the other; the rest only because the window is short. A one-sided survival over a short run is therefore no evidence of stability. Survival is imposed in ensembles as a condition on both runs, `backward:fate: {equals: intact}` and `forward:fate: {equals: intact}` (Ensembles and inference).

A run of decades only catches the fastest instabilities. Near the stability boundary the time to disruption grows roughly exponentially with $(r_{\mathrm{p,out}}/a_{\mathrm{in}})^{3/2}$ ($r_{\mathrm{p,out}}$ the outer periastron distance) and scatters by one to two orders of magnitude between neighbouring initial phases, a signature of chaos (Hayashi et al. 2022, 2023; Mushkin & Katz 2020); for strongly inclined orbits (60°–150°) ZLK oscillations shorten it further, and retrograde coplanar systems are the most stable. A configuration consistent with an observed system must survive its age – usually millions of outer orbits – so the criteria and MEGNO are the complementary tests for members close to the boundary.

What changes the stability. Without dissipation and with constant masses the stability of a hierarchy does not change by itself; slow processes move it across the boundary (Table 10).

Table 10. Processes that move a hierarchy across the stability boundary. *Destabilising* and *Stabilising* list the processes with the literature; the rows are not paired.

Destabilising	Stabilising
mass loss of the inner binary (winds, supernova): the hierarchy weakens (Perets & Kratter 2012; Toonen et al. 2022)	tidal shrinking of an inner binary, also through ZLK cycles with tidal friction (Fabrycky & Tremaine 2007)
expansion of a star: Roche-lobe overflow, mass transfer, common envelope	precession by tides, rotation and relativity that quenches ZLK cycles
ZLK cycles driving a high eccentricity	dissipation in gas during formation (damped eccentricities, aligned orbits)
encounters and the Galactic tide on wide orbits (Kaib et al. 2013; Grishin & Perets 2022)	accretion and migration in the formation disc (Tokovinin & Moe 2020)
sweeping through resonances when periods change	retrograde coplanar configurations (Hayashi et al. 2022)

All processes of the table except supernovae are modelled in SWINGERsim; gas damping is parametric.

6.7 Scattering experiments

`swingersim scatter` studies dynamical formation: a binary (or a single star) meets another binary (or a single star). Each encounter is set up as in Hut & Bahcall (1983) and FEWBODY (Fregeau et al. 2004): isotropic orientations and random phases of the binaries, fixed or thermal eccentricities ($f(e) = 2e$), the centres of mass on a hyperbola with velocity at infinity v_∞ and impact parameter b (b^2 uniform up to $b_{\max}$; with a maximum periastron q, $b_{\max}^2 = q^2 + 2GMq/v_\infty^2$ from gravitational focusing). The critical velocity v_c is the one at which the total energy is zero, $\frac{1}{2}\mu_\infty v_c^2 = \sum Gm_1m_2/(2a)$ (μ_∞ the reduced mass of the two incoming objects, the sum over the binaries). The integration (IAS15, point masses, optional 1PN, collisions when radii are given) starts at a multiple of the largest semi-major axis and ends when the top-level objects found by the hierarchy detection are pairwise unbound, receding and farther apart than the start distance, with every bound sub-system of three or more stars stable by the MA01 criterion. A system that stays bound counts as captured once it has been stable for 10 of its outer orbits; otherwise the run stops at its time or computing limit as *unresolved*.

The outcomes follow Antognini & Thompson (2016): *flyby*, *exchange*, *single ionization*, *exchange + ionization*, *triple + single*, *full ionization*, *bound 2+2* or *bound 3+1* (a captured quadruple) and *collision*. Cross-sections are $\sigma = \pi b_{\max}^2 \times$ the fraction of the outcome. With tidal capture, every close periastron passage of two stars removes the energy of the tides it raises (Press & Teukolsky 1977), $E = \sum GM_\mathrm{o}^2/R\,[(R/R_p)^6 T_2(\eta) + (R/R_p)^8 T_3(\eta)]$ with $\eta = (M/(M+M_\mathrm{o}))^{1/2}(R_p/R)^{3/2}$, where M, R are the mass and radius of the perturbed star, M_o the mass of the other, R_p the periastron distance and T_2, T_3 the polytrope fits of Portegies Zwart & Meinen (1993). The relative velocity is reduced at periastron, so unbound stars can be captured into binaries: two 0.6 $\mathrm{M}_\odot$ stars ($n = 3$, 0.65 $\mathrm{R}_\odot$) meeting at 10 km s^{-1} are captured within 2.37 stellar radii, $n = 1.5$ stars within 3.33 radii, close to the about 3.4 of McMillan et al. (1987). Gas drag during the encounter is not modelled.

6.8 Statistical escape theory

When three stars interact resonantly (no hierarchy, a “democratic” phase) the outcome is practically random, and Stone & Leigh (2019) showed that its distribution is the phase-space volume allowed by the conserved energy and angular momentum (their eq. 2, with the strong-interaction radius $R = 2a_B(1+e_B)$, a_B and e_B the elements of the final binary). `swingersim escape-theory` evaluates this integral numerically for each of the three possible escapers and

gives the escape probabilities and the distributions of the binary's energy, eccentricity and orientation, in a second instead of thousands of N-body runs. It reproduces the limits of the paper (energy distribution $\propto |E_B|^{-7/2}$ for zero angular momentum and $|E_B|^{-4}$ near the maximum; their eccentricity distributions) and agrees with SWINGERsim's own scatterings: for resonant encounters of a 1 + 0.6 $M_\odot$ binary with a 0.8 $M_\odot$ star the lightest star escapes in 75 % of the N-body runs and in 77 % in the theory. The theory covers only resonant three-body encounters; prompt exchanges and fly-bys lie outside it.

6.9 Implementation

External code and data. The MLP classifiers are the trained networks of Vynatheya et al. (2023), used as published; the N-body integrations of the ghost test, chaos maps and scattering use REBOUND/IAS15 and its MEGNO (Rein & Liu 2012; Rein & Spiegel 2015). **Own code.** The MA01 and V22 criteria for every nested triple of any tree; the decomposition of large hierarchies into sub-quadruples and the computation of the classifiers' input parameters; the ghost test; the chaos maps and their classification; the fate classification; the scattering set-up, termination and outcome classification; the tidal-capture energy loss with the Portegies Zwart & Meinen (1993) fits; and the numerical evaluation of the Stone & Leigh (2019) phase-space integral.

Chapter 7

Observables

Nobody watches a multiple star for a million years. Astronomers have snapshots: the brightness of the system over nights to years, the times at which its eclipses happen, velocities from spectra. This chapter describes how SWINGERsim turns the orbits and stars of the previous chapters into these measurements: eclipses and their detectability, light curves, radial velocities, times of minima and O−C diagrams, pulsating stars as clocks, and dust discs as occulters. All observables are computed from the N-body backend, because they need the position of every star on its orbit.

7.1 Eclipse geometry

A binary eclipses when, seen along the line of sight, the disk of one star passes over the disk of the other. For spherical stars of radii R_1, R_2 on a circular orbit of separation a the condition is

$$|\cos i| < \frac{R_1 + R_2}{a},$$

where i is the inclination to the sky (Hierarchies and orbits). For an eccentric orbit a is replaced by the separation at conjunction. For an isotropic orientation the probability that a binary eclipses is $(R_1 + R_2)/a$; for the two binaries of V994 Her this gives 37 % and 40 %, as found by Vokrouhlický (2016) (see *Testing, Part I*).

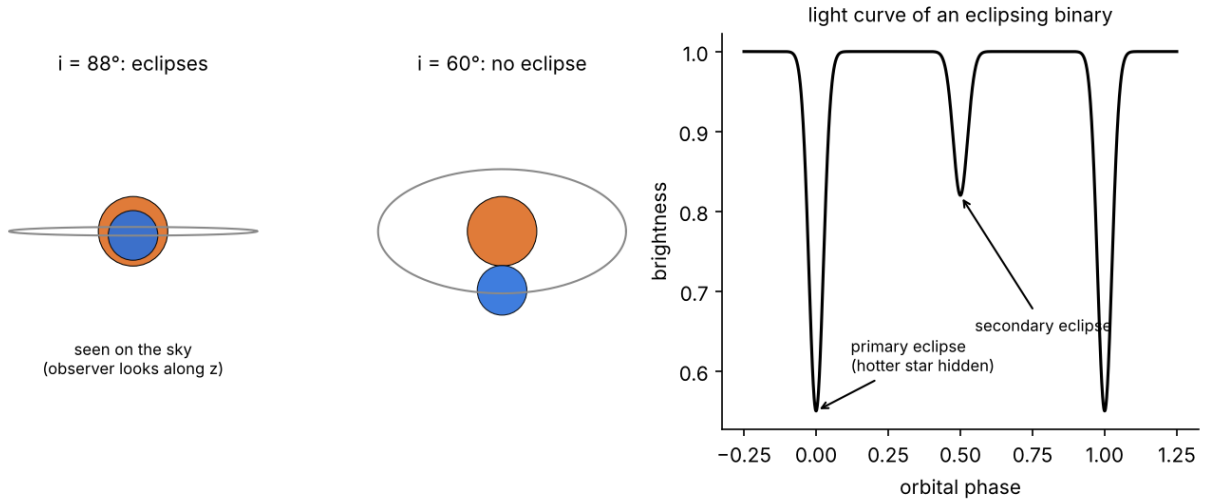


Figure 12. An eclipsing binary. Left and middle: the binary seen on the sky (the observer looks along z); the grey ellipse is the projected orbit of the blue star around the orange one. At $i = 88°$ (left) the projected orbit is nearly a line and the disks overlap at conjunction; at $i = 60°$ (middle) they pass each other without overlapping. Right: the light curve of the eclipsing case, brightness (normalised to 1 outside eclipse) against orbital phase. The deeper minimum at phase 0 is the primary eclipse, when the hotter star is hidden; the shallower one at phase 0.5 the secondary eclipse.

Figure 12 shows the two cases. The depth of each minimum is the fraction of the total light that is hidden, which depends on the overlap of the disks and on the surface brightness of the hidden star: hiding the hotter star costs more light, so the primary minimum is deeper.

During a run SWINGERsim reports for every binary of two stars the fraction of the time in which the condition above holds (spherical stars, circular-orbit form, as in Vokrouhlický 2016, Sect. 3.3), with the radii changing as the stars evolve; for 2+2 systems it also gives the fraction in which both binaries eclipse at once – the defining property of doubly eclipsing systems. Ensembles report whether a binary eclipses at $t = 0$ and when its eclipses start or end.

7.1.1 Geometric eclipses and detectable eclipses

The geometric condition says only that the disks overlap. Whether an observer *detects* the eclipse depends on its depth in the total light. In a multiple star the light of all other stars dilutes the eclipses of a binary. If a binary alone would lose the fraction d of its own light at minimum, and the other stars contribute the fraction l_3 of the total light (the *third light*), the observed depth in magnitudes is

$$\Delta m = -2.5 \log_{10} \left[1 - (1 - l_3)\, d\right].$$

A grazing eclipse (d small) in a bright multiple (l_3 close to 1) can therefore be far below the photometric precision even though the geometric condition holds.

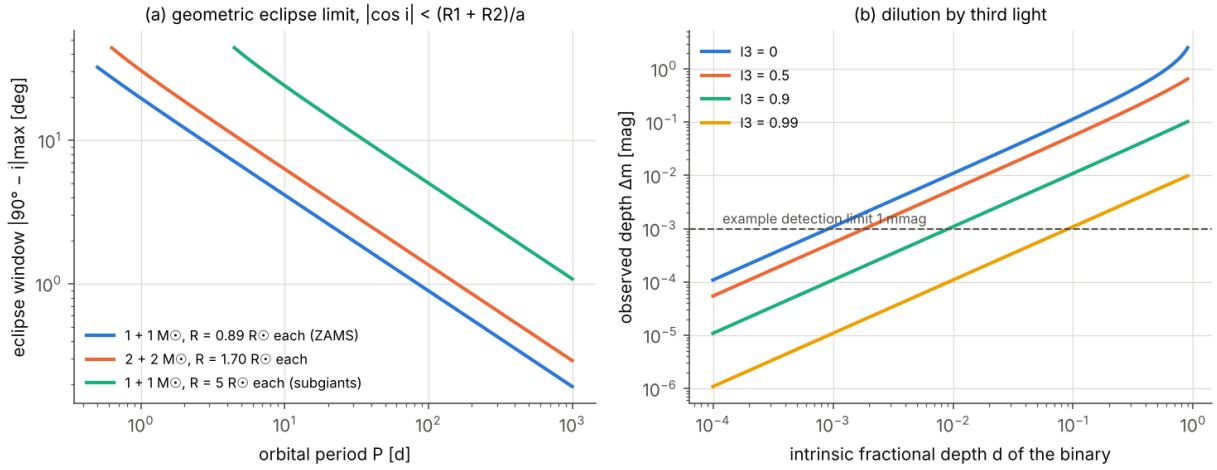


Figure 13. Eclipse limit and detectability (new figure for this volume, computed from the two equations above). (a) The largest tilt from edge-on, $|90^\circ - i|_{\max} = \arcsin[(R_1 + R_2)/a]$, for which a circular binary still eclipses, against its period P in days (both axes logarithmic). Blue: two 1 $M_\odot$ ZAMS stars of 0.89 $R_\odot$; orange: two 2 $M_\odot$ stars of 1.70 $R_\odot$; green: two 1 $M_\odot$ subgiants of 5 $R_\odot$ (shown only where the stars are well detached). (b) Observed depth Δm in magnitudes against the intrinsic fractional depth d of the binary (both logarithmic), for third light l_3 = 0 (blue), 0.5 (orange), 0.9 (green) and 0.99 (yellow). The dashed line marks an example detection limit of 1 mmag.

Figure 13 separates the two questions. Panel (a): a pair of Sun-like stars with a period of 1 d eclipses within about 20° of edge-on, at 10 d within about 4°, at 100 d within less than 1°; larger stars widen the window in proportion to $R_1 + R_2$. This is why long-period binaries are rarely eclipsing and why a precessing binary gains and loses its eclipses (Hierarchies and orbits). Panel (b): for small d the observed depth is $\Delta m \approx 1.086\,(1 - l_3)\,d$, so each line is the blue one shifted down by the factor $1 - l_3$. An intrinsic depth of 1 % appears as 11 mmag alone, 5 mmag with l_3 = 0.5, 1.1 mmag with l_3 = 0.9 and 0.11 mmag with l_3 = 0.99 – the last one below the example detection limit. When SWINGERsim is used to predict when eclipses become visible or disappear, the synthetic light curve (below), not the geometric condition, decides.

7.2 Light curves

`swingersim lightcurve` computes the light an observer records, from the N-body positions of all stars. Three levels of fidelity are available (Table 11).

Table 11. Light-curve models. *Model* names the level; *Stars* the shapes; *Physics* what is included; *Use* where it is appropriate. All levels share the positions, the light-travel time and the normalisation (1 = all stars outside eclipse).

Model	Stars	Physics	Use
spheres (default)	limb-darkened spheres	exact occultations of any number of stars, analytic ellipsoidal, reflection and beaming terms	detached systems, R/R_L well below 1
meshes	Roche surfaces, rotating stars, contact binaries; all stars meshes	gravity and limb darkening per surface element, occultation along every ray	close, semi-detached and contact binaries, fast rotators
PHOEBE hybrid	PHOEBE 2 for each inner binary	Roche geometry, model atmospheres, irradiation of PHOEBE; mutual eclipses of different binaries as spheres	selected solutions, cross-checks, training sets

7.2.1 Spheres: intensity, limb darkening and passbands

The surface brightness of a star in a passband is that of a black body integrated over the passband with photon-counting weights, $w = \int B_\lambda(T_{\rm eff})\, S(\lambda)\, \lambda\, d\lambda$, where B_λ is the Planck function and $S(\lambda)$ the passband response. Available passbands are Johnson–Cousins U, B, V, R, I (Bessell 1990), TESS (Ricker et al. 2015) and Gaia G, BP, RP (Riello et al. 2021). Model atmospheres change the surface-brightness ratio of stars of different temperature by a few per cent; the PHOEBE hybrid uses model atmospheres where this matters.

The intensity across the disk follows the four-coefficient limb-darkening law of Claret (2000),

$$\frac{I(\mu)}{I(1)} = 1 - \sum_{k=1}^{4} a_k \left(1 - \mu^{k/2}\right),$$

where μ is the cosine of the angle between the line of sight and the surface normal and a_k are the coefficients; linear and quadratic laws are special cases, and coefficients can be given per star. Below 15 000 K the coefficients and the gravity-darkening exponents come from Claret & Bloemen (2011), interpolated in $T_{\rm eff}$, $\log g$ and [M/H]; for TESS and Gaia the nearest Johnson–Cousins band of those tables is used (TESS and RP: I; G: R; BP: V). From 15 000 to 55 000 K the coefficients come from the non-LTE TLUSTY atmospheres (OSTAR2002, BSTAR2006) of Reeve & Howarth (2016); where both grids apply, the disk-averaged intensities agree to 0.2–0.7 %. Beyond the gravity-darkening grid of Claret & Bloemen (O stars above about 39 000 K) the exponent follows von Zeipel's law in the band, $y = \frac{1}{4}\, x e^x/(e^x - 1)$, $x = hc/(\lambda k T_{\rm eff})$. Stars outside the tables (for example cool AGB stars) take the nearest tabulated node, with a warning.

Occultations. Every star is occulted by all stars nearer to the observer whose disks overlap it. The hidden flux is $\int_0^R I(r)\, \alpha(r)\, 2\pi r\, dr$, where $\alpha(r)$ is the fraction of the circle of radius r around the centre of the star covered by the union of the occulting disks; the radial integral is computed with Gauss–Legendre quadrature between the radii at which a disk edge crosses (as in ellc, Maxted 2016). This is exact for any number of spheres, so mutual eclipses of the two binaries of a 2+2 system and triple overlaps need no special treatment. It agrees with the analytic model of Mandel & Agol (2002), as implemented in batman (Kreidberg 2015), to 10^{-8}.

Proximity effects of the stars of two-star binaries, in the small-distortion approximation: the ellipsoidal variation $-\alpha_{\rm ell}\, q\, (R/d)^3 (2\cos^2\psi - \sin^2 i)$ with $\alpha_{\rm ell} = 0.15(15+u)(1+\tau)/(3-u)$ (Morris 1985; Morris & Naftilan 1993), where q is the mass ratio, d the separation, ψ the angle between the line of centres and the line of sight, u the linear limb-darkening and τ the gravity-darkening coefficient; Doppler beaming $-(3-\alpha_\nu)\, v_z/c$ with the spectral index α_ν (Loeb & Gaudi 2003); and reflection of the companion's light by a Lambert sphere with Bond albedo 0.5 for convective and 1 for radiative envelopes (Rucinski 1969). The ellipsoidal term underestimates the variation by 3, 7, 11 and 17 % at R/a = 0.10, 0.15, 0.20 and 0.25 compared with the Roche model of ellc; closer pairs need meshes.

Exposure and noise. Finite exposures are modelled by averaging the model over sub-exposures (exposure time and number of sub-points per exposure); Gaussian white noise can be added.

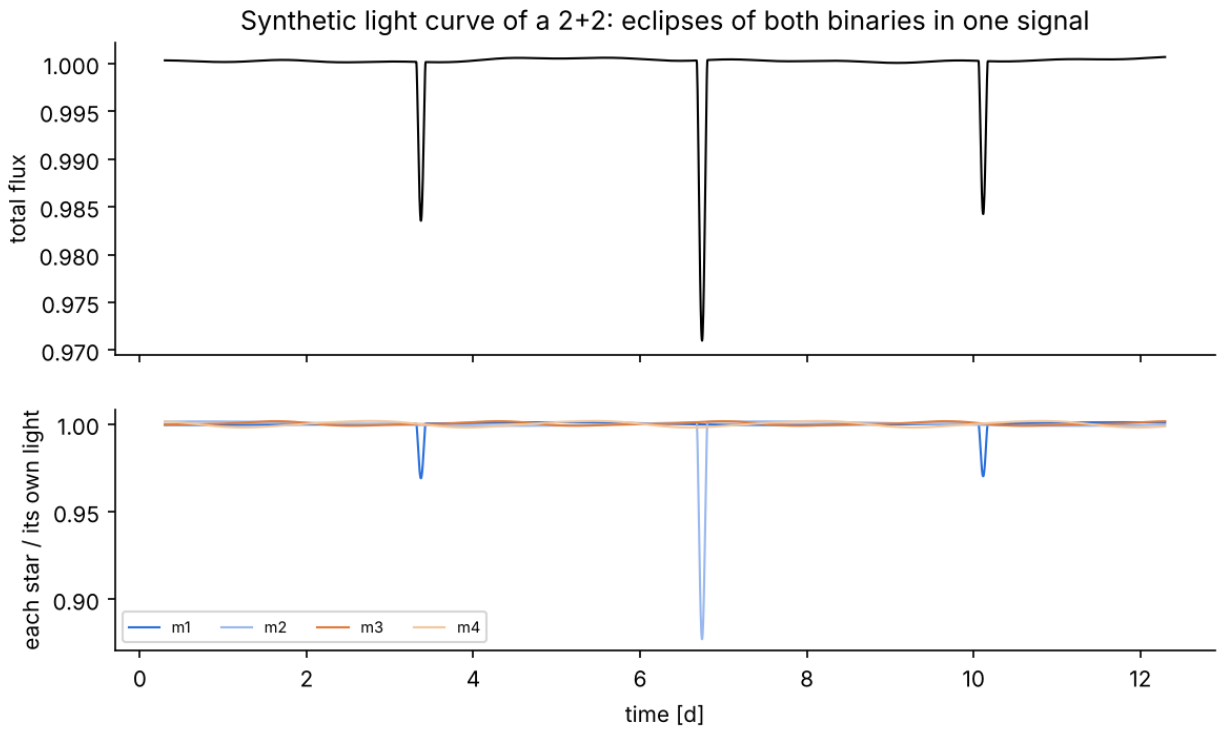


Figure 14. Synthetic light curve of a 2+2 system computed with spheres. Top: total flux (normalised to 1 outside eclipses) against time in days. Bottom: the flux of each star divided by its own unocculted flux, for m1 (dark blue), m2 (light blue), m3 (orange) and m4 (light orange). The eclipses of binary A (m1 and m2 hidden in turn) appear in the total light diluted by the light of B; the small slow wiggles are the ellipsoidal and reflection variations.

Figure 14 shows how the signal of a multiple star is built. The bottom panel separates the stars: m2 loses about 12 % of its own light at its eclipse, but in the total light (top) this becomes a 3 % dip, because the other three stars keep shining. This dilution is the third light of the previous section, and it is why a model of the whole system, not of one binary, is needed to interpret eclipse depths.

7.2.2 Light-travel time

Every star is seen at its own *emission time*: the light observed at time t left a star at depth z (positive away from the observer, barycentre at $z = 0$) at $t_e = t - z/c$. Every two-star binary is integrated separately to the emission time of its centre of mass, $t_e = t - z_{\rm com}(t_e)/c$ (solved iteratively), and only the offsets of its two stars from the centre are corrected to first order, $-\mathbf{v}\,\Delta z/c$ (the Rømer delay inside the binary); other stars use $\mathbf{x}(t) - \mathbf{v}\,z/c$. A first-order correction of the whole delay is not enough when z/c is a sizeable fraction of the binary period: for QZ Car ($z/c \approx 0.2$ d, P = 6 d) it would shift the eclipses by about 200 s.

7.2.3 Distorted stars

Stars can be given a Roche shape (`stars.<m>.shape: roche`, both stars of a circular synchronised binary), the oblate shape of a rigidly rotating star (`shape: rotating`), or an orbit can be a contact binary with fill-out factor f (`orbits.<X>.contact_fillout`). The binary model uses the Roche potential (Kopal 1959; Wilson & Devinney 1971)

$$\Omega(x,y,z) = \frac{1}{r_1} + q\left(\frac{1}{r_2} - x\right) + \frac{1+q}{2}\,(x^2+y^2),$$

in units of the separation, with star 1 at the origin, star 2 at $(1,0,0)$, r_1, r_2 the distances to the

two stars, $q = M_2/M_1$ and z along the orbit normal. The inner critical surface $\Omega_{\text{in}} = \Omega(\text{L1})$ bounds the Roche lobes, the outer one Ω_{out} (through L2 or L3) the common envelope; the fill-out factor is $f = (\Omega_{\text{in}} - \Omega)/(\Omega_{\text{in}} - \Omega_{\text{out}})$, negative for detached and between 0 and 1 for contact binaries (Lucy & Wilson 1979; Rucinski 1973). A rotating star follows the Roche model $1/r + (4/27)\, w^2 r^2 \sin^2\theta = 1$ in units of its polar radius, with $w = \Omega/\Omega_{\text{crit}}$ and θ the colatitude (Maeder 2009); its equator reaches 1.5 times the polar radius at $w = 1$. The configured radius is the volume-equivalent radius.

Each surface is sampled on 64 × 128 directions. An element emits $I = I_{\text{ref}}\,(g/\bar{g})^y\, L(\mu)$, with the local effective gravity g, its surface mean $\bar{g}$, the band gravity-darkening exponent y (von Zeipel 1924; Claret & Bloemen 2011) and the limb darkening $L(\mu)$; a contact binary's two lobes share one effective temperature. An element is hidden when its normal points away from the observer, when a sphere lies in front of it (anti-aliased by its projected size), or when its line of sight passes through another body or the other lobe of its own Roche body. When any star is a mesh, all stars are meshes. Accuracy: 3×10^{-4} of the flux against the exact sphere eclipses; the amplitude of a contact binary agrees with PHOEBE 2.5 to 0.004 mag (see *Testing, Part I*). The error of spheres against Roche surfaces grows steeply with the filling of the lobe: about 0.5 ppt rms at R/R_L = 0.3, 2 ppt at 0.5, 5 ppt at 0.7, 9 ppt at 0.85 and 17 ppt at 0.95, and 15–45 ppt for contact binaries. The Roche model assumes a circular synchronised orbit; eccentric distorted binaries need the PHOEBE hybrid.

7.2.4 The PHOEBE hybrid

With the PHOEBE engine the light of every inner binary of two stars comes from PHOEBE 2 (Prša et al. 2016; Horvat et al. 2018; Conroy et al. 2020): Roche geometry, gravity darkening, irradiation, Castelli–Kurucz model atmospheres or black bodies, and the light-travel time inside the binary. The multiple system comes from SWINGERsim:

1. in chunks of one binary period the N-body integration gives the osculating elements of the binary at the chunk centre and the motion of its centre of mass along the line of sight; period and time of periastron come from the mean longitude over one orbit, because the osculating Keplerian period of a tidally distorted binary is wrong by minutes per orbit;
2. PHOEBE computes the binary through the chunk at the emission times;
3. stars outside two-star binaries shine with their passband luminosity;
4. eclipses between stars of different binaries are taken from the sphere model and subtracted.

The conventions are converted (z away from the observer; PHOEBE's argument of periastron is that of the primary, `per0` = ω + 180°). With the same physics the hybrid reproduces the sphere model to a few 10^{-4}. PHOEBE needs 0.1–0.2 s per point and binary, about 10^4 times more than the sphere model, and is meant for selected solutions. Rotation is synchronous, Doppler beaming is absent in PHOEBE 2.5, and discs are ignored. The native treatment of multiple systems by Brož et al. (2025) is an alternative when all light must come from one code.

7.3 Radial velocities

The radial velocity of star k is its velocity along the line of sight relative to the barycentre plus the systemic velocity γ: $\text{RV}_k = \gamma + v_{z,k}$, positive away from the observer. A velocity measured at time t_{obs} belongs to the emission time t_e with $t_e + z_k(t_e)/c = t_{\text{obs}}$; SWINGERsim finds it by Newton steps until the mismatch is below 1 ms. The light time of the star (up to tens of minutes in a wide outer orbit) is therefore included, which matters for the steep velocity curves of close binaries (a semi-amplitude of 100 km s^{-1} in a 3-day orbit changes by 0.15 km s^{-1} per minute).

Centre of light and the Rossiter–McLaughlin effect. A spectrograph measures the Doppler shift of the light of a star – the flux-weighted mean line-of-sight velocity of its visible surface – not the velocity of its centre of mass. For spherical, slowly rotating stars the two agree. A tidally distorted star is brighter and larger on some sides and its synchronous

rotation adds surface velocity, so its centre of light moves on a different orbit (tens of km s^{-1} in contact binaries; Wilson & Sofia 1976); during an eclipse the covered part of a rotating star is missing from the mean, which shifts the measured velocity (the Rossiter–McLaughlin effect; Rossiter 1924; McLaughlin 1924). With `fit.rv.centre: light` the model velocity is the flux-weighted line-of-sight velocity of the visible surface elements of the meshes above – Roche stars corotate with their orbit, other stars rotate with their spin – in the passband `fit.rv.band`. The offset from the centre-of-mass velocity is added to the exact centre-of-mass velocity, so the light-travel time and the dynamics stay those of the N-body model. Irradiation and beaming are not included in the weights. For a contact binary (q = 0.42, fill-out 0.3) the offsets reach ±34 and ±52 km s^{-1} and agree with PHOEBE 2 to 2 % of their amplitude; a fit of such velocities with the centre of mass gives masses too large by several per cent.

7.4 Times of minima and O−C diagrams

If a binary were alone, its eclipses would follow a linear *ephemeris*, $T_n = T_0 + nP$, with the cycle number n. Observers compare each measured time of minimum (O) with the time computed from such an ephemeris (C) and plot the difference O−C against time. A straight line means a wrong period; anything else means that something moves the binary or changes its orbit.

Synthetic timings. `swingersim timings` integrates the system (N-body with the configured physics) and finds every eclipse of each inner binary: the minima of the projected separation of its two stars, sampled 40 times per binary period and refined with the two-body motion between samples. A conjunction is an eclipse when the projected separation is smaller than $R_1 + R_2$ (the ratio is reported as the impact parameter). A *primary* eclipse is an eclipse of the binary's first star (m1 of A, m3 of B), i.e. the first star is the farther one. Observed times include the light-travel time of the binary's centre of mass, $t_{\mathrm{obs}} = t + z_{\mathrm{com}}/c$; the dynamical delays come from the integration itself. A linear ephemeris fitted to the primary eclipses gives the O−C; secondary eclipses use half-integer epochs.

7.4.1 Light-travel-time effect

The binary orbits the barycentre of the whole system, so it is sometimes farther from the observer and sometimes closer. Its eclipses arrive late and early in turn: the *light-travel-time effect* (LTTE). For an outer orbit of period P_3, eccentricity e_3, argument of periastron ω_3 and true anomaly ν_3, the delay is (Irwin 1952)

$$\Delta_{\mathrm{LTTE}} = \frac{a_{12}\sin i_3}{c}\left[\frac{1-e_3^2}{1+e_3\cos\nu_3}\sin(\nu_3+\omega_3) + e_3\sin\omega_3\right],$$

where $a_{12}\sin i_3$ is the projected semi-major axis of the binary's orbit around the barycentre. Its semi-amplitude is $K = (a_{12}\sin i_3/c)\sqrt{1-e_3^2\cos^2\omega_3}$ and the mass function $f(m_3) = (a_{12}\sin i_3)^3/P_3^2$ (au, yr, $\mathrm{M}_\odot$). The N-body timings contain this term exactly, together with every other effect.

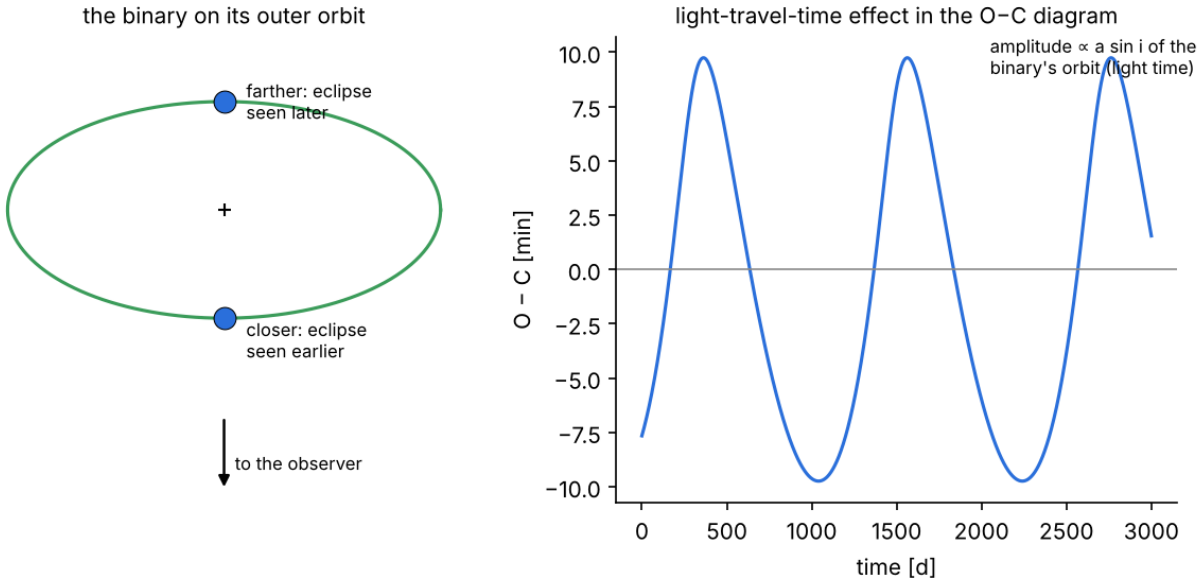


Figure 15. The light-travel-time effect. Left: a binary (blue dot) on its outer orbit (green ellipse) around the barycentre (cross); the arrow points to the observer. On the far side of the orbit its eclipses are seen later, on the near side earlier. Right: the resulting O−C in minutes against time in days, a periodic curve with the outer period and an amplitude equal to the light time across the binary's projected orbit (here about 10 min; illustrative).

Figure 15 links the geometry to the diagram. The O−C curve repeats with the outer period; its semi-amplitude is $a_{12} \sin i_3/c$ (about 8.3 min per au), and its shape departs from a sine for an eccentric outer orbit. In a 2+2 quadruple the two binaries orbit each other, so their LTTE curves are in anti-phase with amplitudes in the inverse ratio of their masses; in compact systems, with outer periods of a year or less, the dynamical perturbations of the period add a part that is not in anti-phase and can be larger than the LTTE.

7.4.2 Apsidal motion in the O−C diagram

In an eccentric binary the time between primary and secondary minimum depends on the orientation of the orbit. When the periastron turns (apsidal motion), the primary and secondary minima move in opposite directions in the O−C diagram, with the apsidal period U.

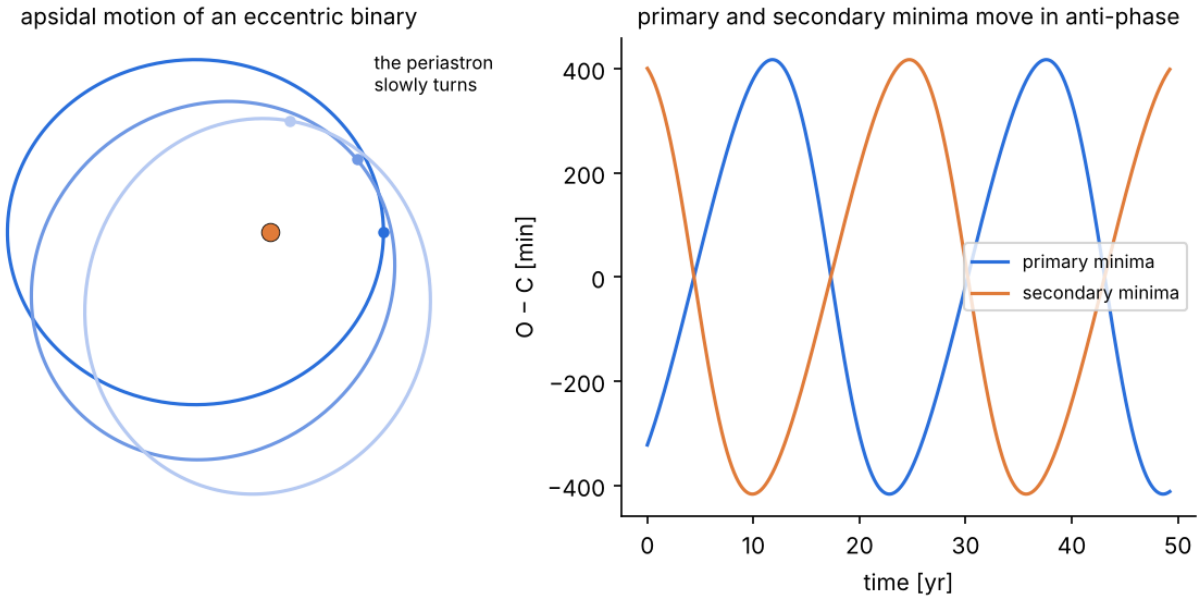


Figure 16. Apsidal motion (illustrative). Left: an eccentric orbit (blue ellipses, shades for successive times) around the orange star; the dots mark the periastron, which slowly turns. Right: O−C in minutes against time in years for the primary (blue) and secondary (orange) minima: two curves in anti-phase with the apsidal period, here about 25 yr.

Figure 16 shows the signature used to measure $\dot{\omega}$. The amplitude of the two curves grows with the eccentricity, and their anti-phase distinguishes apsidal motion from the LTTE, which shifts both minima together. The analytic form used in the stage-1 fit is given in chapter Fitting.

7.4.3 Phenomenological minima

Observers do not fit physics to an eclipse; they measure its time of minimum, depth and width with a phenomenological profile. `swingersim minima` reduces the synthetic light curve in the same way, with the profile of Mikulášek (2015, eq. 13),

$$\Delta m(t) = A\,(1 + Cx^2)\left\{1 - \left[1 - e^{1-\cosh x}\right]^{\Gamma}\right\}, \qquad x = \frac{t - t_{\min}}{D},$$

with central depth A (mag), width D, kurtosis Γ and correction C, on a quadratic baseline (the light of the other stars, the ellipsoidal and reflection variation, which is symmetric about the minimum to second order, and a slow drift). The eclipses are located with the synthetic timings; the window around each is 2.5 times its duration (at most half the period). A *shape* (A, D, Γ, C) is fitted per binary and kind (primary, secondary) to all its eclipses together, optionally per season (which follows the slow change of depth and width as the orbit precesses); then the time of every eclipse is fitted with that shape and free depth, width and baseline, its error from the covariance matrix scaled by the residuals. By default every binary is measured on its own light plus the mean light of the others (ideal disentangling); with the blended option the total light is used as observed, eclipses of other binaries in the window are masked, and their proximity variation shifts the minima (in a CzeV4315-like example the scatter of B's minima grows from 0.2 s to 8 s).

The difference between a minimum and the barycentric conjunction is physical. Light-travel time inside the binary delays the eclipse of the farther star: the secondary minimum of a circular orbit comes $(P/\pi c)(K_2 - K_1) = 2a \sin i\,(m_1 - m_2)/(Mc)$ after half the period (Kaplan 2010; K_1, K_2 the velocity semi-amplitudes), reproduced to 0.4 %. For a noise-free curve the Mikulášek profile fits detached eclipses to about 1 % of their depth.

7.5 Pulsating stars as clocks

Many components of multiple systems pulsate: δ Scuti and γ Doradus stars near 1.5–2.5 $M_\odot$, β Cephei and slowly pulsating B (SPB) stars, RR Lyrae stars, Cepheids and long-period variables on the AGB. With `stars.<m>.pulsation` a star's own flux varies as

$$\Delta m(t) = \sum_k A_k \sin\left[2\pi\left(\frac{t_e}{P_k} + \phi_k\right)\right],$$

with amplitudes A_k, periods P_k and phases ϕ_k of the modes, evaluated at the star's *emission time* t_e. An orbiting pulsator is therefore a clock: its phases lag by the light time across its orbits, the phase modulation of Murphy et al. (2014). For a circular orbit the delays are $\tau(t) = (a_1 \sin i/c)\sin(2\pi t/P_{\rm orb} + \varphi)$ (a_1 the semi-major axis of the pulsator's orbit around the barycentre); in a multiple system the delays of the inner orbit add to those of the outer one. SWINGERsim reproduces delays measured from a light curve sampled like Kepler data to 3×10^{-7}.

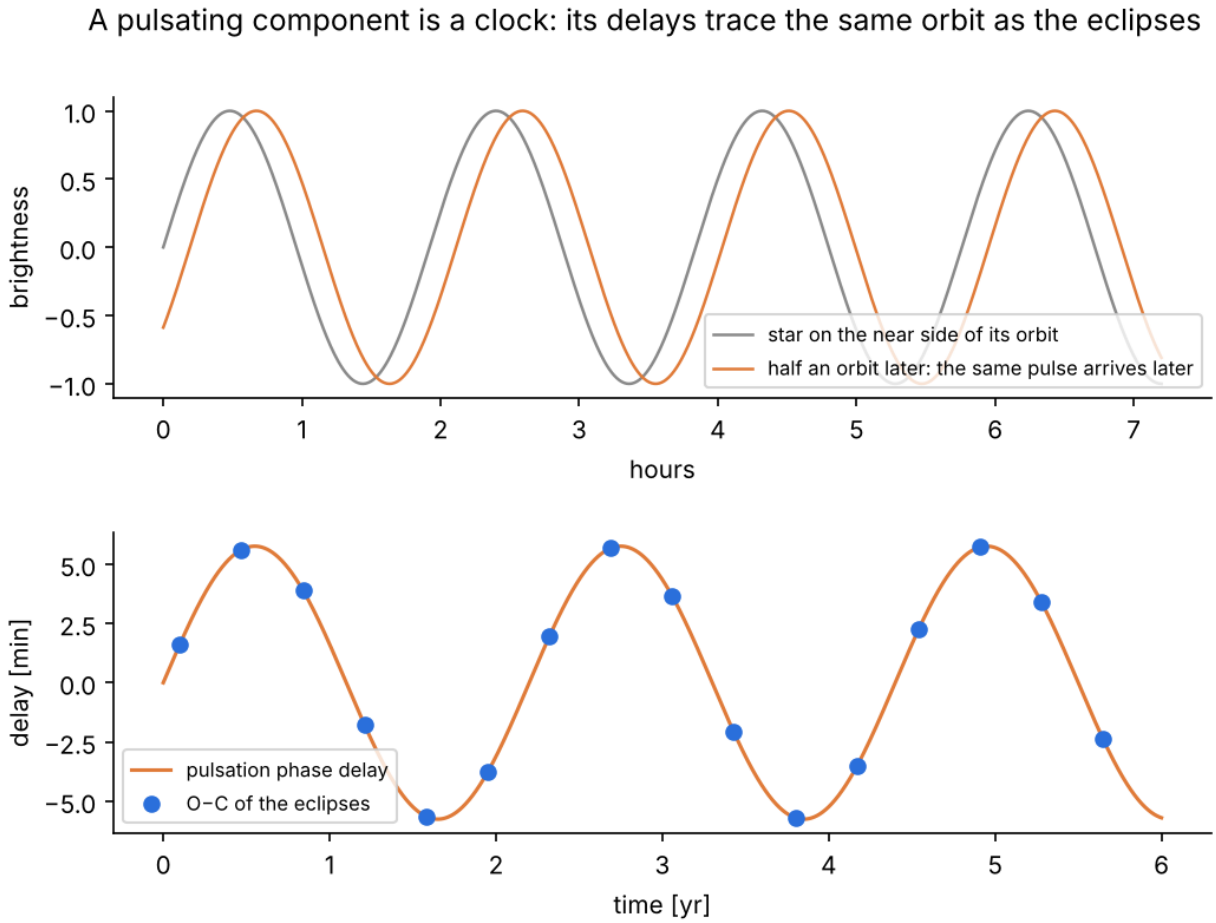


Figure 17. A pulsating component as a clock (illustrative). Top: brightness of the pulsator (arbitrary units) against time in hours, on the near side of its orbit (grey) and half an orbit later (orange): the same pulses arrive later. Bottom: the delay of the pulsation phase in minutes against time in years (orange line) and the O−C of the eclipses of the same binary (blue dots): both trace the same orbital light-time curve.

Figure 17 shows why a pulsator in an eclipsing binary carries two clocks that must agree: the eclipses sample the light-time curve once or twice per binary period, the pulsations continuously.

Periods follow from the stellar model (current mass, radius, luminosity, $T_{\rm eff}$) unless `period_d` is given (Table 12).

Table 12. Pulsation types and the source of their periods. *Type* is the value of `pulsation.type`; *Period relation* the formula used; *Source* the reference and its range.

Type	Period relation	Source
`lpv`	non-linear fundamental-mode period–mass–radius relation, broken power law with saturation	Trabucchi et al. (2021), eqs. 1–2; O-rich AGB, 0.6–7 $M_\odot$
`delta_scuti`	$Q = P\sqrt{\bar\rho/\bar\rho_\odot}$ = 0.0329 d (F), 0.0251, 0.0203, 0.0172 d (1st–3rd overtone)	Breger (1979); consistent with Poro et al. (2024)
`rr_lyrae`	$\log P_0 = 11.627 + 0.823\log L - 3.506\log T_{\rm eff} - 0.582\log M$; $P_1/P_0 = 0.744$	Bono et al. (1996) as quoted by Alcock et al. (1997)
`cepheid`	$\log P = a + b\log T_{\rm eff} + c\log M + d\log L$ (F), first overtone separately	Bono, Castellani & Marconi (2000), Z = 0.004–0.02
`beta_cep`	radial fundamental mode, Q = 0.033 d	Handler (2013)
`spb`, `gamma_dor`	series $P_0 + k\,\Delta P$ of gravity modes; P_0, ΔP given by the user	Wu, Li & Deng (2018)
`manual`	`period_d` or Q given	–

Every type warns when the star's $T_{\rm eff}$ lies outside its class (a plausibility check). `swingersim pulsation` also reports dP/dt from the evolving stellar model, the evolutionary period change that O−C diagrams of pulsators measure. MIST stars in the middle of the δ Scuti strip lie on the observed period–luminosity relation of Poro et al. (2024) to 0.1 mag, and MIST Cepheids on their blue loops match the period–radius relation of Anderson et al. (2016) to 0.01 dex. Amplitudes are fixed per band, mode shapes are sinusoids (no combination frequencies or non-sinusoidal RR Lyrae and Mira shapes), and pulsation does not change the radius used for eclipses.

7.6 Discs as occulters

Some eclipses are caused by a dark disc of dust: the two-year eclipses of ε Aur by a disc around its companion, the eclipses of EE Cep whose depth changes from cycle to cycle as its disc precesses, or KH 15D, a young binary disappearing behind the edge of a precessing circumbinary ring. A disc in SWINGERsim is a slab: a cylinder of outer radius $r_{\rm out}$, optional hole $r_{\rm in}$ and full thickness H, centred on a star or on the centre of mass of an orbit. For every line of sight from a point of a star towards the observer the length L inside the slab is computed exactly, and the transmission is $e^{-\tau L/H}$, with τ the optical depth perpendicular to the disc (opaque without τ). An edge-on disc is a dark band of height H, an inclined one an ellipse with a rim. The stellar disk is sampled on 4000 equal-area points weighted by the limb darkening; a sharp edge crossing a star is resolved to about 10^{-3} of its flux. A disc can be fixed on the sky, tilted from an orbit, precess at a given rate, or precess as a rigid body under the torque of its binary (`dynamics: rigid`; Zanazzi & Lai 2018, eq. 3; Larwood et al. 1996 for circumstellar discs), with polar alignment around eccentric binaries (Farago & Laskar 2010; Martin & Lubow 2017). Discs can shine as flat reprocessing discs, $T = (2/3\pi)^{1/4}(R_*/r)^{3/4}T_*$ (Chiang & Goldreich 1997), and scatter starlight with an albedo and a Henyey–Greenstein phase function (Henyey & Greenstein 1941). Not modelled: the disc's mass and gravity on the orbits, forward scattering through optically thin parts, warps.

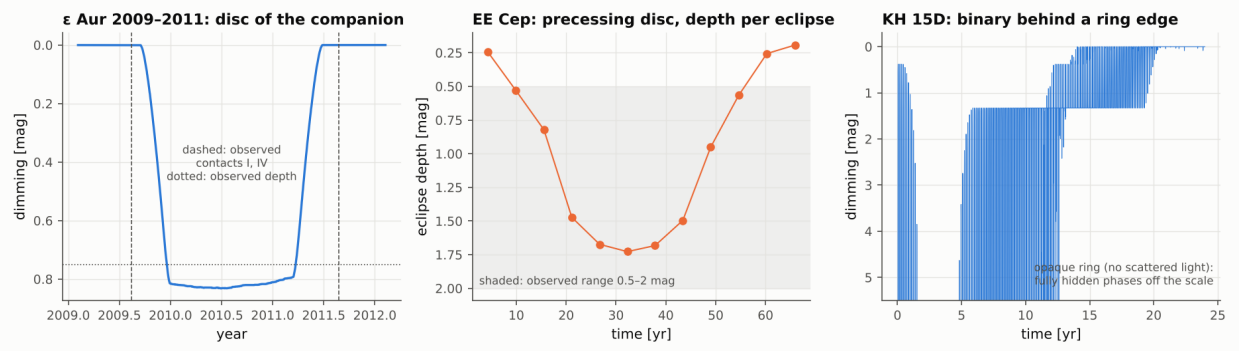


Figure 18. Light curves of disc-eclipsing systems computed with SWINGERsim. Left: ε Aur 2009–2011, dimming in magnitudes against year; dashed vertical lines: the observed contacts I and IV; dotted horizontal line: the observed depth. Middle: EE Cep, depth of every eclipse in magnitudes against time in years over one precession of its disc (illustrative model; shaded: the observed range 0.5–2 mag). Right: KH 15D, dimming in magnitudes against time in years as the edge of its opaque ring sweeps over the binary orbit (illustrative model; phases in which the binary is fully hidden lie off the scale).

Figure 18 shows three regimes. ε Aur: an opaque slab passing in front of a star gives a flat-bottomed eclipse whose contacts and depth the model matches. EE Cep: the precessing disc changes the depth from eclipse to eclipse, so the depths trace the precession. KH 15D: the ring edge progressively covers the orbit of the binary, so the fraction of each orbit in which a star is hidden changes over years. For the illustrative EE Cep disc (0.8 au) the torque of the B star gives a precession period of 183 yr against the observed 62–67 yr; a disc of about 1.6 au would match – the precession constrains the size of the disc.

7.7 What is not modelled

Stellar spots, combination frequencies of pulsations, the light and gravity of disc material beyond the slab model, and astrometry: SWINGERsim computes no photocentre motion and no Gaia astrometric signal. Relative and absolute positions of the stars on the sky are available in the run output and could serve for such a model, but no astrometric observable is computed or fitted.

7.8 Implementation

External libraries and data. Limb-darkening and gravity-darkening tables of Claret & Bloemen (2011) and the TLUSTY coefficients of Reeve & Howarth (2016); passband curves of Bessell (1990), TESS (Ricker et al. 2015) and Gaia DR3 (Riello et al. 2021); PHOEBE 2 (Prša et al. 2016; Conroy et al. 2020) in the optional hybrid. batman, ellc and PHOEBE serve only as references in the tests. **Own code.** The black-body passband integration and the table interpolation; the exact occultation of any number of limb-darkened spheres; the light-travel-time solution per binary; the analytic proximity terms; the Roche and rotating-star surfaces, their meshes, gravity darkening and ray-traced occultations including self-occultation of contact binaries; the coupling to PHOEBE (chunks, element conversion, mutual eclipses); radial velocities at emission time and the centre-of-light velocities on the meshes; the synthetic eclipse timings and O−C; the phenomenological minima with the Mikulášek profile; the pulsation periods and the phase modulation; and the disc geometry, transmission, light and rigid precession.

Chapter 8

Fitting observed data

The previous chapters run a model forward: from orbits and stars to light curves, velocities and times of minima. This chapter describes the inverse step – adjusting the model to measured data – which SWINGERsim performs in a sequence of steps called the *fit chain*. The chain starts from an analytic description of every eclipsing pair (stage 1), optionally anchors published elements to the measured minima, and ends with one N-body model of the whole system fitted to all data at once (stage 2), with robust error treatment, optional Markov-chain Monte Carlo (MCMC) uncertainties and predictions of future minima.

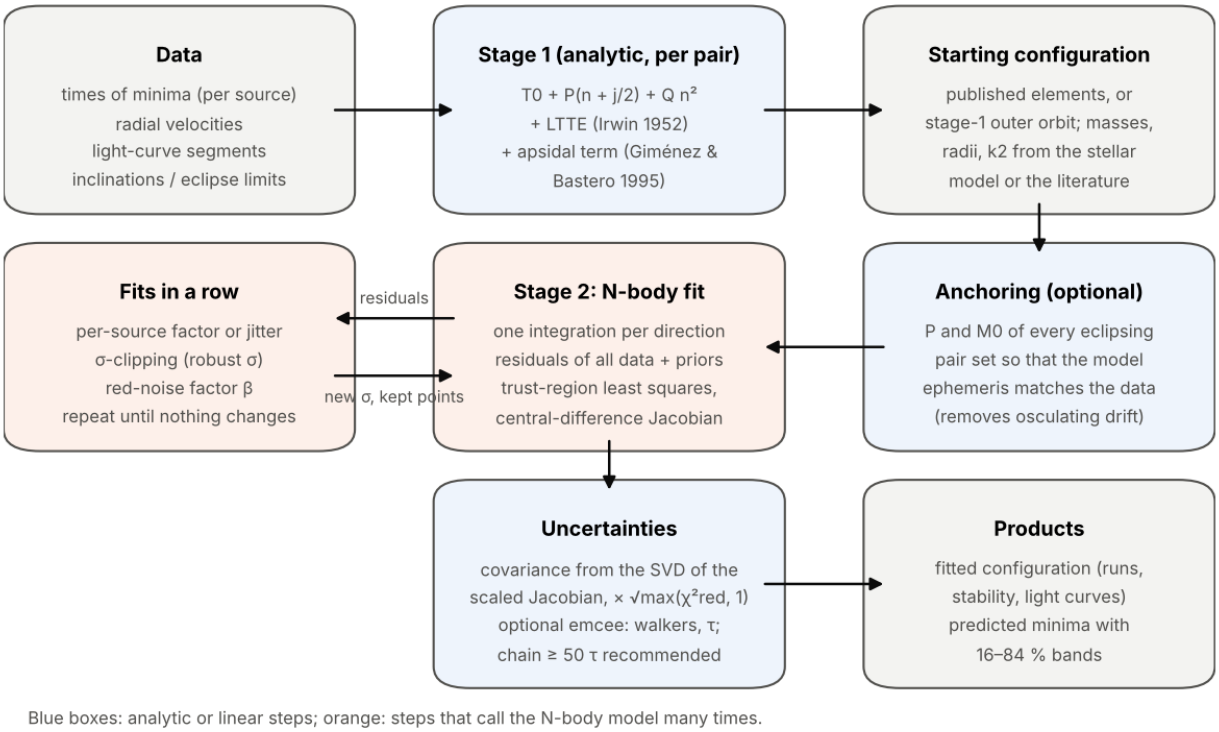


Figure 19. The fit chain (new figure for this volume). Boxes are steps, arrows the flow of information. Top row: the data (times of minima by source, radial velocities, light-curve segments, inclinations or eclipse limits at given epochs) enter stage 1, an analytic model per pair, whose outer orbit can provide the starting configuration together with published elements and stellar parameters. Right: optional anchoring of the period and phase of each eclipsing pair. Middle: stage 2, the least-squares fit of one N-body model to all data, alternates with the “fits in a row” (left) that rescale the errors per source, reject outliers and estimate the red-noise factor, until nothing changes. Bottom: uncertainties from the covariance matrix and optionally from emcee, and the products – a fitted configuration usable by every other command and predicted minima with their bands. Blue boxes are analytic or linear steps; orange boxes call the N-body model many times.

Figure 19 is the map of this chapter. Each box corresponds to one of the following sections, and the loop between the two orange boxes is the robust part of the fit: the error model is not fixed in advance but estimated from the residuals of the previous fit.

8.1 Data

The fit uses up to four kinds of data, all on one time axis (BJD in the TDB system; tables in HJD or JD in UTC are converted with the sky coordinates of the target):

- **Times of minima** (table `fit.minima`): pair, type (primary or secondary), time, error, source (instrument, observer or paper) and method. The *source* matters: errors and outliers are treated per source.
- **Radial velocities** (`fit.rv`): star, time, velocity, error, source; the model velocity is computed at the emission time and, optionally, as the centre-of-light velocity (Observables).
- **Light curves** (`fit.lc`): segments of normalised flux with errors, for example the windows around eclipses or binned sectors, in a given passband and exposure time.
- **Inclinations against time** (`fit.inclination`): inclinations of an eclipsing pair measured at several epochs, or one-sided constraints (Inclination data).

A *pair* is normally a binary of two stars, but it can be any orbit: the conjunctions of a wider orbit – the centres of mass of its two components aligned along the line of sight – are the centres of third-body eclipse events, so a tertiary eclipse can be fitted like a binary minimum.

8.2 Stage 1: analytic O−C model per pair

Stage 1 is the analytic O−C model that the SQUADRA group fits with OCFit (Gajdoš & Parimucha 2019), one pair at a time or several pairs with one outer orbit:

$$T(n,j) = T_0 + P\,(n + j/2) + Q\,n^2 + \Delta_{\mathrm{LTTE}}(t) + D_j(\omega_n),$$

with the cycle number n, the minimum type j (0 primary, 1 secondary), the reference epoch T_0, the period P, the quadratic coefficient Q (a constant period change, $Q = \frac{1}{2}P\dot{P}$), the light-travel-time term of an outer orbit (Observables) and the apsidal term D_j. The terms are chosen per pair (`fit.stage1.terms`, `pairs`).

Apsidal term. The apsidal motion of an eccentric pair, $\omega_n = \omega_0 + \dot{\omega}\,n$, enters through the time of minimum projected separation. At $i = 90°$ this is the conjunction $\nu + \omega = \pi/2$ (primary) or $3\pi/2$ (secondary), and

$$D_j = \frac{P}{2\pi}\left[M(\nu_j) + \omega - \left(\frac{\pi}{2} + j\pi\right)\right],$$

the time from the mean to the true conjunction ($M(\nu)$ the mean anomaly belonging to the true anomaly ν). This is the exact form of the series of Giménez & Bastero (1995) used by OCFit; for $i < 90°$ the minimum of the projected separation is found numerically (the $e\cot^2 i$ terms of Giménez & Garcia-Pelayo 1983). The apsidal period is $U = 2\pi P/\dot{\omega}$.

Shared outer orbit. With `shared_outer` the pairs share one outer orbit (common P_3, periastron time T_3 and e_3, the periastron of the second pair opposite to the first) with separate $a\sin i$; the ratio of the $a\sin i$ values gives the mass ratio of the pairs, $m_A/m_B = (a\sin i)_B/(a\sin i)_A$.

Method. A weighted least-squares fit started from a grid of circular LTTE fits over the outer period (and of the anti-phase of primary and secondary minima for the apsidal term); times are fitted relative to a reference epoch near the middle of the data so that the derivatives keep their precision. Errors come from the covariance matrix, scaled by $\sqrt{\chi^2_{\mathrm{red}}}$ when the reduced χ^2 exceeds 1, or from an MCMC run (below). Outliers can be rejected and errors

rescaled per source as in stage 2. The model agrees with OCFit's series to 10^{-3} s at $e = 0.3$, and a fit of SWINGERsim's own N-body timings of a 2+2 recovers the Keplerian $a \sin i$ of both pairs to 0.3 % and the outer period to 4×10^{-5} (see *Testing, Part I*).

Limits. Stage 1 treats each pair as a Keplerian binary on a Keplerian outer orbit. It cannot describe dynamical perturbations, precession or the coupling of the pairs in a compact system; its outer orbit and $a \sin i$ are a starting point for stage 2.

8.3 Anchoring to the present

Published photodynamical solutions give osculating elements at an epoch. Started from them, the N-body model of SWINGERsim has a mean inner period that differs slightly from the one measured in the minima (the elements are osculating, the period seen in the minima is a mean; the other code may differ in physics), and the computed minima drift away from the measured ones over the years. *Anchoring* (`fit.stage2.anchor`) removes this drift before any comparison: the period P and the mean anomaly M_0 of every eclipsing pair are adjusted so that the mean linear ephemeris of the model's eclipses equals that of the measured minima. A coarse step scales P by $P_{\mathrm{data}}/P_{\mathrm{model}}$ and shifts M_0 by the delay of the model eclipse nearest $t = 0$; a fine step fits a straight line through measured − model at the epochs of the minima of each pair (unbiased by uneven sampling of a curved O−C), repeated until no minimum moves by more than 0.1 min. Everything else stays as configured. Stage 2 removes the same difference by construction, because it fits the osculating elements that reproduce the minima; anchoring provides a good starting point and is useful alone when only a prediction is wanted.

8.4 Stage 2: the joint N-body model

Stage 2 (`fit.stage2`) fits one N-body model of the whole system to all data at once. The model is the same as that of a normal run (Newtonian gravity, 1PN relativity, tides, imposed period changes as configured), so the light-travel-time effect, apsidal motion, dynamical perturbations and precession come out of one model, and the fitted configuration can be run directly (stability, evolution forward and backward, light curves, predictions).

Model times. For every measured minimum the system is integrated to shortly before the expected conjunction; the two-body motion of the pair from that state gives the nearest conjunction of the right type (minimum projected separation; primary when the first component is the farther one); the N-body integration continues to it and refines it. The observed time adds the light time of the pair's centre of mass, $t_{\mathrm{obs}} = t + z_{\mathrm{cm}}/c$. Minima before $t = 0$ come from a second, backward integration. All requests of all data types (minima, velocities at their emission times, inclinations at their epochs) go into one queue ordered by time, so one model evaluation costs one integration over the span of the data in each direction. The times agree with those of the synthetic timings at 400 samples per period to 0.06 s.

8.4.1 Free parameters

Table 13 lists the parameters that can be fitted.

Table 13. Parameters that stage 2 can fit (`fit.stage2.free`). *Name* is the form in the configuration; *Meaning* the quantity; *Default bounds* the box within which the optimiser searches unless `bounds` gives another. Everything not listed as free stays as configured.

Name	Meaning	Default bounds
`orbits.<o>.P`	period [d]	0.5–2 × start
`orbits.<o>.M0`, `omega`, `Omega`	mean anomaly, argument of periastron, node [deg]	none
`orbits.<o>.e`	eccentricity	0–0.95

Name	Meaning	Default bounds
`orbits.<o>.ecosw`, `esinw`	$e\cos\omega$, $e\sin\omega$ (replace e, ω for small e)	±0.95, $e \leq 0.95$
`orbits.<o>.i`	inclination [deg]	0–180
`orbits.<o>.Pdot`	imposed period change [d yr^{-1}]	none
`masses.<m>`	mass [$M_\odot$]	0.05–20 × start
`stars.<m>.R`, `Teff`	radius [$R_\odot$], effective temperature [K]	0.2–5 ×, 0.5–2 × start
`stars.<m>.apsidal_k2`, `rg2`	structure constants	10^{-5}–0.74, 0.005–0.4
`rv.gamma`, `rv.offset.<source>`	systemic velocity, zero point of a source [km s^{-1}]	none

Fitting k_2 (or the radii) turns the apsidal motion into a constraint on stellar structure; fitting `Pdot` lets a parabola in the O−C be fitted together with the LTTE (Mass transfer and evolution). A rotation of the whole system about the line of sight changes none of the data, so with the node of every orbit free only their differences are measured; SWINGERsim then holds the node of the outermost orbit at its configured value and says so.

What the data constrain. Minima alone determine the periods, phases, eccentricities and arguments of periastron of all orbits and, through the light-time amplitude and the dynamical perturbations, the mass ratios. The inclinations of the eclipsing pairs and the absolute masses need light curves and radial velocities; without them they should be fixed at the values of a light-curve or spectroscopic solution or constrained by priors. Mutual inclinations and nodes are constrained only in compact systems with a measurable dynamical (not light-time) signal.

8.4.2 Likelihood

The fit minimises

$$\chi^2 = \sum_k \left(\frac{d_k - m_k(\mathbf{p})}{\sigma_k} \right)^2 + \sum_j \left(\frac{p_j - \mu_j}{s_j} \right)^2 ,$$

where d_k are the data (times, velocities, fluxes, inclinations), $m_k(\mathbf{p})$ the model at the parameters $\mathbf{p}$, σ_k the errors after the corrections below, and the second sum the Gaussian priors on parameters p_j with means μ_j and widths s_j. This is $-2\ln$ of a Gaussian likelihood times Gaussian priors, up to a constant. Angles in priors are wrapped to ±180°. A data point for which the model does not exist (no conjunction of that type, a star beyond its track) counts $10^3\sigma$, so the optimiser moves away from such regions.

Light curves enter with a calibration per segment, solved linearly at every evaluation and not part of the free parameters:

$$F_{\rm obs}(t) = a\,F_{\rm model}(t) + b + c\,(t - t_{\rm mid}) + d\,(t - t_{\rm mid})^2,$$

where a scales the eclipse depths (dilution by third light or a different normalisation), b is the zero point and c, d describe slow trends (polynomial degree 0–2); with `scale: false` the depths must follow from the stars as given. The degrees of freedom of the fit are reduced by these coefficients. The model light curve uses the exposure time of the data. Star shapes are chosen per light curve: with `shapes: auto` a star of a circular two-star binary gets its Roche surface when its volume radius exceeds a fraction `roche_threshold` (default 0.5) of its Roche lobe – spheres err by about 2 ppt there (Observables) – and stays a sphere otherwise.

Deformed stars in the fit. Meshes are expensive. With `fidelity: mapped` (default) the fit runs on a coarse mesh (24 × 48 directions) plus the difference *accurate* − *coarse* evaluated at the current solution with the accurate mesh (64 × 128); the difference is recomputed at

each new solution until it changes by less than 0.3σ (at most five passes). This *output space mapping* converges to the solution of the accurate model with one accurate evaluation per pass; χ^2 and the light curves reported at the solution are those of the accurate mesh.

8.4.3 Inclination data

A table of inclinations of an eclipsing pair measured at several epochs (light-curve solutions of separate data sets, as published for V907 Sco by Zasche et al. 2023a) is compared with the inclination of the pair's orbit in the N-body model at the same times. Eclipses measure only $|\cos i|$, so the model inclination is folded to 0°–90°, and so are measured values above 90° (Hierarchies and orbits). A row can also be one-sided:

- `upper` / `lower`: $i_{\mathrm{model}} \leq i$ or $\geq i$ (for example "about 75° or less" from a light curve without eclipses);
- `no_eclipse`: the pair did not eclipse at that epoch, i.e. i_{model} lies below the eclipse limit $\arccos[(R_1+R_2)/r_{\mathrm{conj}}]$ of the model, with the model radii and the separation r_{conj} at the nearer conjunction;
- `eclipse`: the pair did eclipse, i_{model} above that limit.

A one-sided row contributes $(\mathrm{bound} - i_{\mathrm{model}})/\sigma$ only when the model violates it (a *hinge*) and nothing otherwise; σ sets how sharply. Satisfied one-sided rows are not counted as data in the degrees of freedom. A lower limit `min_sigma_deg` on the errors represents the systematics of the light-curve solutions.

8.4.4 Optimiser and covariance

χ^2 is minimised by a trust-region reflective least-squares method with bounds (SciPy `least_squares`), optionally with a robust loss (`soft_l1`, `huber`, `cauchy`, which reduce the influence of residuals beyond `f_scale` σ). The Jacobian comes from **central differences**, two N-body integrations per free parameter and iteration, computed in parallel; forward differences are biased by the curvature of the residuals (for example the cumulative effect of a period change). At a bound, or where the model fails on one side (a star beyond the end of its track, an unstable orbit), a one-sided difference is used and the parameter is reported as lying at the edge of the model's validity, where its error is not valid. The optimiser stops when χ^2 changes by less than 10^{-5} relative or 0.05 absolute per iteration (far below the change of 1 that moves a parameter by one σ), or after `max_iterations` (default 30).

The covariance matrix is computed from the singular-value decomposition of the column-scaled Jacobian rather than by inverting $J^{\mathsf{T}}J$, which squares the condition number: with parameters of very different scale (a period known to 10^{-5} d, angles to 0.1°) and correlations near 0.999, a pseudo-inverse of $J^{\mathsf{T}}J$ silently drops the weakest direction and gives errors far too small. Directions with a singular value below 10^{-12} of the largest are reported as unconstrained. The errors of the data rows are scaled by $\sqrt{\max(\chi^2_{\mathrm{red}}, 1)}$; the priors keep their widths. Strongly correlated parameters ($|r| > 0.95$) are listed in the report – typically M_0, ω and P of an outer orbit covered by less than about two periods – because their linear errors hold only while the χ^2 valley is straight.

A closure test on TIC 120362137 (synthetic minima at the 147 epochs of the real TESS minima, 12 free parameters perturbed) converges from $\chi^2 = 1.9 \times 10^8$ to 107 ($\chi^2_{\mathrm{red}} = 0.79$) in 331 model evaluations and recovers all 12 parameters within 1.5σ of the linear errors (see *Testing, Part I*).

8.5 Errors per source, jitter and outliers

Times of minima from different sources have errors of different quality. The quoted errors of old photographic, visual or survey minima are often far below their real scatter, and even formal errors of TESS minima are below what any model reproduces (reduced χ^2 of 86–1432 in a test on SQUADRA data). Left as they are, a few hundred old minima with optimistic

errors outweigh the precise ones. SWINGERsim therefore estimates the error model from the residuals in **fits in a row**: fit, correct the errors and reject outliers, fit again from the solution, until neither the rejected points nor the corrections change by more than 2 % (at most `refits` fits, default 4). Minima, velocities (groups per source) and light curves (groups per file and segment) are treated alike.

Error correction per source (`error_scale: per_source`; `global` uses one correction for all minima). Sources with fewer than `min_source_minima` (default 5) points share one group. Each group s gets either

- a **factor** (`error_model: scale`), $\sigma_k \to f_s\,\sigma_k$ with $f_s = \sqrt{\chi^2_s/\nu_s} \geq 1$, or
- a **jitter** added in quadrature (`error_model: jitter`), $\sigma_k \to \sqrt{\sigma_k^2 + j_s^2}$, with $j_s \geq 0$ the root of $\sum_{k\in s} r_k^2/(\sigma_k^2 + j_s^2) = \nu_s$,

where χ^2_s is the contribution of the group, r_k its residuals and ν_s its degrees of freedom: the free parameters are shared among the groups in proportion to their points (a light-curve segment also subtracts its calibration coefficients). A factor keeps the relative weights within a source; a jitter acts as an error floor – model systematics and timing systematics add in quadrature to the measurement error – which suits sources of very precise minima. Per source the errors are never reduced below the quoted ones.

Outlier rejection (`clip_sigma`, e.g. 5). A point is left out when its normalised residual exceeds `clip_sigma` robust standard deviations of its group, $1.4826\,\mathrm{median}|r_k/\sigma_k|$ (at least 1, so that a source that fits better than its errors loses no good points). The worst points go first, at most `max_clip_fraction` (default 10 %) of all; a point left out earlier comes back when the present model fits it within 0.8 × the threshold. Points left out remain in the outputs.

Red noise (`fit.lc.red_noise`). Residuals of light curves are often correlated in time (instrumental trends, oscillations, spots), and white-noise errors then underestimate the parameter errors. The *red-noise factor* β of the time-averaging method (Pont, Zucker & Queloz 2006; Winn et al. 2008) compares the scatter of residuals averaged in bins of N points with the white-noise expectation,

$$\beta_N = \frac{\sigma_N}{\sigma_1 N^{-1/2}\sqrt{M/(M-1)}},$$

where σ_1 is the scatter of the unbinned residuals, σ_N that of the bin means and M the number of bins; β is the median over bin widths of 20, 40, 60 and 120 min, at least 1. The errors of each segment are multiplied by β in the fits in a row.

8.6 MCMC uncertainties

The linear errors rely on a locally linear model. In a long, curved valley of χ^2 – typical for an outer orbit covered by less than about two periods – they come out too small. With `sampler: emcee` SWINGERsim samples the posterior around the least-squares solution with the affine-invariant ensemble sampler of Goodman & Weare (2010) as implemented in emcee (Foreman-Mackey et al. 2013).

- **Posterior.** $\ln p = -\frac{1}{2}[\chi^2_{\mathrm{data}}/\max(\chi^2_{\mathrm{red}}, 1) + \chi^2_{\mathrm{prior}}]$ inside the bounds, so the data errors are inflated as for the linear errors while the priors keep theirs. For orbits fitted with $e\cos\omega$ and $e\sin\omega$ a Jacobian $1/e$ is added: a flat prior in $(e\cos\omega, e\sin\omega)$ is a prior $\propto e$ that pushes nearly circular orbits away from $e = 0$ (Lucy & Sweeney 1971; Ford 2006; Eastman, Gaudi & Agol 2013), and the correction makes the prior flat in e and ω.
- **Walkers** – the parallel chains of the ensemble sampler – number at least $2n + 2$ for n free parameters, and even (emcee moves half of them at a time). The number does not depend on the computer, so the same seed gives the same chain on any number of cores. They start in a ball of half the covariance matrix around the solution, i.e. spread along the valley.

- **Steps.** Each walker makes `mcmc_steps` steps (default 300); the first half is discarded as burn-in. Errors are half of the 16–84 % range of the samples.
- **Autocorrelation time.** The integrated autocorrelation time τ is the number of steps after which a chain forgets its state; a chain of N steps contains about N/τ independent samples per walker. SWINGERsim reports τ of every parameter, its maximum and the ratio steps/τ, and warns when the chain after burn-in is shorter than 10 τ.

Convergence rule of thumb. A chain at least 50 τ long is the guidance of the emcee authors (Foreman-Mackey et al. 2013) for a reliable estimate of τ and of the posterior quantiles. With `mcmc_tau_target: 50` the chain is extended by a quarter of its length at a time until it reaches 50 τ or `mcmc_max_steps`. Every model evaluation is an N-body integration, so 50 τ can be expensive. **When the chain is short**, report it as such: give the number of walkers, steps, the discarded burn-in, $\tau_{\max}$, steps/τ and the acceptance fraction; quote the MCMC intervals as indicative (they are usually too narrow, because the walkers have not yet explored the tails), and quote the linear errors and the list of strongly correlated parameters alongside.

8.7 Predictions

With `fit.output.predict` the fitted model predicts the eclipses of the chosen pairs for a number of years ahead (from now, or from the last measured minimum if it is later). Each predicted minimum carries an uncertainty band: parameter sets are drawn from the emcee samples (when the fit ran emcee) or from the covariance matrix, each set is integrated once (in parallel), and the band is the 16–84 % range of the times of the same eclipse across the sets. Only eclipses that occur in the fitted model (impact parameter below 1) are listed. Bands from the covariance matrix inherit its linear approximation; in curved valleys the MCMC samples give the more honest band.

8.8 Limits

- The fit finds a local minimum near the starting point; a poor start (a wrong outer period) can end in a wrong minimum. Stage 1, anchoring and published solutions provide the start.
- Linear errors are reliable only in straight χ^2 valleys; MCMC errors only for converged chains.
- Systematics of the model (missing physics, fixed stellar parameters) enter the error corrections per source, not the parameters; a jitter that is large compared with the quoted errors is a sign of missing physics.
- The light-curve model inherits the limits of chapter Observables: no spots, Roche shapes only for circular synchronised binaries.

8.9 Implementation

External libraries. SciPy's `least_squares` (trust-region reflective) as optimiser; emcee (Foreman-Mackey et al. 2013) as MCMC sampler and for the autocorrelation times; REBOUND/REBOUNDx in every model evaluation; Astropy for time-system conversions.
Own code. The stage-1 model (LTTE, exact apsidal term, shared outer orbit, start grids); anchoring; the stage-2 model with the event queue that serves minima, velocities and inclinations from one integration per direction; the parameter mapping, bounds, $e\cos\omega/e\sin\omega$ handling and the fixed outermost node; the central-difference Jacobian with one-sided steps at bounds and at the edge of model validity; the SVD covariance; the fits in a row with per-source factors and jitters, robust clipping with re-inclusion and the red-noise factor; the light-curve calibration and the output-space-mapped mesh fidelity; the hinge rows of one-sided inclination data; the extension of emcee chains to a target number of τ and the $1/e$ prior correction; and the predictions with their bands.

Chapter 9

Ensembles and inference

The parameters of a real multiple star are never all known. Photometry fixes the inclination of an eclipsing binary but not the orientation of its node on the sky; the outer orbit is often known only from a timing signal; masses carry uncertainties. A single simulation answers “what if”; to answer “how likely”, SWINGERsim runs **Monte Carlo ensembles** – many members, each a full run with parameters drawn from distributions – and conditions them on measurements. This chapter describes the ensembles, the observation weights and their interpretation as Bayesian quantities, and the auxiliary machine-learning tools built on ensembles.

SWINGERsim is aimed at **specific systems**: a configuration describes one observed multiple star, and an ensemble explores the uncertainty of that one system. Emulators and simulation-based inference (end of the chapter) are auxiliary tools that make such ensembles cheaper to use; they are not population-synthesis machinery.

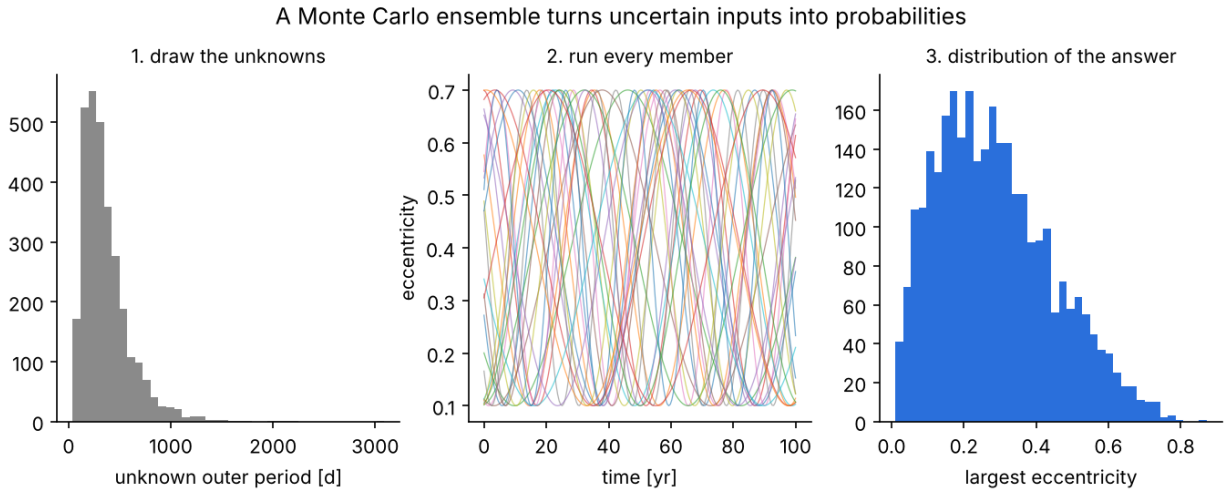


Figure 20. A Monte Carlo ensemble in three steps (illustrative). Left: histogram of an unknown parameter drawn for the members (here an outer period in days; vertical axis: number of members). Middle: the eccentricity of the inner binary against time in years for a sample of members, one coloured line each – every member is a full run. Right: histogram of the answer to the question asked (here the largest eccentricity reached; vertical axis: number of members). The distribution on the right, not any single run, is the result.

Figure 20 shows the logic: uncertain inputs (left) become a bundle of possible histories (middle) and then a distribution of the quantity of interest (right). Every statement drawn from an ensemble is a statement about a fraction or a quantile of such a distribution, with a statistical error.

9.1 What is varied

The configuration describes the nominal system; `ensemble.vary` maps parameters (`masses.m1`, `orbits.out.P`, `orbits.B.Omega`, `stars.m2.R`, ...) to distributions, which form the **prior** of the ensemble. Table 14 lists the distributions.

Table 14. Distributions of `ensemble.vary`. *Form* is how it is written; *Distribution* what is drawn; *Typical use* the parameters it suits. Every distribution can be truncated with `min`, `max` (a value outside is drawn again).

Form	Distribution	Typical use
a number	fixed value	–
`uniform: [a, b]`	uniform between a and b	eccentricities, phases
`loguniform: [a, b]`	uniform in $\log x$	periods, separations spanning decades
`normal: [μ, σ]`	Gaussian	measured masses, periods, radii
`isotropic` or `isotropic: [i1, i2]`	$\cos i$ uniform (between the limits)	inclination of an orbit of unknown orientation
`angle`	uniform in 0°–360°	nodes, arguments of periastron, mean anomalies
`choice: [x1, x2, …]`	one of the values with equal probability	discrete alternatives
`mirror: true` (with an inclination)	$i \to 180° - i$ with probability 1/2	sense of revolution of an eclipsing binary

Angles in the sky frame. For an orbit of unknown orientation, an isotropic inclination together with a uniform node gives an isotropic distribution of its normal, and the mutual inclinations follow from the geometry (Hierarchies and orbits). For a measured eclipsing binary, i is drawn around its measured value, and `mirror: true` adds the unknown sense of revolution; only the relative sense of two orbits matters, so the first binary can be kept at $i < 90°$. In a population study (for example Pejcha et al. 2013) the reference plane is the outer orbit instead.

Requirements. `require` redraws members that fail a criterion: `stable_ma01`, `stable_v22` (Stability and fates) or `hierarchical` (every inner apocentre inside the pericentre of the orbit around it). A requirement truncates the prior; results are then conditional on it. `ma01_unstable_keep` and `ma01_window` shape training sets for stability classifiers (as Vynatheya et al. 2023 kept 0.3 of the MA01-unstable draws for 2+2 and 0.2 for 3+1).

Directions. With `directions: [forward, backward]` every member is integrated both ways from the same present state: the backward run tells for how long the system has looked as it does now (for example since when both binaries eclipse), the forward run what comes next, and survival in both directions is a strong constraint (Stability and fates).

Summary and reproducibility. Every member is reduced to one row: fate, eccentricity extremes, mutual inclinations and their ranges, eclipse fractions and eclipse changes, apsidal periods, Roche-lobe filling, stability verdicts (MA01, V22, optionally the MLP classifier and the ghost test) and the present rates of change. The members depend only on the configuration and `seed`, so an ensemble can be extended or rerun member by member with another backend.

9.1.1 Rates of change today

The most informative measurements of a precessing system are its present rates of change, for example of the inclination of an eclipsing binary measured from changing eclipse depths (Zasche et al. 2025). SWINGERsim computes `<orbit>_didt0_deg_yr`, the rate of the sky inclination, and `<orbit>_dtiltdt0_deg_yr`, the rate of the *tilt* $|i - 90°|$, by a linear fit from $t = 0$

over two periods of the perturbing orbit (at least six outputs) or over `run.rate_window_yr`, so that the oscillations of the osculating elements with the outer period average out. For an eclipsing binary the tilt rate is the useful one: eclipses that deepen mean a negative tilt rate whatever the unknown sense of revolution, whereas the sign of di/dt depends on it. The forward run gives a one-sided rate forward, the backward run backward; an observer measures a rate over a baseline centred on now, so the conditioning uses by default the *central* rate, the mean of the two. When an inclination oscillates quickly the one-sided rates can differ by several degrees per year; `rate_window_yr` should be about half the observing baseline.

9.2 Statistical error of a fraction

A fraction p estimated from n members has a binomial uncertainty of about $\sqrt{p(1-p)/n}$: ±2.5 % for p = 0.3 and n = 400. SWINGERsim reports the Wilson score interval (Wilson 1927) at 68 %, which stays inside 0–1 also for fractions near 0 or 1. Quantiles (median, 16 % and 84 %) are reported for continuous columns. Members with a mild hierarchy are often chaotic: two members with almost the same parameters end differently, so only distributions over many members are meaningful.

9.3 Conditioning on observations

Without measurements of how the system changes, forward and backward ensembles of an observed system are statistically alike. Measurements break this symmetry. Every member i gets the weight of a Gaussian likelihood

$$w_i = \exp\left[-\frac{1}{2}\sum_k \left(\frac{x_{ik} - x_k^{\mathrm{obs}}}{\sigma_k}\right)^2\right],$$

where x_{ik} is the value of summary column k for member i (for example the tilt rate of B or an apsidal period), and $x_k^{\mathrm{obs}} \pm \sigma_k$ the measurement (`ensemble.observed`, entries `{value, sigma}`). Hard limits (`{min, max}`, or `{equals: …}` for text columns such as the fate) set $w_i = 0$ for members that contradict what is known for certain – for example `backward:m1_roche_fill_max: {max: 1}` (the stars of an observed detached binary have not filled their Roche lobes in the past) or `backward:fate: {equals: intact}`. A member without a finite value of an observed column gets weight 0. Column names can carry `forward:`, `backward:` or `central:` to say from which run the value comes.

All statistics of the report are then weighted, both directions use the weight of the same member, and the drawn parameters are listed before (prior) and after (conditioned) the weighting. Weighted quantiles are computed from the cumulative weights.

Interpretation. This is *importance sampling* of the posterior distribution with the drawn distributions as the prior: by Bayes' theorem $p(\theta \mid x^{\mathrm{obs}}) \propto L(x^{\mathrm{obs}} \mid \theta)\,\pi(\theta)$, and members drawn from the prior π and weighted by the likelihood L represent the posterior. Two assumptions are built in: the measurement errors are Gaussian and independent, and the simulated summary values are taken without model error (the scatter between members represents the unknown parameters, not a noise of the model).

Effective number of members. Weighting concentrates the information in fewer members. The effective number (Kish 1965)

$$n_{\mathrm{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2}$$

replaces n in the statistical errors (the Wilson intervals are computed with it). If it is small (below about 20), the measurement selects only a small part of the ensemble and the results

are noisy: more members should be drawn, or the prior narrowed around the region that fits.

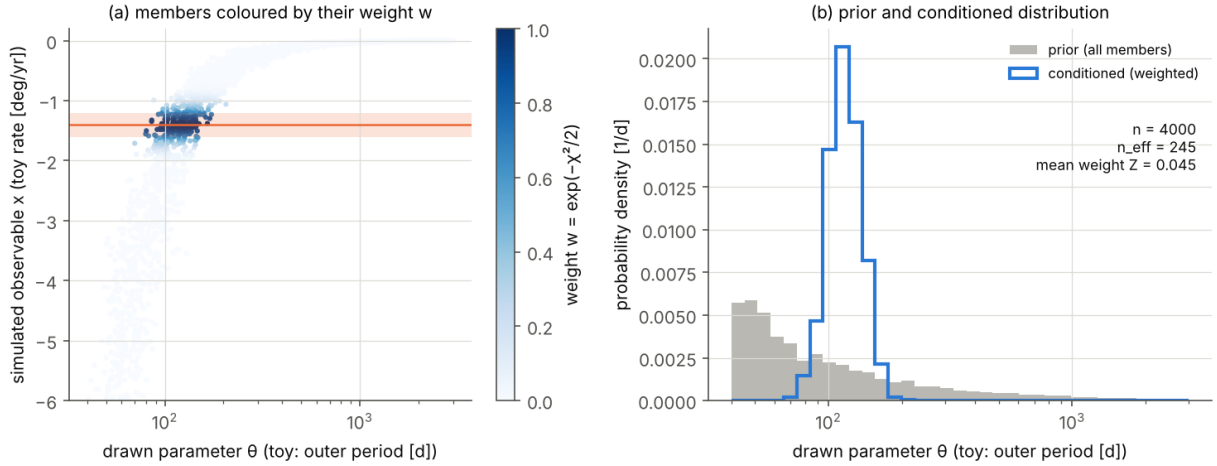


Figure 21. Conditioning by importance weights in a toy model (new figure for this volume; synthetic numbers, not a real system). A parameter θ – here playing the role of an outer period in days – is drawn log-uniformly for 4000 members; each member's observable is a power law of θ with 30 % random scatter (standing for the unknown angles). (a) Simulated observable against θ, each member a dot coloured by its weight w (white 0, dark blue 1); the orange line and band are the measurement -1.4 ± 0.2. (b) Probability density of θ for the prior (grey histogram, all members with equal weight) and after conditioning (blue outline, weighted). The text gives n, n_{eff} and the mean weight Z.

Figure 21 shows what the weights do. In panel (a) only members whose simulated value falls inside the orange band receive a high weight; because of the scatter they come from a range of θ, not one value. Panel (b) shows the result: the broad prior (grey) becomes a narrow conditioned distribution (blue) around θ ≈ 100–150. Of the 4000 members about 245 carry the weight (n_{eff}), so the conditioned statistics have the precision of an ensemble of about 245 members, not 4000.

9.3.1 The mean weight as Bayesian evidence

The mean weight of an ensemble,

$$Z = \frac{1}{n}\sum_{i=1}^{n} w_i ,$$

is a Monte Carlo estimate of the integral of the likelihood over the prior, $\int L(x^{\mathrm{obs}} \mid \theta)\,\pi(\theta)\,d\theta$, i.e. of the *Bayesian evidence* (marginal likelihood) of the hypothesis that the ensemble represents, up to the constant normalisation $\prod_k (2\pi\sigma_k^2)^{-1/2}$ of the Gaussian likelihood. Members removed by hard limits count with $w = 0$; members redrawn by `require` do not count at all, so a requirement is part of the hypothesis. Its statistical error is about $\mathrm{std}(w)/\sqrt{n}$, and it is noisy when n_{eff} is small.

Comparing hypotheses. For two hypotheses H_1, H_2 – for example a 2+2 quadruple and a 2+(2+1) quintuple, or two ranges of an unknown outer period – conditioned on **the same observations with the same errors and the same hard limits**, the normalisation cancels and the ratio

$$B_{12} = \frac{Z_1}{Z_2}$$

is the *Bayes factor*: the factor by which the data change the odds of H_1 against H_2. With equal prior odds it is the posterior odds. In the application to CzeV4315 in *Testing*

SWINGERsim, Part IV (one binary with a measured inclination rate, the other with persistent eclipses), a 2+(2+1) model reached $Z = 0.022$ with $n_{\mathrm{eff}} = 46$ (1500 N-body members). A 2+2 model with the 119-d period as its mutual orbit gave no member above a weight of 10^{-12} among 1000, i.e. $Z \lesssim 10^{-3}$. The Bayes factor in favour of the quintuple is therefore at least about 20, formally 10^{13}. When one hypothesis has (almost) no matching member, quote the upper bound $Z < 1/n$, not the formal value. Three cautions apply: (1) the evidence depends on the prior – a broad prior that wastes members on regions excluded by the data lowers Z, which is the Bayesian penalty for vagueness, but the comparison is only as meaningful as the priors are physically motivated; (2) a small n_{eff} makes Z itself uncertain by a large factor; (3) Z from a secular ensemble can be overestimated when members that fit are chaotic or unstable on longer time scales, which the secular backend does not see – an N-body check of the best members is advisable. The ensemble report prints n_{eff} and Z with its error $\mathrm{std}(w)/\sqrt{n}$, and it flags the case of fewer than two matching members, for which the upper bound applies.

Reporting. A conditioned result should state the prior (the `vary` distributions and requirements), the observations and their errors, n, n_{eff}, the weighted median and the 16–84 % range of each quantity, fractions with their Wilson intervals, and, when hypotheses are compared, both values of Z with their n_{eff}.

9.4 Emulators

An ensemble maps the parameters of its members to summary columns, at a cost of seconds to hours per member. A neural *emulator* learns this map from a finished ensemble and evaluates it in microseconds. The networks are multi-layer perceptrons (scikit-learn, as in Vynatheya et al. 2022, 2023); inputs and regression targets are standardised (optionally after $\log_{10}$), angles can be mapped to $(\cos\varphi, \sin\varphi)$, and a known power law can be divided out of a target. A *deep ensemble* of networks with different seeds (default 5; Lakshminarayanan et al. 2017) gives the spread of the prediction; 20 % of the members are held out and give the scores. Two uses:

- **Inference.** Draw 10^6–10^7 parameter sets from the ensemble's distributions, predict the observed columns and weight every draw with the likelihood of the measurements, in which the variance of the emulator (spread of the deep ensemble plus its held-out residual) is added to that of the measurement. This is the conditioning above with far more members; precise measurements no longer collapse n_{eff}.
- **Classification.** A binary column (the ghost-test verdict, `fate==intact`) is learned by a classifier that returns a probability – for example the stability of a quadruple from its scale-free parameters, like the classifiers of Vynatheya et al. (2022, 2023).

An emulator is only as good as its ensemble: outside the sampled ranges and where members are sparse it extrapolates, which the deep-ensemble spread shows only partly.

9.5 Simulation-based inference

Neural posterior estimation (NPE; Papamakarios & Murray 2016; Greenberg et al. 2019), with the `sbi` package (Tejero-Cantero et al. 2020), trains a conditional density $q(\theta \mid x)$ on the pairs of drawn parameters θ and simulated observables x of an ensemble. Measurement errors enter as Gaussian noise added to the simulated observables, and hard limits select the training members. Afterwards the posterior for any measured x is sampled in under a second, without new simulations. It answers the same question as the importance weights, with the ensemble's distributions as the prior; weighting needs no training but its n_{eff} collapses for precise measurements, while NPE interpolates between members. A trained network can be wrong where the ensemble has few members, so it is checked against the weighting and by a coverage test: the fraction of held-out members whose true values lie inside their 68 % intervals (ideally 0.68), and simulation-based calibration (Talts et al. 2018), in which the rank of the true value among posterior samples is uniform for a calibrated posterior.

9.6 Implementation

External libraries. scikit-learn multi-layer perceptrons for the emulators and classifiers; the `sbi` package (Tejero-Cantero et al. 2020) with PyTorch for neural posterior estimation; the trained classifiers of Vynatheya et al. (2023) when installed. **Own code.** The drawing of members (all distributions, truncation, mirroring, requirements, training-set shaping) with deterministic seeds; the parallel execution with resumption; the reduction of every run to its summary row (including the central rates of change); the Wilson intervals; the observation weights with Gaussian terms, hard limits and direction prefixes; n_{eff} and weighted quantiles; the emulator layer (input and target transforms, deep ensembles, retraining of networks stuck on a plateau, emulator variance in the likelihood); and the SBI interface with noise augmentation, coverage and calibration tests.

Chapter 10

Accuracy and limits

This chapter collects what the previous chapters say about accuracy and scope: how precisely the numerical methods solve their equations, which physics is not modelled, in which regimes the models stop being valid, and how every piece of physics was validated.

10.1 Numerical accuracy

Table 15 lists the numerical accuracy of the main methods – the error with which SWINGERsim solves its own equations, not the error of the physics.

Table 15. Numerical accuracy. *Method* is the step; *Measure* the quantity compared; *Accuracy* the achieved level; *Chapter* where the method is described.

Method	Measure	Accuracy	Chapter
elements → state → elements	elements after the round trip	10^{-12}	Hierarchies and orbits
N-body integration (IAS15)	relative energy error, Newtonian	$\sim 10^{-14}$	Dynamics
N-body with 1PN	1PN Hamiltonian	10^{-13}–10^{-11}	Dynamics
forward and backward run	initial elements recovered	10^{-12}	Dynamics
secular backends	relative tolerance of DOP853; conserved H checked	10^{-10} (default)	Dynamics
SSE interpolation	radius against direct SSE (95th percentile)	0.8 % MS, 1.7 % HG, 4 % GB	Stars
MIST interpolation	radius at the end of the MS, ages (twice the grid spacing)	1 %, 1–2 %	Stars
sphere occultations	flux against Mandel & Agol (2002)	10^{-8}	Observables
surface meshes (64 × 128)	flux against exact sphere eclipses	3×10^{-4}	Observables
disc occultation (4000 points)	flux at a sharp edge	$\sim 10^{-3}$	Observables
synthetic minima in the fit	times against dense timings	0.06 s	Fitting
imposed period change	deviation from the parabola	< 0.05 s	Mass transfer and evolution

Method	Measure	Accuracy	Chapter
radial velocities	emission-time mismatch	< 1 ms	Observables

Numerical accuracy is rarely the limiting factor. The limits come from the physics and from the data.

10.2 Regime limits

- **Secular backends** require strong hierarchies ($\varepsilon_{\rm SA}$ well below 0.01), no mean-motion resonances and no close encounters; in chaotic regions they agree with the N-body solution only for a limited time (Dynamics).
- **Chaos** limits every single run to a few Lyapunov times, forward and backward; beyond, only ensembles are meaningful (Dynamics).
- **Stellar models**: SSE stops at helium ignition and covers 0.1–8 $M_\odot$; MIST covers 0.1–300 $M_\odot$ up to the departure from the AGB (massive stars: carbon burning); remnants are not evolved; the masses and radii of evolved massive stars are model-dependent at the level of tens of per cent (Stars).
- **Structure constants** from polytropes are rough (a factor of about three for 2–3 $M_\odot$ stars); Claret's grids cover 0.2–8 $M_\odot$ (Claret 2023) and up to 125 $M_\odot$ at one metallicity (Claret 2004).
- **Tides**: the equilibrium-tide dissipation is uncertain by large factors; main-sequence circularisation of old clusters is not reproduced (Tides and rotation).
- **Mass transfer** follows BSE recipes for circular binaries; helium stars and wind accretion are not modelled (Mass transfer and evolution).
- **Light curves**: Roche shapes only for circular synchronised binaries; the small-distortion terms of the sphere model err by 3–17 % in the ellipsoidal amplitude at R/a = 0.10–0.25 and should be replaced by meshes for close pairs (Observables).
- **Stability tools** refer to about 100 outer orbits; the MLP classifiers are approximations outside their training set (Stability and fates).
- **Fits** find local minima; their linear errors hold in straight χ^2 valleys and MCMC errors only for converged chains (Fitting).

10.3 What is not modelled

Table 16 lists the effects that lie outside the scope of SWINGERsim, with what replaces them where anything does.

Table 16. Effects outside the scope of SWINGERsim. *Effect* is the physics; *Consequence* what its absence means for the results and what replaces it where something does.

Effect	Consequence
hydrodynamics (gas flows, common-envelope hydrodynamics, collisions as fluid events)	gas enters only parametrically (damping of orbits during formation); common envelopes and mergers through energy and mass recipes
magnetohydrodynamics (dynamos, resolved magnetic fields)	magnetism enters through parametrised braking and activity laws calibrated on MHD and observations
gas drag in scattering experiments	tidal capture is included, drag is not
eccentric or asynchronous distorted binaries in the mesh light curves	Roche geometry assumes circular synchronised orbits; the PHOEBE hybrid handles eccentric distorted binaries
post-AGB stars, white dwarfs, neutron stars, black holes as evolving stars	they can be point masses; a common-envelope core is a point mass of constant mass

Effect	Consequence
helium stars, wind accretion	a donor stripped to its core ends the history; winds are isotropic and leave the system
main-sequence tidal dissipation beyond the equilibrium tide and the two-layer estimate of inertial waves	main-sequence circularisation periods of old clusters too short (M67: 7.5 d instead of 12.1 d; Barker 2022 explains M67 with inertial waves computed from full stellar profiles)
stellar spots in the light curves	activity cycles enter the eclipse timings, spots not the light
pulsation shapes, amplitudes from the structure, gravity-mode spacings from the structure	modes are sinusoids of fixed amplitude; SPB and γ Dor stars need P_0 and ΔP from the user
disc mass and gravity, warps, forward scattering in thin parts	discs are massless slabs or rigidly precessing planes
astrometry (photocentre motion, Gaia astrometric orbits)	positions are in the output, but no astrometric observable is computed or fitted
supernovae and natal kicks	not modelled

10.4 The validation principle

Every physics module of SWINGERsim is compared with published results before it is used: analytic formulae, published numerical results, other codes (REBOUND/REBOUNDx examples, MSE, PHOEBE, BSE/COSMIC, MESA, batman, ellc) and observed systems. Each comparison is one number with a tolerance fixed before the comparison, and its verdict is recorded: passed, explained (outside the tolerance for a documented reason that is not an error of SWINGERsim – for example published numbers that contradict each other, or a stated limitation of the physical model), or failed. The tolerance is not widened to hide a disagreement. The one-sentence summaries in the previous chapters (“reproduces … to …”) are the outcomes of these comparisons; the comparisons themselves, with their sources, inputs and results, are documented in *Testing SWINGERsim, Part I: Validation against the literature*. Parts II–IV of the testing documents apply the full chain to published multiple stars, to eclipsing binaries with changing inclinations and to special cases.

10.5 Implementation

The accuracy figures of Table 15 are measured by SWINGERsim’s own test suite and benchmark collection (reproduction of published results), which are part of the code. The integrators whose round-off behaviour sets the floor are external: IAS15 of REBOUND (Rein & Spiegel 2015) and DOP853 of SciPy.

Chapter 11

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